

# User Manual

for S32 MCL Driver

Document Number: UM11MCLASR4.4 Rev0000R4.0.2 Rev. 1.0

|                                                               |           |
|---------------------------------------------------------------|-----------|
| <b>1 Revision History</b>                                     | <b>2</b>  |
| <b>2 Introduction</b>                                         | <b>3</b>  |
| 2.1 Supported Derivatives . . . . .                           | 3         |
| 2.2 Overview . . . . .                                        | 4         |
| 2.3 About This Manual . . . . .                               | 4         |
| 2.4 Acronyms and Definitions . . . . .                        | 5         |
| 2.5 Reference List . . . . .                                  | 5         |
| <b>3 Driver</b>                                               | <b>6</b>  |
| 3.1 Requirements . . . . .                                    | 6         |
| 3.2 Driver Design Summary . . . . .                           | 6         |
| 3.3 Hardware Resources . . . . .                              | 7         |
| 3.4 Deviations from Requirements . . . . .                    | 7         |
| 3.5 Driver Limitations . . . . .                              | 7         |
| 3.6 Driver usage and configuration tips . . . . .             | 8         |
| 3.6.1 MCAL MCL DMA migration guide to RTD MCL . . . . .       | 8         |
| 3.6.2 Channel State Machine . . . . .                         | 12        |
| 3.7 Runtime errors . . . . .                                  | 16        |
| 3.8 Symbolic Names Disclaimer . . . . .                       | 17        |
| <b>4 Tresos Configuration Plug-in</b>                         | <b>18</b> |
| 4.1 Module Mcl . . . . .                                      | 21        |
| 4.2 Container MclGeneral . . . . .                            | 22        |
| 4.3 Parameter MclEnableDevErrorDetect . . . . .               | 22        |
| 4.4 Parameter Mcl_VersionInfoApi . . . . .                    | 23        |
| 4.5 Parameter MclEnableUserModeSupport . . . . .              | 23        |
| 4.6 Parameter MclEnableMulticoreSupport . . . . .             | 24        |
| 4.7 Parameter MclEnableVirtualAddressMappingSupport . . . . . | 24        |
| 4.8 Reference MclEcucPartitionRef . . . . .                   | 24        |
| 4.9 Container MclDma . . . . .                                | 25        |
| 4.10 Parameter MclEnableDma . . . . .                         | 25        |
| 4.11 Parameter MclEnableCrc . . . . .                         | 26        |
| 4.12 Container MclCache . . . . .                             | 26        |
| 4.13 Parameter MclEnableCache . . . . .                       | 27        |
| 4.14 Container MclFtmCommon . . . . .                         | 27        |
| 4.15 Parameter Mcl_FtmCommonTimebase . . . . .                | 27        |
| 4.16 Container MclConfig . . . . .                            | 28        |
| 4.17 Container MclVirtualMemorySection . . . . .              | 28        |
| 4.18 Parameter MclVirtualAddressStart . . . . .               | 29        |
| 4.19 Parameter MclVirtualAddressEnd . . . . .                 | 29        |

|      |           |                                           |    |
|------|-----------|-------------------------------------------|----|
| 4.20 | Parameter | MclPhysicalAddressStart                   | 30 |
| 4.21 | Parameter | MclPhysicalAddressEnd                     | 30 |
| 4.22 | Container | dmaLogicInstance_ConfigType               | 31 |
| 4.23 | Parameter | dmaLogicInstance_IdName                   | 31 |
| 4.24 | Parameter | dmaLogicInstance_hwId                     | 32 |
| 4.25 | Parameter | dmaLogicInstance_enDebug                  | 32 |
| 4.26 | Parameter | dmaLogicInstance_enRoundRobin             | 33 |
| 4.27 | Parameter | dmaLogicInstance_enHaltAfterError         | 34 |
| 4.28 | Parameter | dmaLogicInstance_enChLinking              | 34 |
| 4.29 | Parameter | dmaLogicInstance_enGIMasterIdReplication  | 35 |
| 4.30 | Reference | dmaLogicInstance_EcucPartitionRef         | 35 |
| 4.31 | Container | dmaLogicInstance_DmaCrc                   | 36 |
| 4.32 | Parameter | dmaLogicInstance_DmaCrcEnSwapBit          | 36 |
| 4.33 | Parameter | dmaLogicInstance_DmaCrcEnSwapByte         | 37 |
| 4.34 | Parameter | dmaLogicInstance_DmaCrcEnGlobal           | 37 |
| 4.35 | Container | dmaLogicChannel_Type                      | 38 |
| 4.36 | Parameter | dmaLogicChannel_LogicName                 | 38 |
| 4.37 | Parameter | dmaLogicChannel_HwInstId                  | 39 |
| 4.38 | Parameter | dmaLogicChannel_HwChId                    | 39 |
| 4.39 | Parameter | dmaLogicChannel_InterruptCallback         | 40 |
| 4.40 | Parameter | dmaLogicChannel_ErrorInterruptCallback    | 40 |
| 4.41 | Parameter | dmaLogicChannel_EnableGlobalConfig        | 41 |
| 4.42 | Parameter | dmaLogicChannel_EnableTransferConfig      | 41 |
| 4.43 | Parameter | dmaLogicChannel_EnableScatterGather       | 42 |
| 4.44 | Parameter | dmaLogicChannel_EnableCrcConfig           | 42 |
| 4.45 | Reference | dmaLogicChannel_EcucPartitionRef          | 43 |
| 4.46 | Container | dmaLogicChannel_ConfigType                | 43 |
| 4.47 | Container | dmaLogicChannel_GlobalConfigType          | 44 |
| 4.48 | Container | dmaLogicChannelConfig_GlobalControlType   | 44 |
| 4.49 | Parameter | dmaGlobalControl_enMasterIdReplication    | 45 |
| 4.50 | Parameter | dmaGlobalControl_enBufferedWrites         | 45 |
| 4.51 | Container | dmaLogicChannelConfig_GlobalRequestType   | 46 |
| 4.52 | Parameter | dmaGlobalRequest_enDmamuxTrigger          | 46 |
| 4.53 | Parameter | dmaGlobalRequest_enDmamuxSource           | 46 |
| 4.54 | Parameter | dmaGlobalRequest_Dmamux0HwRequest         | 47 |
| 4.55 | Parameter | dmaGlobalRequest_Dmamux1HwRequest         | 48 |
| 4.56 | Parameter | dmaGlobalRequest_Dmamux2HwRequest         | 49 |
| 4.57 | Parameter | dmaGlobalRequest_Dmamux3HwRequest         | 50 |
| 4.58 | Parameter | dmaGlobalRequest_enDmaRequest             | 51 |
| 4.59 | Container | dmaLogicChannelConfig_GlobalInterruptType | 52 |

|                                                                                     |    |
|-------------------------------------------------------------------------------------|----|
| 4.60 Parameter dmaGlobalInterrupt_enDmaErrorInterrupt . . . . .                     | 52 |
| 4.61 Container dmaLogicChannelConfig_GlobalPriorityType . . . . .                   | 52 |
| 4.62 Parameter dmaGlobalPriority_GroupPriority . . . . .                            | 54 |
| 4.63 Parameter dmaGlobalPriority_LevelPriority . . . . .                            | 54 |
| 4.64 Parameter dmaGlobalPriority_enPreemption . . . . .                             | 55 |
| 4.65 Parameter dmaGlobalPriority_disPreempt . . . . .                               | 55 |
| 4.66 Container dmaLogicChannel_TransferConfigType . . . . .                         | 56 |
| 4.67 Container dmaLogicChannelConfig_TransferControlType . . . . .                  | 56 |
| 4.68 Parameter dmaLogicChannelConfig_enDmaMajorInterrupt . . . . .                  | 57 |
| 4.69 Parameter dmaLogicChannelConfig_enDmaHalfMajorInterrupt . . . . .              | 57 |
| 4.70 Parameter dmaLogicChannelConfig_disDmaAutoHwReq . . . . .                      | 58 |
| 4.71 Parameter dmaLogicChannelConfig_bandwidthControl . . . . .                     | 58 |
| 4.72 Parameter dmaLogicChannelConfig_ScatterGatherAddressType . . . . .             | 59 |
| 4.73 Parameter dmaLogicChannelConfig_DestinationStoreAddressType . . . . .          | 59 |
| 4.74 Container dmaLogicChannelConfig_TransferSourceType . . . . .                   | 59 |
| 4.75 Parameter dmaLogicChannelConfig_SourceAddressType . . . . .                    | 61 |
| 4.76 Parameter dmaLogicChannelConfig_SourceSignedOffsetType . . . . .               | 61 |
| 4.77 Parameter dmaLogicChannelConfig_SourceLastAddressAdjustmentType . . . . .      | 62 |
| 4.78 Parameter dmaTransferConfig_TransferSizeType . . . . .                         | 62 |
| 4.79 Parameter dmaLogicChannelConfig_SourceModuloType . . . . .                     | 63 |
| 4.80 Container dmaLogicChannelConfig_TransferDestinationType . . . . .              | 63 |
| 4.81 Parameter dmaLogicChannelConfig_DestinationAddressType . . . . .               | 64 |
| 4.82 Parameter dmaLogicChannelConfig_DestinationSignedOffsetType . . . . .          | 64 |
| 4.83 Parameter dmaLogicChannelConfig_DestinationLastAddressAdjustmentType . . . . . | 64 |
| 4.84 Parameter dmaTransferConfig_TransferSizeType . . . . .                         | 65 |
| 4.85 Parameter dmaLogicChannelConfig_DestinationModuloType . . . . .                | 65 |
| 4.86 Container dmaLogicChannelConfig_TransferMinorLoopType . . . . .                | 66 |
| 4.87 Parameter dmaLogicChannelConfig_enSourceOffset . . . . .                       | 66 |
| 4.88 Parameter dmaLogicChannelConfig_enDestinationOffset . . . . .                  | 67 |
| 4.89 Parameter dmaLogicChannelConfig_OffsetValueType . . . . .                      | 67 |
| 4.90 Parameter dmaLogicChannelConfig_enMinorLoopLinkCh . . . . .                    | 68 |
| 4.91 Parameter dmaLogicChannelConfig_MinorLoopSizeType . . . . .                    | 68 |
| 4.92 Reference dynamic_dmaLogicChannelConfig_MinorLoopLinkChValueType . . . . .     | 69 |
| 4.93 Container dmaLogicChannelConfig_TransferMajorLoopType . . . . .                | 69 |
| 4.94 Parameter dmaLogicChannelConfig_enMajorLoopLinkCh . . . . .                    | 69 |
| 4.95 Parameter dmaLogicChannelConfig_MajorLoopCountType . . . . .                   | 70 |
| 4.96 Reference dynamic_dmaLogicChannelConfig_MajorLoopLinkChValueType . . . . .     | 70 |
| 4.97 Container dmaLogicChannel_ScatterGatherConfigType . . . . .                    | 71 |
| 4.98 Container dmaLogicChannelConfig_ScatterGatherArrayType . . . . .               | 71 |
| 4.99 Container dmaLogicChannelConfig_ScatterGatherElementConfigType . . . . .       | 72 |

|       |                                                                                      |    |
|-------|--------------------------------------------------------------------------------------|----|
| 4.100 | Parameter dmaLogicChannelConfig_ScatterGatherElementNameType . . . . .               | 72 |
| 4.101 | Parameter dmaLogicChannelConfig_LastElementLink_ScatterGatherType . . . . .          | 73 |
| 4.102 | Parameter dmaLogicChannelConfig_enScatterGatherConfig . . . . .                      | 73 |
| 4.103 | Reference dynamic_dmaLogicChannelConfig_BasicElementLink_ScatterGatherType . . . . . | 74 |
| 4.104 | Container dmaLogicChannelConfig_TransferControlType . . . . .                        | 74 |
| 4.105 | Parameter dmaLogicChannelConfig_enStart . . . . .                                    | 75 |
| 4.106 | Parameter dmaLogicChannelConfig_enDmaMajorInterrupt . . . . .                        | 75 |
| 4.107 | Parameter dmaLogicChannelConfig_enDmaHalfMajorInterrupt . . . . .                    | 75 |
| 4.108 | Parameter dmaLogicChannelConfig_disDmaAutoHwReq . . . . .                            | 76 |
| 4.109 | Parameter dmaLogicChannelConfig_bandwidthControl . . . . .                           | 76 |
| 4.110 | Parameter dmaLogicChannelConfig_ScatterGatherAddressType . . . . .                   | 77 |
| 4.111 | Parameter dmaLogicChannelConfig_DestinationStoreAddressType . . . . .                | 77 |
| 4.112 | Container dmaLogicChannelConfig_TransferSourceType . . . . .                         | 78 |
| 4.113 | Parameter dmaLogicChannelConfig_SourceAddressType . . . . .                          | 78 |
| 4.114 | Parameter dmaLogicChannelConfig_SourceSignedOffsetType . . . . .                     | 79 |
| 4.115 | Parameter dmaLogicChannelConfig_SourceLastAddressAdjustmentType . . . . .            | 79 |
| 4.116 | Parameter dmaTransferConfig_TransferSizeType . . . . .                               | 80 |
| 4.117 | Parameter dmaLogicChannelConfig_SourceModuloType . . . . .                           | 80 |
| 4.118 | Container dmaLogicChannelConfig_TransferDestinationType . . . . .                    | 81 |
| 4.119 | Parameter dmaLogicChannelConfig_DestinationAddressType . . . . .                     | 81 |
| 4.120 | Parameter dmaLogicChannelConfig_DestinationSignedOffsetType . . . . .                | 81 |
| 4.121 | Parameter dmaLogicChannelConfig_DestinationLastAddressAdjustmentType . . . . .       | 82 |
| 4.122 | Parameter dmaTransferConfig_TransferSizeType . . . . .                               | 82 |
| 4.123 | Parameter dmaLogicChannelConfig_DestinationModuloType . . . . .                      | 83 |
| 4.124 | Container dmaLogicChannelConfig_TransferMinorLoopType . . . . .                      | 83 |
| 4.125 | Parameter dmaLogicChannelConfig_enSourceOffset . . . . .                             | 84 |
| 4.126 | Parameter dmaLogicChannelConfig_enDestinationOffset . . . . .                        | 84 |
| 4.127 | Parameter dmaLogicChannelConfig_OffsetValueType . . . . .                            | 85 |
| 4.128 | Parameter dmaLogicChannelConfig_enMinorLoopLinkCh . . . . .                          | 85 |
| 4.129 | Parameter dmaLogicChannelConfig_MinorLoopSizeType . . . . .                          | 85 |
| 4.130 | Reference dynamic_dmaLogicChannelConfig_MinorLoopLinkChValueType . . . . .           | 86 |
| 4.131 | Container dmaLogicChannelConfig_TransferMajorLoopType . . . . .                      | 86 |
| 4.132 | Parameter dmaLogicChannelConfig_enMajorLoopLinkCh . . . . .                          | 87 |
| 4.133 | Parameter dmaLogicChannelConfig_MajorLoopCountType . . . . .                         | 87 |
| 4.134 | Reference dynamic_dmaLogicChannelConfig_MajorLoopLinkChValueType . . . . .           | 88 |
| 4.135 | Container dmaLogicChannel_CrcConfigType . . . . .                                    | 88 |
| 4.136 | Parameter dmaLogicChannel_HwCrcChId . . . . .                                        | 89 |
| 4.137 | Parameter dmaLogicChannel_CrcMode . . . . .                                          | 89 |
| 4.138 | Parameter dmaLogicChannel_CrcPolynomial . . . . .                                    | 90 |
| 4.139 | Parameter dmaLogicChannel_CrcInitialValue . . . . .                                  | 90 |

|                                                        |           |
|--------------------------------------------------------|-----------|
| 4.140 Parameter dmaLogicChannel_CrcEnInitSel . . . . . | 91        |
| 4.141 Parameter dmaLogicChannel_CrcEnLogic . . . . .   | 91        |
| 4.142 Container CommonPublishedInformation . . . . .   | 92        |
| 4.143 Parameter ArReleaseMajorVersion . . . . .        | 92        |
| 4.144 Parameter ArReleaseMinorVersion . . . . .        | 93        |
| 4.145 Parameter ArReleaseRevisionVersion . . . . .     | 93        |
| 4.146 Parameter ModuleId . . . . .                     | 94        |
| 4.147 Parameter SwMajorVersion . . . . .               | 94        |
| 4.148 Parameter SwMinorVersion . . . . .               | 95        |
| 4.149 Parameter SwPatchVersion . . . . .               | 95        |
| 4.150 Parameter VendorId . . . . .                     | 96        |
| <b>5 Module Index . . . . .</b>                        | <b>97</b> |
| 5.1 Software Specification . . . . .                   | 97        |
| <b>6 Module Documentation . . . . .</b>                | <b>98</b> |
| 6.1 CACHE IP Driver . . . . .                          | 98        |
| 6.1.1 Detailed Description . . . . .                   | 98        |
| 6.1.2 Enum Reference . . . . .                         | 98        |
| 6.1.3 Function Reference . . . . .                     | 99        |
| 6.2 MCL Driver . . . . .                               | 104       |
| 6.2.1 Detailed Description . . . . .                   | 104       |
| 6.2.2 Data Structure Documentation . . . . .           | 107       |
| 6.2.3 Macro Definition Documentation . . . . .         | 112       |
| 6.2.4 Enum Reference . . . . .                         | 119       |
| 6.2.5 Function Reference . . . . .                     | 124       |
| 6.3 DMA IP Driver . . . . .                            | 136       |
| 6.3.1 Detailed Description . . . . .                   | 136       |
| 6.3.2 Data Structure Documentation . . . . .           | 138       |
| 6.3.3 Types Reference . . . . .                        | 146       |
| 6.3.4 Enum Reference . . . . .                         | 146       |
| 6.3.5 Function Reference . . . . .                     | 151       |
| 6.4 FTM IP Driver . . . . .                            | 161       |
| 6.4.1 Detailed Description . . . . .                   | 161       |
| 6.4.2 Function Reference . . . . .                     | 161       |



## Chapter 1

### Revision History

| Revision | Date       | Author       | Description                                             |
|----------|------------|--------------|---------------------------------------------------------|
| 1.0      | 30.06.2023 | NXP RTD Team | S32 Real-Time Drivers AUTOSAR 4.4 Version 4.0.2 Release |

## Chapter 2

### Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This User Manual describes NXP Semiconductor MCL for *platform*. MCL driver configuration parameters and deviations from the specification are described in Driver chapter of this document. MCL driver requirements and APIs are described in the MCL driver software specification document.

### 2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32g274a\_bga525
- s32g254a\_bga525
- s32g233a\_bga525
- s32g234m\_bga525
- s32g378a\_bga525
- s32g379a\_bga525
- s32g398a\_bga525
- s32g399a\_bga525
- s32g338m\_bga525
- s32g339m\_bga525
- s32g358a\_bga525
- s32g359a\_bga525
- s32r45\_bga780

All of the above microcontroller devices are collectively named as S32.



## 2.2 Overview

**AUTOSAR (AUTomotive Open System ARchitecture)** is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

## 2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

## 2.4 Acronyms and Definitions

| Term  | Definition                               |
|-------|------------------------------------------|
| API   | Application Programming Interface        |
| ASM   | Assembler                                |
| BSMI  | Basic Software Make file Interface       |
| CAN   | Controller Area Network                  |
| C/CPP | C and C++ Source Code                    |
| CS    | Chip Select                              |
| CTU   | Cross Trigger Unit                       |
| DEM   | Diagnostic Event Manager                 |
| DET   | Development Error Tracer                 |
| DMA   | Direct Memory Access                     |
| ECU   | Electronic Control Unit                  |
| FIFO  | First In First Out                       |
| LSB   | Least Significant Bit                    |
| MCU   | Micro Controller Unit                    |
| MIDE  | Multi Integrated Development Environment |
| MSB   | Most Significant Bit                     |
| N/A   | Not Applicable                           |
| RAM   | Random Access Memory                     |
| SIU   | Systems Integration Unit                 |
| SWS   | Software Specification                   |
| VLE   | Variable Length Encoding                 |
| XML   | Extensible Markup Language               |

## 2.5 Reference List

| #  | Title                   | Version                                 |
|----|-------------------------|-----------------------------------------|
| 1  | S32G2 Reference Manual  | Rev. 7, February 2023                   |
| 2  | S32G3 Reference Manual  | Rev. 3 , Feb 2023                       |
| 3  | S32R45 Reference Manual | Rev. 3, 12/2021                         |
| 4  | S32G2 Errata Document   | Mask Set Errata for Mask 0P77B v3.0     |
| 5  | S32G2 Errata Document   | Mask Set Errata for Mask 1P77B v1.0     |
| 6  | S32G3 Errata Document   | Mask Set Errata for Mask 1P72B v2.1     |
| 7  | S32R45 Errata Document  | Mask Set Errata for Mask P57D, Rev. 2.0 |
| 8  | S32G2 Data Sheet        | Rev 6, Dec 2022                         |
| 9  | S32G3 Data Sheet        | Rev 2, RC, Feb 2023                     |
| 10 | VR5510 Data Sheet       | Rev 5, April 2022                       |
| 11 | S32R45 Data Sheet       | Rev. 3 — 11/2022                        |

## Chapter 3

### Driver

- [Requirements](#)
- [Driver Design Summary](#)
- [Hardware Resources](#)
- [Deviations from Requirements](#)
- [Driver Limitations](#)
- [Driver usage and configuration tips](#)
- [Runtime errors](#)
- [Symbolic Names Disclaimer](#)

### 3.1 Requirements

Requirements for this driver are detailed in the Driver Software Specification document (See Table [Reference List](#) ).

For CDD: MCL Driver is a Complex Device Driver (CDD), so there are no requirements regarding this module.

It has vendor-specific requirements and implementation.

### 3.2 Driver Design Summary

The Mcl Driver controls the DMA(Direct Memory Access) and CACHE modules of the S32 device. It provides the following features:

- Configuration and initialization of the DMA.
- Configuration and initialization of the CACHE.
- Handling of the DMA interrupt requests.
- DMA Normal Transfer Mode and Scatter/Gather Mode.

### 3.3 Hardware Resources

The Mcl Driver consists of:

1. A DMA Peripheral which has 2 Hardware Instance with 32 Hardware Channels.
2. A CACHE Memory included into the ARM Core Peripherals.

### 3.4 Deviations from Requirements

The driver deviates from the MCL Driver software specification in some places. The table below identifies the requirements that are not implemented or out of scope for the MCL Driver.

| Term | Definition            |
|------|-----------------------|
| N/S  | Out of scope          |
| N/I  | Not implemented       |
| N/F  | Not fully implemented |

Below table identifies the requirements that are not fully implemented, implemented differently or out of scope for the MCL driver.

| Requirement       | Status | Description                                                                                                                                                                                                                                                                        | Notes                                                                                                                               |
|-------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| CPR_RTD_00190.mcl | N/S    | All modules which need to call Dem module shall provide a configuration parameter for disabling all calls of Dem_SetEventStatus.<br>If this parameter is activated, no call of Dem_SetEventStatus must be performed. Per default, the call of Dem_SetEventStatus shall be allowed. | Only K1xx support this requirements ( <a href="https://jira.sw.nxp.com/browse/AAI-767">https://jira.sw.nxp.com/browse/AAI-767</a> ) |

### 3.5 Driver Limitations

The Mcl Driver has the following limitations:

- When DMA is used with CACHE enabled, the user shall Invalidate/Clean the CACHE in order to synchronize the transfered data.
- When using CACHE Invalidate and Clean functionalities, the user shall take into consideration that all variables that reside in the cache, are affected.
- The Cache Invalidate function need to be called before Cache Enable function is called (make sure that Cache wasn't enabled before).

- The Cache Clean function need to be called before Cache Disable function is called.
- When the DMA transfer has errors, user must call `Mcl_SetDmaChannelCommand/Dma_Ip_SetLogicChannelCommand` function to clear error state and error status.
- When using Virtual Address Mapping feature, the user shall handle the linker file to avoid the memory address is invalid.
- The DMA Driver shall have the Source Address and Destination Address configured at runtime. These two parameters are not available in the configurator.
- In Scatter/Gather Mode, the Logic Channel will not execute the interrupt callback in a specific setup:
  - Setup:
    - TCD0 : Interrupt enabled and Mcl callback set;
    - TCD1 : Interrupt disabled.
  - Interrupt handler implementation:
    - Step 1: Check interrupt event;
    - Step 2: Clear interrupt flag;
    - Step 3: Check if interrupt is enabled;
    - Step 4: Call the callback function.

But in this case, when TCD0 is completed an interrupt occurs and the handler will execute. At the same time TCD1 is also loaded. When the handler checks if interrupt is enabled (Step 3), the value found in the register is 0 (interrupt disabled), thus the callback function is not called.

Note:

- `DMA_CRC` documentation: S32G2 Reference Manual Rev. 4 is not correct. Only S32G3 Reference Manual is correct. This information was confirmed by Documentation team.
- End of package feature: This feature is not enabled on S32G2 and S32G3.
- Recommendation: Please avoid placing the stack into a cacheable area. Rationale: If the cache policy is write-back and the stack is placed in a cacheable area, synchronization issues might appear. The application is responsible to handle such coherency issue.

## 3.6 Driver usage and configuration tips

### 3.6.1 MCAL MCL DMA migration guide to RTD MCL

### 3.6.1.1 Introduction

The RTD MCL DMA Driver brings a Generic Interface, to help the User in application development across multiple SoCs. The Generic Interface consists of software functions that are fully configurable using User defined configurations. The Generic Interface is structured into four function groups:

1. Set Command Functions
2. Get Status Functions
3. Set List Functions
4. Get Information (Parameter) Functions

The Set Command Functions shall trigger actions specific to the invoked Logic Entity. For example, the Logic Dma Instance shall be commanded to: “Stop execution”, “Stop execution with error signaling”, “Pause execution” or “Resume execution”.

The Get Status Functions shall read the status specific to the invoked Logic Entity. For example, the Logic Dma Instance shall return: “Hardware Errors specific to the IP”, “The Active Channel Id” and “Active Status”.

The Set List Functions shall configure a user defined list of settings for the invoked Logic Entity. For example, the Logic Dma Channel shall be configurable with the Transfer List of Parameters like: “Source Address”, “← Destination Address”, “Source Signed Offset”, “Destination Signed Offset”, “Major Loop Count”, etc... For the specific ScatterGather mode, the additional Logic Element parameter specifies the Software TCD that shall be loaded with the ScatterGather List of Parameters.

The Get Parameter Functions shall return specific parameter value. For example, the Logic Dma Channel shall return the “Destination Address” value, which contains the last accessed destination memory location.

### 3.6.1.2 TRESOS Configuration

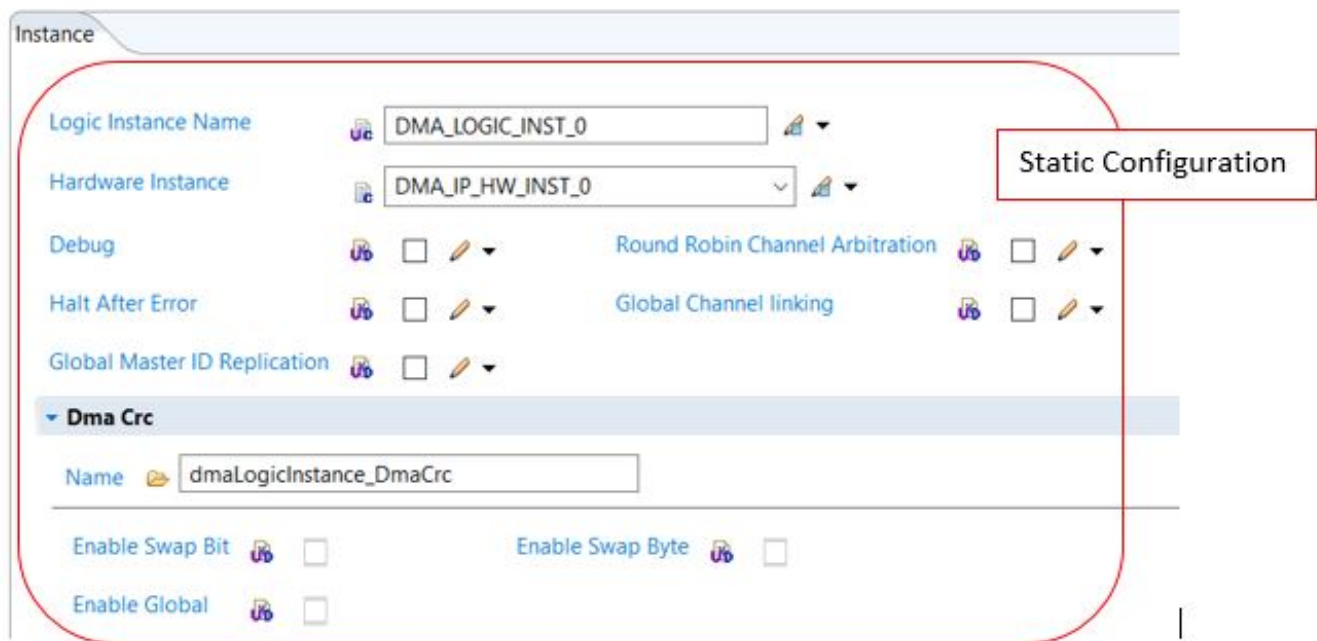


Figure 3.1 DMA Logic Instance

The DMA Logic Instance configuration presented in Figure 3.1 is static, thus it can't be changed dynamically during runtime. The "Logic Instance Name" represents the Handler(Tag) used with the DMA API. The Handler(Tag) is set by the User for the specific Application; for example the "DMA\_LOGIC\_INSTANCE\_0" shall be changed to "DMA\_LOGIC\_INST\_COMMUNICATION".

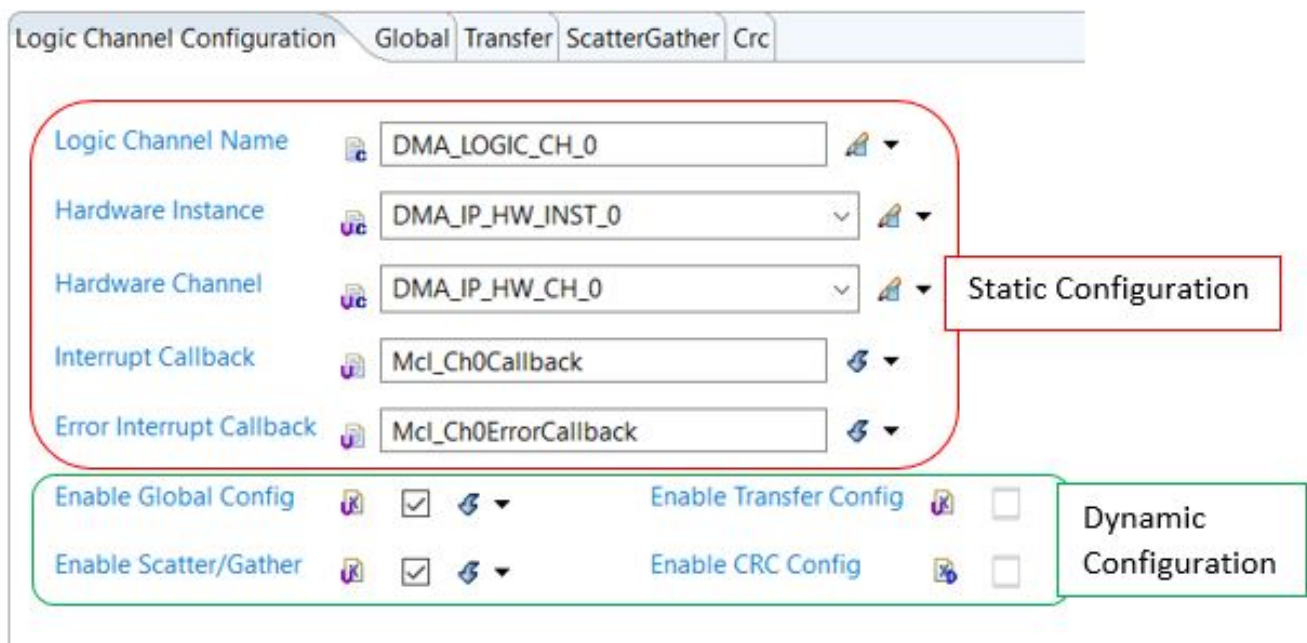


Figure 3.2 DMA Logic Channel

The DMA Logic Channel configuration in Figure 3.2 is composed of static and dynamic settings.

1. The static configuration is needed in order for the DMA Logic Channel to be created and used. The static configuration allocates resources for the DMA Logic Channel basic configuration. The static configuration can't be changed during runtime. The "Logic Channel Name" represents the Handler(Tag) used with the DMA API. The Handler(Tag) is set by the User based on the Application; For example the "DMA\_LOGIC\_CH\_0" shall be changed to "DMA\_CH\_CAN0\_RX" or "DMA\_LOGIC\_CH\_SPI2\_TX".
2. The dynamic configuration can be set in TRESOS and shall be automatically loaded in the DMA Logic Channel at initialization time. By enabling a checkbox, the respective configuration option is enabled and system memory is allocated during generation. The dynamic configuration can be set during runtime by using the specific API. By disabling a checkbox, the respective configuration option is disabled and system memory is not allocated during generation.

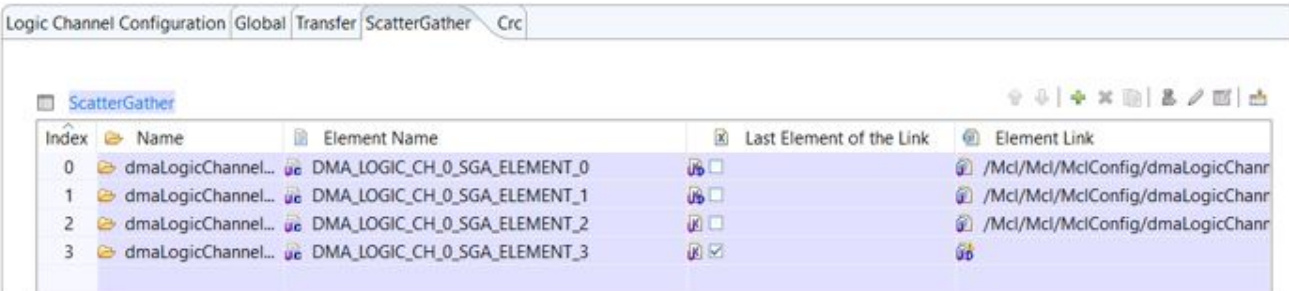
2.1. The "Enable Global Config" contains settings outside the Transfer Control Descriptor (TCD).

2.2. The "Enable Transfer Config" contains settings of the Transfer Control Descriptor (TCD).

2.3. The "Enable Scatter/Gather" contains settings that extend the "Transfer Config" by creating Software Transfer Control Descriptors (STCDs).

2.4. The "Enable CRC Config" contains settings for the DMA Logic Channel CRC computation functionality.

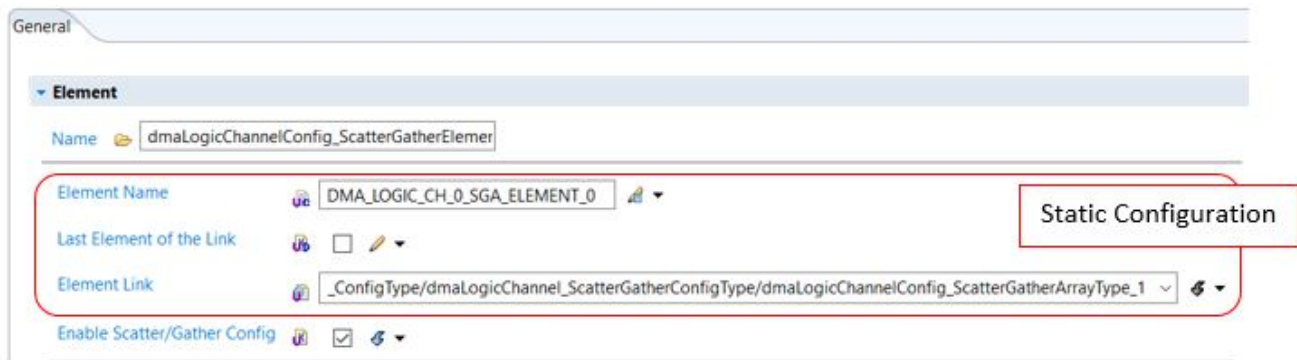
Note: The DMA Logic Channel can be configured in "Transfer Mode" or "Scatter/Gather Mode", thus only one of the 2 configurations can be set at any time. During runtime, the DMA Logic Channel can be configured between the 2 modes by calling the "SetTransfer" or "SetScatterGather" API.



| Index | Name               | Element Name                 | Last Element of the Link            | Element Link                     |
|-------|--------------------|------------------------------|-------------------------------------|----------------------------------|
| 0     | dmaLogicChannel... | DMA_LOGIC_CH_0_SGA_ELEMENT_0 | <input type="checkbox"/>            | /Mcl/Mcl/MclConfig/dmaLogicChanr |
| 1     | dmaLogicChannel... | DMA_LOGIC_CH_0_SGA_ELEMENT_1 | <input type="checkbox"/>            | /Mcl/Mcl/MclConfig/dmaLogicChanr |
| 2     | dmaLogicChannel... | DMA_LOGIC_CH_0_SGA_ELEMENT_2 | <input type="checkbox"/>            | /Mcl/Mcl/MclConfig/dmaLogicChanr |
| 3     | dmaLogicChannel... | DMA_LOGIC_CH_0_SGA_ELEMENT_3 | <input checked="" type="checkbox"/> |                                  |

**Figure 3.3 DMA Logic Channel – ScatterGather Element List**





**Figure 3.4 DMA Logic Channel – ScatterGather Element Configuration**

The DMA Logic Channel ScatterGather Element list (Figure 3.3) shall be loaded with the required number of elements. Each element represents a STCD and is part of a linked list of elements.

The DMA Logic Channel ScatterGather Element Configuration (Figure 3.4) contains the static configuration of the element.

1. The “Element Name” represents the Handler (Tag) (static configuration).
2. The “Last Element of the Link” sets the element as the last link element (static configuration).
3. The “Element Link” points to the next element of the link (static configuration).
4. The “Enable Scatter/Gather Config” enables the element configuration.

Note1: Element can’t be added during runtime. The elements are allocated resources during generation (System memory for STCDs and configuration if set). The element linkage can’t be changed during runtime.

Note2: The DMA Logic Channel ScatterGather Element List can be configured to contain multiple independent chained lists. During runtime, the Logic Channel can be assigned a different Chained List.

Note3: Each Element is allocated 32 bytes of memory space aligned to 32 bytes, representing the STCD. Additionally, if the configuration is enabled from TRESOS, additional 60 bytes are allocated.

## 3.6.2 Channel State Machine

### 3.6.2.1 MCL Driver

The MCL DMA Driver runs based on the presented State Machine Diagram.

The MCL DMA State Machine applies to the MCL DMA Channels and it contains 5 States.

The State Machine supports the following 21 Transitions:

T0: Hardware Reset  
 T1: Mcl\_Init() with DMA Channel no Transfer or Scatter/Gather generated configurations  
 T2: Mcl\_DeInit()  
 T3: Mcl\_SetDmaChannelTransferList()  
 T4: Mcl\_Init() with DMA Channel Transfer generated configuration  
 T5: Mcl\_SetDmaChannelScatterGatherList() + Mcl\_SetDmaChannelScatterGatherConfig()  
 T6: Mcl\_SetDmaChannelTransferList()  
 T7: Mcl\_SetDmaChannelScatterGatherList() + Mcl\_SetDmaChannelScatterGatherConfig()  
 T8: Mcl\_Init() with DMA Channel Scatter/Gather generated configuration  
 T9: Detection of channel error  
 T10: Mcl\_SetDmaInstanceCommand(CHANNEL, MCL\_DMA\_CH\_ACK\_ERROR)  
 T11: Detection of channel error  
 T12: Mcl\_DeInit()  
 T13: Mcl\_DeInit()  
 T14: Detection of channel error  
 T15: Detection of channel error  
 T16: Mcl\_DeInit()  
 T17: Mcl\_DeInit()  
 T18: Mcl\_SetDmaChannelCommand(CHANNEL, MCL\_DMA\_CH\_ACK\_ERROR)  
 T19: Mcl\_SetDmaChannelTransferList()  
 T20: Mcl\_SetDmaChannelScatterGatherList() + Mcl\_SetDmaChannelScatterGatherConfig()  
 T21: Detection of channel error

When the DMA Driver detects a software or hardware error for the Logic Channel, it shall enter the "MCL\_DMA\_CH\_ERROR\_STATE".

To exit from the "MCL\_DMA\_CH\_ERROR\_STATE", the application shall use the specified transitions.

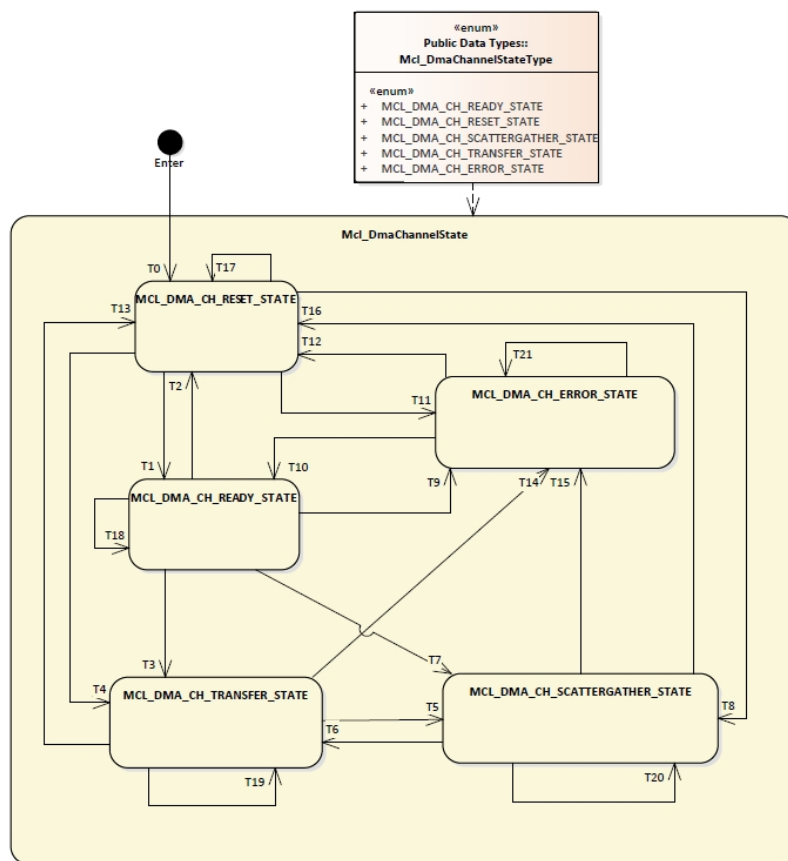


Figure 3.5 MCL DMA Channel State Machine

### 3.6.2.2 DMA Driver

The DMA IP Driver runs based on the presented State Machine Diagram.

The DMA State Machine applies to the DMA Logic Channels and it contains 5 States.

The State Machine supports the following 21 Transitions:

T0: Hardware Reset

T1: Dma\_Ip\_LogicChannelInit() with no Transfer or Scatter/Gather generated configurations

T2: Dma\_Ip\_LogicChannelDeinit()

T3: Dma\_Ip\_SetLogicChannelTransferList()

T4: Dma\_Ip\_LogicChannelInit() with Transfer generated configuration

T5: Dma\_Ip\_SetLogicChannelScatterGatherList() + Dma\_Ip\_SetLogicChannelScatterGatherConfig()

T6: Dma\_Ip\_SetLogicChannelTransferList()

T7: Dma\_Ip\_SetLogicChannelScatterGatherList() + Dma\_Ip\_SetLogicChannelScatterGatherConfig()

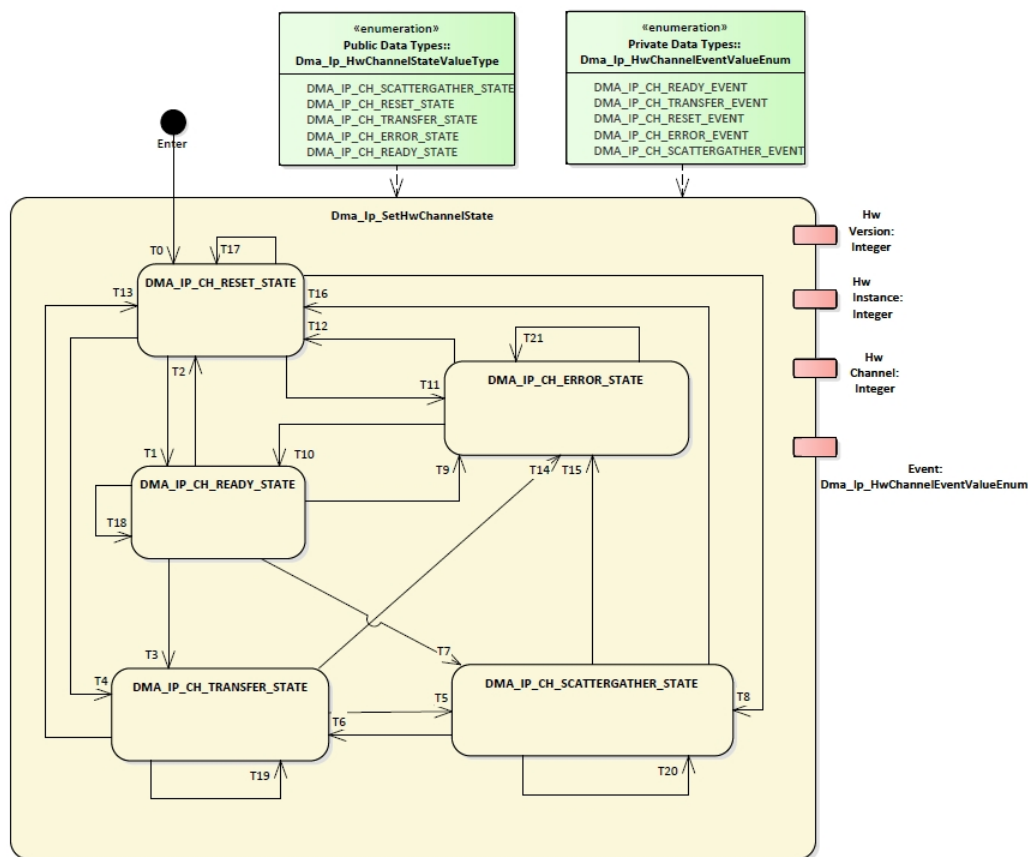
```

T8: Dma_Ip_LogicChannelInit() with Scatter/Gather generated configuration
T9: Detection of channel error
T10: Dma_Ip_SetLogicChannelCommand(CHANNEL, DMA_IP_CH_CLEAR_ERROR)
T11: Detection of channel error
T12: Dma_Ip_LogicChannelDeinit()
T13: Dma_Ip_LogicChannelDeinit()
T14: Detection of channel error
T15: Detection of channel error
T16: Dma_Ip_LogicChannelDeinit()
T17: Dma_Ip_LogicChannelDeinit()
T18: Dma_Ip_SetLogicChannelCommand(CHANNEL, DMA_IP_CH_CLEAR_ERROR)
T19: Dma_Ip_SetLogicChannelTransferList()
T20: Dma_Ip_SetLogicChannelScatterGatherList() + Dma_Ip_SetLogicChannelScatterGatherConfig()
T21: Detection of channel error

```

When the DMA Driver detects a software or hardware error for the Logic Channel, it shall enter the "DMA\_IP\_↔CH\_ERROR\_STATE".

To exit from the "DMA\_IP\_CH\_ERROR\_STATE", the application shall use the specified transitions.



**Figure 3.6 DMA IP Channel State Machine**

### 3.7 Runtime errors

The driver generates the following DET errors at runtime.

| Function                      | Error Code                        | Condition triggering the error                      |
|-------------------------------|-----------------------------------|-----------------------------------------------------|
| Mcl_Init()                    | MCL_E_UNINIT                      | API is called with a NULL pointer as parameter.     |
| Mcl_DeInit()                  | MCL_E_PARAM_CONFIG                | API is called with invalid configuration parameter. |
| Mcl_DeInit()                  | MCL_E_UNINIT                      | API is called with a NULL pointer as parameter.     |
| Mcl_SetDmaInstanceCommand()   | MCL_DET_DMA_INSTANCE↔<br>_COMMAND | API is called with invalid instance command.        |
| Mcl_GetDmaInstanceStatus()    | MCL_E_INVALID_INSTANCE            | API is called with invalid instance.                |
| Mcl_SetDmaChannelCommand()    | MCL_E_INVALID_CHANNEL             | API is called with invalid channel.                 |
| Mcl_SetDmaChannelCommand()    | MCL_E_INVALID_COMMAND             | API is called with invalid command.                 |
| Mcl_GetDmaChannelStatus()     | MCL_E_INVALID_CHANNEL             | API is called with invalid channel.                 |
| Mcl_SetDmaChannelGlobalList() | MCL_E_INVALID_CHANNEL             | API is called with invalid channel.                 |

| Function                                               | Error Code              | Condition triggering the error        |
|--------------------------------------------------------|-------------------------|---------------------------------------|
| <a href="#">Mcl_SetDmaChannelGlobalList()</a>          | MCL_E_INVALID_PARAMETER | API is called with invalid parameter. |
| <a href="#">Mcl_SetDmaChannelTransferList()</a>        | MCL_E_INVALID_CHANNEL   | API is called with invalid channel.   |
| <a href="#">Mcl_SetDmaChannelTransferList()</a>        | MCL_E_INVALID_PARAMETER | API is called with invalid parameter. |
| <a href="#">Mcl_SetDmaChannelScatterGatherList()</a>   | MCL_E_INVALID_CHANNEL   | API is called with invalid channel.   |
| <a href="#">Mcl_SetDmaChannelScatterGatherList()</a>   | MCL_E_INVALID_PARAMETER | API is called with invalid parameter. |
| <a href="#">Mcl_GetDmaChannelParam()</a>               | MCL_E_INVALID_CHANNEL   | API is called with invalid channel.   |
| <a href="#">Mcl_GetDmaChannelParam()</a>               | MCL_E_INVALID_PARAMETER | API is called with invalid parameter. |
| <a href="#">Mcl_SetDmaChannelScatterGatherConfig()</a> | MCL_E_INVALID_CHANNEL   | API is called with invalid channel.   |
| <a href="#">Mcl_SetDmaChannelCrcList()</a>             | MCL_E_INVALID_CHANNEL   | API is called with invalid channel.   |
| <a href="#">Mcl_CacheEnable()</a>                      | MCL_E_INVALID_INSTANCE  | API is called with invalid instance.  |
| <a href="#">Mcl_CacheDisable()</a>                     | MCL_E_INVALID_INSTANCE  | API is called with invalid instance.  |
| <a href="#">Mcl_CacheInvalidate()</a>                  | MCL_E_INVALID_INSTANCE  | API is called with invalid instance.  |
| <a href="#">Mcl_CacheClean()</a>                       | MCL_E_INVALID_INSTANCE  | API is called with invalid instance.  |
| <a href="#">Mcl_CacheInvalidateByAddr()</a>            | MCL_E_INVALID_INSTANCE  | API is called with invalid instance.  |
| <a href="#">Mcl_CacheCleanByAddr()</a>                 | MCL_E_INVALID_INSTANCE  | API is called with invalid instance.  |
| <a href="#">Mcl_CacheCleanByAddr()</a>                 | MCL_E_INVALID_INSTANCE  | API is called with invalid instance.  |

### 3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like:

```
#define <Mip>Conf_<Container_ShortName>_<Container_ID>
```

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

## Chapter 4

### Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the driver. All the parameters are described below.

- Module [Mcl](#)
  - Container [MclGeneral](#)
    - \* Parameter [MclEnableDevErrorDetect](#)
    - \* Parameter [Mcl\\_VersionInfoApi](#)
    - \* Parameter [MclEnableUserModeSupport](#)
    - \* Parameter [MclEnableMulticoreSupport](#)
    - \* Parameter [MclEnableVirtualAddressMappingSupport](#)
    - \* Reference [MclEcucPartitionRef](#)
    - \* Container [MclDma](#)
      - Parameter [MclEnableDma](#)
      - Parameter [MclEnableCrc](#)
    - \* Container [MclCache](#)
      - Parameter [MclEnableCache](#)
    - \* Container [MclFtmCommon](#)
      - Parameter [Mcl\\_FtmCommonTimebase](#)
  - Container [MclConfig](#)
    - \* Container [MclVirtualMemorySection](#)
      - Parameter [MclVirtualAddressStart](#)
      - Parameter [MclVirtualAddressEnd](#)
      - Parameter [MclPhysicalAddressStart](#)
      - Parameter [MclPhysicalAddressEnd](#)
    - \* Container [dmaLogicInstance\\_ConfigType](#)
      - Parameter [dmaLogicInstance\\_IdName](#)
      - Parameter [dmaLogicInstance\\_hwId](#)
      - Parameter [dmaLogicInstance\\_enDebug](#)
      - Parameter [dmaLogicInstance\\_enRoundRobin](#)
      - Parameter [dmaLogicInstance\\_enHaltAfterError](#)
      - Parameter [dmaLogicInstance\\_enChLinking](#)
      - Parameter [dmaLogicInstance\\_enGIMasterIdReplication](#)
      - Reference [dmaLogicInstance\\_EcucPartitionRef](#)

- Container [dmaLogicInstance\\_DmaCrc](#)
- Parameter [dmaLogicInstance\\_DmaCrcEnSwapBit](#)
- Parameter [dmaLogicInstance\\_DmaCrcEnSwapByte](#)
- Parameter [dmaLogicInstance\\_DmaCrcEnGlobal](#)
- \* Container [dmaLogicChannel\\_Type](#)
  - Parameter [dmaLogicChannel\\_LogicName](#)
  - Parameter [dmaLogicChannel\\_HwInstId](#)
  - Parameter [dmaLogicChannel\\_HwChId](#)
  - Parameter [dmaLogicChannel\\_InterruptCallback](#)
  - Parameter [dmaLogicChannel\\_ErrorInterruptCallback](#)
  - Parameter [dmaLogicChannel\\_EnableGlobalConfig](#)
  - Parameter [dmaLogicChannel\\_EnableTransferConfig](#)
  - Parameter [dmaLogicChannel\\_EnableScatterGather](#)
  - Parameter [dmaLogicChannel\\_EnableCrcConfig](#)
  - Reference [dmaLogicChannel\\_EcucPartitionRef](#)
  - Container [dmaLogicChannel\\_ConfigType](#)
  - Container [dmaLogicChannel\\_GlobalConfigType](#)
  - Container [dmaLogicChannelConfig\\_GlobalControlType](#)
  - Parameter [dmaGlobalControl\\_enMasterIdReplication](#)
  - Parameter [dmaGlobalControl\\_enBufferedWrites](#)
  - Container [dmaLogicChannelConfig\\_GlobalRequestType](#)
  - Parameter [dmaGlobalRequest\\_enDmamuxTrigger](#)
  - Parameter [dmaGlobalRequest\\_enDmamuxSource](#)
  - Parameter [dmaGlobalRequest\\_Dmamux0HwRequest](#)
  - Parameter [dmaGlobalRequest\\_Dmamux1HwRequest](#)
  - Parameter [dmaGlobalRequest\\_Dmamux2HwRequest](#)
  - Parameter [dmaGlobalRequest\\_Dmamux3HwRequest](#)
  - Parameter [dmaGlobalRequest\\_enDmaRequest](#)
  - Container [dmaLogicChannelConfig\\_GlobalInterruptType](#)
  - Parameter [dmaGlobalInterrupt\\_enDmaErrorInterrupt](#)
  - Container [dmaLogicChannelConfig\\_GlobalPriorityType](#)
  - Parameter [dmaGlobalPriority\\_GroupPriority](#)
  - Parameter [dmaGlobalPriority\\_LevelPriority](#)
  - Parameter [dmaGlobalPriority\\_enPreemption](#)
  - Parameter [dmaGlobalPriority\\_disPreempt](#)
  - Container [dmaLogicChannel\\_TransferConfigType](#)
  - Container [dmaLogicChannelConfig\\_TransferControlType](#)
  - Parameter [dmaLogicChannelConfig\\_enDmaMajorInterrupt](#)
  - Parameter [dmaLogicChannelConfig\\_enDmaHalfMajorInterrupt](#)
  - Parameter [dmaLogicChannelConfig\\_disDmaAutoHwReq](#)
  - Parameter [dmaLogicChannelConfig\\_bandwidthControl](#)
  - Parameter [dmaLogicChannelConfig\\_ScatterGatherAddressType](#)
  - Parameter [dmaLogicChannelConfig\\_DestinationStoreAddressType](#)
  - Container [dmaLogicChannelConfig\\_TransferSourceType](#)
  - Parameter [dmaLogicChannelConfig\\_SourceAddressType](#)
  - Parameter [dmaLogicChannelConfig\\_SourceSignedOffsetType](#)
  - Parameter [dmaLogicChannelConfig\\_SourceLastAddressAdjustmentType](#)



- Parameter [dmaTransferConfig\\_TransferSizeType](#)
- Parameter [dmaLogicChannelConfig\\_SourceModuloType](#)
- Container [dmaLogicChannelConfig\\_TransferDestinationType](#)
- Parameter [dmaLogicChannelConfig\\_DestinationAddressType](#)
- Parameter [dmaLogicChannelConfig\\_DestinationSignedOffsetType](#)
- Parameter [dmaLogicChannelConfig\\_DestinationLastAddressAdjustmentType](#)
- Parameter [dmaTransferConfig\\_TransferSizeType](#)
- Parameter [dmaLogicChannelConfig\\_DestinationModuloType](#)
- Container [dmaLogicChannelConfig\\_TransferMinorLoopType](#)
- Parameter [dmaLogicChannelConfig\\_enSourceOffset](#)
- Parameter [dmaLogicChannelConfig\\_enDestinationOffset](#)
- Parameter [dmaLogicChannelConfig\\_OffsetValueType](#)
- Parameter [dmaLogicChannelConfig\\_enMinorLoopLinkCh](#)
- Parameter [dmaLogicChannelConfig\\_MinorLoopSizeType](#)
- Reference [dynamic\\_dmaLogicChannelConfig\\_MinorLoopLinkChValueType](#)
- Container [dmaLogicChannelConfig\\_TransferMajorLoopType](#)
- Parameter [dmaLogicChannelConfig\\_enMajorLoopLinkCh](#)
- Parameter [dmaLogicChannelConfig\\_MajorLoopCountType](#)
- Reference [dynamic\\_dmaLogicChannelConfig\\_MajorLoopLinkChValueType](#)
- Container [dmaLogicChannel\\_ScatterGatherConfigType](#)
- Container [dmaLogicChannelConfig\\_ScatterGatherArrayType](#)
- Container [dmaLogicChannelConfig\\_ScatterGatherElementConfigType](#)
- Parameter [dmaLogicChannelConfig\\_ScatterGatherElementNameType](#)
- Parameter [dmaLogicChannelConfig\\_LastElementLink\\_ScatterGatherType](#)
- Parameter [dmaLogicChannelConfig\\_enScatterGatherConfig](#)
- Reference [dynamic\\_dmaLogicChannelConfig\\_BasicElementLink\\_ScatterGatherType](#)
- Container [dmaLogicChannelConfig\\_TransferControlType](#)
- Parameter [dmaLogicChannelConfig\\_enStart](#)
- Parameter [dmaLogicChannelConfig\\_enDmaMajorInterrupt](#)
- Parameter [dmaLogicChannelConfig\\_enDmaHalfMajorInterrupt](#)
- Parameter [dmaLogicChannelConfig\\_disDmaAutoHwReq](#)
- Parameter [dmaLogicChannelConfig\\_bandwidthControl](#)
- Parameter [dmaLogicChannelConfig\\_ScatterGatherAddressType](#)
- Parameter [dmaLogicChannelConfig\\_DestinationStoreAddressType](#)
- Container [dmaLogicChannelConfig\\_TransferSourceType](#)
- Parameter [dmaLogicChannelConfig\\_SourceAddressType](#)
- Parameter [dmaLogicChannelConfig\\_SourceSignedOffsetType](#)
- Parameter [dmaLogicChannelConfig\\_SourceLastAddressAdjustmentType](#)
- Parameter [dmaTransferConfig\\_TransferSizeType](#)
- Parameter [dmaLogicChannelConfig\\_SourceModuloType](#)
- Container [dmaLogicChannelConfig\\_TransferDestinationType](#)
- Parameter [dmaLogicChannelConfig\\_DestinationAddressType](#)
- Parameter [dmaLogicChannelConfig\\_DestinationSignedOffsetType](#)
- Parameter [dmaLogicChannelConfig\\_DestinationLastAddressAdjustmentType](#)
- Parameter [dmaTransferConfig\\_TransferSizeType](#)
- Parameter [dmaLogicChannelConfig\\_DestinationModuloType](#)
- Container [dmaLogicChannelConfig\\_TransferMinorLoopType](#)

- Parameter [dmaLogicChannelConfig\\_enSourceOffset](#)
- Parameter [dmaLogicChannelConfig\\_enDestinationOffset](#)
- Parameter [dmaLogicChannelConfig\\_OffsetValueType](#)
- Parameter [dmaLogicChannelConfig\\_enMinorLoopLinkCh](#)
- Parameter [dmaLogicChannelConfig\\_MinorLoopSizeType](#)
- Reference [dynamic\\_dmaLogicChannelConfig\\_MinorLoopLinkChValueType](#)
- Container [dmaLogicChannelConfig\\_TransferMajorLoopType](#)
- Parameter [dmaLogicChannelConfig\\_enMajorLoopLinkCh](#)
- Parameter [dmaLogicChannelConfig\\_MajorLoopCountType](#)
- Reference [dynamic\\_dmaLogicChannelConfig\\_MajorLoopLinkChValueType](#)
- Container [dmaLogicChannel\\_CrcConfigType](#)
- Parameter [dmaLogicChannel\\_HwCrcChId](#)
- Parameter [dmaLogicChannel\\_CrcMode](#)
- Parameter [dmaLogicChannel\\_CrcPolynomial](#)
- Parameter [dmaLogicChannel\\_CrcInitValue](#)
- Parameter [dmaLogicChannel\\_CrcEnInitSel](#)
- Parameter [dmaLogicChannel\\_CrcEnLogic](#)
- Container [CommonPublishedInformation](#)
  - \* Parameter [ArReleaseMajorVersion](#)
  - \* Parameter [ArReleaseMinorVersion](#)
  - \* Parameter [ArReleaseRevisionVersion](#)
  - \* Parameter [ModuleId](#)
  - \* Parameter [SwMajorVersion](#)
  - \* Parameter [SwMinorVersion](#)
  - \* Parameter [SwPatchVersion](#)
  - \* Parameter [VendorId](#)

## 4.1 Module Mcl

Vendor specific: Configuration of the Mcl (MicroController Library ) module.

Included containers:

- [MclGeneral](#)
- [MclConfig](#)
- [CommonPublishedInformation](#)

| Property                | Value                                   |
|-------------------------|-----------------------------------------|
| type                    | ECUC-MODULE-DEF                         |
| lowerMultiplicity       | 1                                       |
| upperMultiplicity       | 1                                       |
| postBuildVariantSupport | true                                    |
| supportedConfigVariants | VARIANT-POST-BUILD, VARIANT-PRE-COMPILE |

## 4.2 Container McIGeneral

Vendor specific: Configuration of general McI parameters.

Included subcontainers:

- [McIDma](#)
- [McICache](#)
- [McIFtmCommon](#)

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.3 Parameter McIEnableDevErrorDetect

Vendor specific:

Enable/Disable the Development Error Detection (DET).

true: Enabled.

false: Disabled.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: PRE-COMPILE  |
| defaultValue                 | false                            |

## 4.4 Parameter Mcl\_VersionInfoApi

Vendor specific: Enables/Disables the get version info API function

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: PRE-COMPILE  |
| defaultValue                 | false                            |

## 4.5 Parameter MclEnableUserModeSupport

When this parameter is enabled, the MCL module will adapt to run from User Mode, with the following measures:

b) using 'call trusted function' stubs for all internal function calls that access registers requiring supervisor mode.

for more information, please see chapter 5.7 User Mode Support in IM

Note: Implementation Specific Parameter.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: PRE-COMPILE  |
| defaultValue                 | false                            |

## 4.6 Parameter MclEnableMulticoreSupport

Enable/Disable Multicore support

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: PRE-COMPILE  |
| defaultValue                 | false                            |

## 4.7 Parameter MclEnableVirtualAddressMappingSupport

Enable/Disable Virtual Address Mapping support

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: PRE-COMPILE  |
| defaultValue                 | false                            |

## 4.8 Reference MclEcucPartitionRef

The partition shall be used to initialize and de-initialize the Mcl Driver.

| Property | Value              |
|----------|--------------------|
| type     | ECUC-REFERENCE-DEF |

| Property                     | Value                                                     |
|------------------------------|-----------------------------------------------------------|
| origin                       | NXP                                                       |
| lowerMultiplicity            | 0                                                         |
| upperMultiplicity            | 1                                                         |
| postBuildVariantMultiplicity | true                                                      |
| multiplicityConfigClasses    | VARIANT-POST-BUILD: PRE-COMPILE                           |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE                          |
| postBuildVariantValue        | true                                                      |
| valueConfigClasses           | VARIANT-POST-BUILD: PRE-COMPILE                           |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE                          |
| requiresSymbolicNameValue    | False                                                     |
| destination                  | /AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition |

## 4.9 Container MclDma

Vendor specific:

Container for the Dma related configuration parameters.

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.10 Parameter MclEnableDma

Vendor specific: Enable/Disable DMA support.

| Property          | Value                  |
|-------------------|------------------------|
| type              | ECUC-BOOLEAN-PARAM-DEF |
| origin            | NXP                    |
| symbolicNameValue | false                  |
| lowerMultiplicity | 1                      |
| upperMultiplicity | 1                      |

| Property                     | Value                            |
|------------------------------|----------------------------------|
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: PRE-COMPILE  |
| defaultValue                 | false                            |

## 4.11 Parameter MclEnableCrc

Vendor specific: Enable/Disable DMA CRC support.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: PRE-COMPILE  |
| defaultValue                 | false                            |

## 4.12 Container MclCache

Vendor specific:

Container for the CACHE related configuration parameters.

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

### 4.13 Parameter MclEnableCache

Vendor specific:

Enable/Disable all CACHE support.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: PRE-COMPILE  |
| defaultValue                 | false                            |

### 4.14 Container MclFtmCommon

Vendor specific:

Container for the FTM Common related configuration parameters.

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

### 4.15 Parameter Mcl\_FtmCommonTimebase

Enables/Disables the option to set the a common timebase for multiple FTM modules.

Note:



Note This is an Implementation Specific Parameter. Enabling this feature will allow the use of the Mcl\_SelectCommonTimebase API

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: PRE-COMPILE  |
| defaultValue                 | false                            |

### 4.16 Container MclConfig

Vendor specific: This container is the base for a multiple configuration set

Included subcontainers:

- [MclVirtualMemorySection](#)
- [dmaLogicInstance\\_ConfigType](#)
- [dmaLogicChannel\\_Type](#)

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

### 4.17 Container MclVirtualMemorySection

Vendor specific:

Data to configure Virtual address and Physical address.

Included subcontainers:

- None

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF    |
| lowerMultiplicity            | 0                                |
| upperMultiplicity            | Infinite                         |
| postBuildVariantMultiplicity | false                            |
| multiplicityConfigClasses    | VARIANT-POST-BUILD: POST-BUILD   |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE |

## 4.18 Parameter MciVirtualAddressStart

This parameter represents the Virtual Address Start.

The address must be aligned to 4 bytes.

Note: Implementation Specific Parameter.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0                                |
| max                          | 4294967295                       |
| min                          | 0                                |

## 4.19 Parameter MciVirtualAddressEnd

This parameter represents the Virtual Address End.

The address must be aligned to 4 bytes.

Note: Implementation Specific Parameter.

| Property | Value                  |
|----------|------------------------|
| type     | ECUC-INTEGER-PARAM-DEF |
| origin   | NXP                    |

| Property                     | Value                            |
|------------------------------|----------------------------------|
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 255                              |
| max                          | 4294967295                       |
| min                          | 0                                |

## 4.20 Parameter MclPhysicalAddressStart

This parameter represents the Physical Address Start.

The address must be aligned to 4 bytes.

Note: Implementation Specific Parameter.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0                                |
| max                          | 4294967295                       |
| min                          | 0                                |

## 4.21 Parameter MclPhysicalAddressEnd

This parameter represents the Physical Address End.

The address must be aligned to 4 bytes.

Note: Implementation Specific Parameter.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| default Value                | 255                              |
| max                          | 4294967295                       |
| min                          | 0                                |

## 4.22 Container dmaLogicInstance\_ConfigType

Vendor specific: Configuration of a DMA Instance.

Included subcontainers:

- [dmaLogicInstance\\_DmaCrc](#)

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF    |
| lowerMultiplicity            | 0                                |
| upperMultiplicity            | 3                                |
| postBuildVariantMultiplicity | false                            |
| multiplicityConfigClasses    | VARIANT-POST-BUILD: POST-BUILD   |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE |

## 4.23 Parameter dmaLogicInstance\_IdName

Vendor specific:

Logic Instance Name.

| Property | Value                 |
|----------|-----------------------|
| type     | ECUC-STRING-PARAM-DEF |
| origin   | NXP                   |

| Property                     | Value                            |
|------------------------------|----------------------------------|
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-POST-BUILD: POST-BUILD   |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE |
| defaultValue                 | DMA_LOGIC_INST_0                 |

## 4.24 Parameter dmaLogicInstance\_\_hwId

Vendor specific:

Select the Hardware DMA Instance.

NOTE: This is an Implementation Specific Parameter.

| Property                     | Value                                    |
|------------------------------|------------------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF               |
| origin                       | NXP                                      |
| symbolicNameValue            | false                                    |
| lowerMultiplicity            | 1                                        |
| upperMultiplicity            | 1                                        |
| postBuildVariantMultiplicity | N/A                                      |
| multiplicityConfigClasses    | N/A                                      |
| postBuildVariantValue        | true                                     |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE         |
|                              | VARIANT-POST-BUILD: POST-BUILD           |
| defaultValue                 | DMA_IP_HW_INST_0                         |
| literals                     | ['DMA_IP_HW_INST_0', 'DMA_IP_HW_INST_1'] |

## 4.25 Parameter dmaLogicInstance\_\_enDebug

Vendor specific:

DMA\_CR[EDBG].

Enable Debug.

0 - The assertion of the system debug control input is ignored.

1 - The assertion of the system debug control input causes the eDMA to stall the start of a new channel.

Executing channels are allowed to complete.

Channel execution will resume when either the system debug control input is negated or the EDBG bit is cleared.

Note: Implementation Specific Parameter.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-POST-BUILD: POST-BUILD   |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE |
| defaultValue                 | false                            |

## 4.26 Parameter dmaLogicInstance\_\_enRoundRobin

Vendor specific:

DMA\_CR[ERCA].

Enable Round Robin Channel Arbitration.

0 - Fixed-priority arbitration is used for channel selection within each group.

1 - Round-Robin arbitration is used for channel selection within each group.

Note: Implementation Specific Parameter.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-POST-BUILD: POST-BUILD   |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE |
| defaultValue                 | false                            |

## 4.27 Parameter dmaLogicInstance\_\_enHaltAfterError

Vendor specific:

DMA\_CR[HOE] for eDMA2 instances or DMA\_CSR[HAE] for eDMA3 instances.

Halt On/After Error.

0 - Normal operation.

1 - Any error will cause the HALT bit to be set.

Subsequently, all service requests will be ignored until the HALT bit is cleared.

Note: Implementation Specific Parameter.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-POST-BUILD: POST-BUILD   |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE |
| defaultValue                 | false                            |

## 4.28 Parameter dmaLogicInstance\_\_enChLinking

Vendor specific:

DMA\_CR[GCLC].

Global Channel Linking Control.

0 - Channel linking is disabled for all channels.

1 - Channel linking is available and controlled by each channel's link settings.

| Property          | Value                  |
|-------------------|------------------------|
| type              | ECUC-BOOLEAN-PARAM-DEF |
| origin            | NXP                    |
| symbolicNameValue | false                  |

| Property                     | Value                            |
|------------------------------|----------------------------------|
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-POST-BUILD: POST-BUILD   |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE |
| defaultValue                 | false                            |

## 4.29 Parameter dmaLogicInstance\_enGIMasterIdReplication

Vendor specific:

DMA\_CSR[GMRC].Available only for eDMA3 instances.

Global master ID replication control

0 - Master ID replication is disabled for all channels

1 - Master ID replication is available and it is controlled by each channel's CHn\_SBR[EMI] setting.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-POST-BUILD: POST-BUILD   |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE |
| defaultValue                 | false                            |

## 4.30 Reference dmaLogicInstance\_\_EcucPartitionRef

Maps a DMA instance to zero or multiple ECUC partitions to limit the access to this mappable element.

The ECUC partitions referenced are a subset of the ECUC partitions where the MCL driver is mapped to



| Property                     | Value                                                     |
|------------------------------|-----------------------------------------------------------|
| type                         | ECUC-REFERENCE-DEF                                        |
| origin                       | AUTOSAR_ECUC                                              |
| lowerMultiplicity            | 0                                                         |
| upperMultiplicity            | 1                                                         |
| postBuildVariantMultiplicity | false                                                     |
| multiplicityConfigClasses    | VARIANT-PRE-COMPILE: PRE-COMPILE                          |
|                              | VARIANT-POST-BUILD: POST-BUILD                            |
| postBuildVariantValue        | true                                                      |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE                          |
|                              | VARIANT-POST-BUILD: POST-BUILD                            |
| requiresSymbolicNameValue    | False                                                     |
| destination                  | /AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition |

### 4.31 Container dmaLogicInstance\_\_DmaCrc

Vendor specific:

Container for the Dma Crc related configuration parameters.

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

### 4.32 Parameter dmaLogicInstance\_\_DmaCrcEnSwapBit

Vendor specific: Enable Swap Bit

| Property          | Value                  |
|-------------------|------------------------|
| type              | ECUC-BOOLEAN-PARAM-DEF |
| origin            | NXP                    |
| symbolicNameValue | false                  |
| lowerMultiplicity | 1                      |

| Property                     | Value                            |
|------------------------------|----------------------------------|
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

### 4.33 Parameter dmaLogicInstance\_\_DmaCrcEnSwapByte

Vendor specific: Enable Swap Byte

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

### 4.34 Parameter dmaLogicInstance\_\_DmaCrcEnGlobal

Vendor specific: Enable Global

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |

| Property     | Value                          |
|--------------|--------------------------------|
|              | VARIANT-POST-BUILD: POST-BUILD |
| defaultValue | false                          |

## 4.35 Container dmaLogicChannel\_Type

Vendor specific:

Logic Channel Configuration.

Included subcontainers:

- [dmaLogicChannel\\_ConfigType](#)

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF    |
| lowerMultiplicity            | 0                                |
| upperMultiplicity            | 64                               |
| postBuildVariantMultiplicity | false                            |
| multiplicityConfigClasses    | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: PRE-COMPILE  |

## 4.36 Parameter dmaLogicChannel\_LogicName

Vendor specific:

Logic Channel Name.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-STRING-PARAM-DEF            |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | false                            |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: PRE-COMPILE  |
| defaultValue                 | DMA_LOGIC_CH_0                   |

### 4.37 Parameter dmaLogicChannel\_HwInstId

Vendor specific:

Select the Hardware DMA Instance.

NOTE: This is an Implementation Specific Parameter.

| Property                     | Value                                    |
|------------------------------|------------------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF               |
| origin                       | NXP                                      |
| symbolicNameValue            | false                                    |
| lowerMultiplicity            | 1                                        |
| upperMultiplicity            | 1                                        |
| postBuildVariantMultiplicity | N/A                                      |
| multiplicityConfigClasses    | N/A                                      |
| postBuildVariantValue        | true                                     |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE         |
|                              | VARIANT-POST-BUILD: POST-BUILD           |
| defaultValue                 | DMA_IP_HW_INST_0                         |
| literals                     | ['DMA_IP_HW_INST_0', 'DMA_IP_HW_INST_1'] |

### 4.38 Parameter dmaLogicChannel\_HwChId

Vendor specific:

Select the physical eDMA Channel.

NOTE: This is an Implementation Specific Parameter.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF       |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | DMA_IP_HW_CH_0                   |

| Property | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| literals | ['DMA_IP_HW_CH_0', 'DMA_IP_HW_CH_1', 'DMA_IP_HW_CH_↵<br>2', 'DMA_IP_HW_CH_3', 'DMA_IP_HW_CH_4', 'DMA_IP_HW_CH_↵<br>5', 'DMA_IP_HW_CH_6', 'DMA_IP_HW_CH_7', 'DMA_IP_HW_CH_↵<br>8', 'DMA_IP_HW_CH_9', 'DMA_IP_HW_CH_10', 'DMA_IP_HW_CH_↵<br>11', 'DMA_IP_HW_CH_12', 'DMA_IP_HW_CH_13', 'DMA_IP_HW_C↵<br>H_14', 'DMA_IP_HW_CH_15', 'DMA_IP_HW_CH_16', 'DMA_IP_HW↵<br>_CH_17', 'DMA_IP_HW_CH_18', 'DMA_IP_HW_CH_19', 'DMA_IP_H↵<br>W_CH_20', 'DMA_IP_HW_CH_21', 'DMA_IP_HW_CH_22', 'DMA_IP↵<br>_HW_CH_23', 'DMA_IP_HW_CH_24', 'DMA_IP_HW_CH_25', 'DMA_↵<br>IP_HW_CH_26', 'DMA_IP_HW_CH_27', 'DMA_IP_HW_CH_28', 'DM↵<br>A_IP_HW_CH_29', 'DMA_IP_HW_CH_30', 'DMA_IP_HW_CH_31'] |

### 4.39 Parameter dmaLogicChannel\_InterruptCallback

Vendor specific:

User callback function to report that the transfer is half or complete depending on configuration.

NOTE: Use NULL\_PTR w/o quotes. If the used string is different from NULL\_PTR it will be used as the configured function name.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-FUNCTION-NAME-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | NULL_PTR                         |

### 4.40 Parameter dmaLogicChannel\_ErrorInterruptCallback

Vendor specific:

User callback function

NOTE: Use NULL\_PTR w/o quotes. If the used string is different from NULL\_PTR it will be used as the configured function name.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-FUNCTION-NAME-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| default Value                | NULL_PTR                         |

#### 4.41 Parameter dmaLogicChannel\_\_EnableGlobalConfig

Vendor specific: Enable and allocate memory for Global Configuration. Global Configuration can be configured during generation time or during runtime.

Note: If the check box is enabled, then memory space is allocated for the configuration.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| default Value                | false                            |

#### 4.42 Parameter dmaLogicChannel\_\_EnableTransferConfig

Vendor specific: Enable and allocate memory for Transfer Configuration. Transfer Configuration can be configured during generation time or during runtime.

Note: If the check box is enabled, then memory space is allocated for the configuration.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| default Value                | false                            |

#### 4.43 Parameter dmaLogicChannel\_\_EnableScatterGather

Vendor specific: Enable and allocate memory for ScatterGather Transfer Mode.

The ScatterGather Transfer Mode shall allocate memory for each Element, comprised of: Element Linkage and Element Software TCD.

The Element allocation can be done only in the configurator.

The Element Configuration can be further enabled for each individual element.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| default Value                | false                            |

#### 4.44 Parameter dmaLogicChannel\_\_EnableCrcConfig

Vendor specific: Enable and allocate memory for DMA CRC Configuration. DMA CRC Configuration can be configured during generation time or during runtime.

Note: If the check box is enabled, then memory space is allocated for the configuration.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| default Value                | false                            |

#### 4.45 Reference dmaLogicChannel\_\_EcucPartitionRef

Maps a DMA channel to zero or multiple ECUC partitions to limit the access to this mappable element.

The ECUC partitions referenced are a subset of the ECUC partitions where the MCL driver is mapped to

| Property                     | Value                                                        |
|------------------------------|--------------------------------------------------------------|
| type                         | ECUC-REFERENCE-DEF                                           |
| origin                       | NXP                                                          |
| lowerMultiplicity            | 0                                                            |
| upperMultiplicity            | 1                                                            |
| postBuildVariantMultiplicity | false                                                        |
| multiplicityConfigClasses    | VARIANT-PRE-COMPILE: PRE-COMPILE                             |
|                              | VARIANT-POST-BUILD: POST-BUILD                               |
| postBuildVariantValue        | true                                                         |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE                             |
|                              | VARIANT-POST-BUILD: POST-BUILD                               |
| requiresSymbolicNameValue    | False                                                        |
| destination                  | /AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition |

#### 4.46 Container dmaLogicChannel\_\_ConfigType

Vendor specific: Logic Channel Configuration.

Included subcontainers:

- [dmaLogicChannel\\_\\_GlobalConfigType](#)



- [dmaLogicChannel\\_TransferConfigType](#)
- [dmaLogicChannel\\_ScatterGatherConfigType](#)
- [dmaLogicChannel\\_CrcConfigType](#)

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

### 4.47 Container dmaLogicChannel\_GlobalConfigType

Vendor specific:

Logic Channel Global Configuration.

Included subcontainers:

- [dmaLogicChannelConfig\\_GlobalControlType](#)
- [dmaLogicChannelConfig\\_GlobalRequestType](#)
- [dmaLogicChannelConfig\\_GlobalInterruptType](#)
- [dmaLogicChannelConfig\\_GlobalPriorityType](#)

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

### 4.48 Container dmaLogicChannelConfig\_GlobalControlType

Vendor specific:

TCD Global Control.

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

#### 4.49 Parameter dmaGlobalControl\_enMasterIdReplication

Vendor specific: Set the Dma Channel to use the same protection level and system bus ID of the master programming the Dma Channel.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

#### 4.50 Parameter dmaGlobalControl\_enBufferedWrites

Vendor specific: Set the Dma Channel writes to be bufferable.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

## 4.51 Container dmaLogicChannelConfig\_GlobalRequestType

Vendor specific:

TCD Request Control.

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.52 Parameter dmaGlobalRequest\_\_enDmamuxTrigger

Vendor specific: Enable the Dma Channel Mux Trigger.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

## 4.53 Parameter dmaGlobalRequest\_\_enDmamuxSource

Vendor specific: Enable the Dma Channel Mux Source.

| Property | Value                  |
|----------|------------------------|
| type     | ECUC-BOOLEAN-PARAM-DEF |

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | false                                                              |

#### 4.54 Parameter dmaGlobalRequest\_Dmamux0HwRequest

Vendor specific: DMAMUX0 Source

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF                                         |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | DMA_IP_REQ_MUX0_DISABLED                                           |

| Property | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| literals | ['DMA_IP_REQ_MUX0_DISABLED', 'DMA_IP_REQ_MUX0_CANFD0', 'DMA_IP_REQ_MUX0_CANFD1', 'DMA_IP_REQ_MUX0_LINFLEXD0_TX', 'DMA_IP_REQ_MUX0_LINFLEXD0_RX', 'DMA_IP_REQ_MUX0_LINFLEXD1_TX', 'DMA_IP_REQ_MUX0_LINFLEXD1_RX', 'DMA_IP_REQ_MUX0_DSPI0_TX', 'DMA_IP_REQ_MUX0_DSPI0_RX', 'DMA_IP_REQ_MUX0_DSPI0_CMD', 'DMA_IP_REQ_MUX0_DSPI1_TX', 'DMA_IP_REQ_MUX0_DSPI1_RX', 'DMA_IP_REQ_MUX0_DSPI1_CMD', 'DMA_IP_REQ_MUX0_DSPI2_TX', 'DMA_IP_REQ_MUX0_DSPI2_RX', 'DMA_IP_REQ_MUX0_DSPI2_CMD', 'DMA_IP_REQ_MUX0_I2C0_RX', 'DMA_IP_REQ_MUX0_I2C0_TX', 'DMA_IP_REQ_MUX0_I2C1_RX', 'DMA_IP_REQ_MUX0_I2C1_TX', 'DMA_IP_REQ_MUX0_CTU_FIFO0', 'DMA_IP_REQ_MUX0_CTU_FIFO1', 'DMA_IP_REQ_MUX0_CTU_FIFO2', 'DMA_IP_REQ_MUX0_CTU_FIFO3', 'DMA_IP_REQ_MUX0_CTU_CMD_REQ', 'DMA_IP_REQ_MUX0_SUILO_0', 'DMA_IP_REQ_MUX0_SUILO_1', 'DMA_IP_REQ_MUX0_SUILO_4', 'DMA_IP_REQ_MUX0_SUILO_5', 'DMA_IP_REQ_MUX0_SUILO_8', 'DMA_IP_REQ_MUX0_SUILO_9', 'DMA_IP_REQ_MUX0_SARADC0', 'DMA_IP_REQ_MUX0_SUIL1_0', 'DMA_IP_REQ_MUX0_SUIL1_1', 'DMA_IP_REQ_MUX0_SUIL1_4', 'DMA_IP_REQ_MUX0_SUIL1_5', 'DMA_IP_REQ_MUX0_FTM0_CH0', 'DMA_IP_REQ_MUX0_FTM0_CH1', 'DMA_IP_REQ_MUX0_FTM0_CH2', 'DMA_IP_REQ_MUX0_FTM0_CH3', 'DMA_IP_REQ_MUX0_FTM0_CH4', 'DMA_IP_REQ_MUX0_FTM0_CH5', 'DMA_IP_REQ_MUX0_ALWAYS_ON0', 'DMA_IP_REQ_MUX0_ALWAYS_ON1', 'DMA_IP_REQ_MUX0_ALWAYS_ON2', 'DMA_IP_REQ_MUX0_ALWAYS_ON3'] |

## 4.55 Parameter dmaGlobalRequest\_Dmamux1HwRequest

Vendor specific: DMAMUX1 Source

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF       |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | DMA_IP_REQ_MUX1_DISABLED         |

| Property | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| literals | ['DMA_IP_REQ_MUX1_DISABLED', 'DMA_IP_REQ_MUX1_CANFD2', 'DMA_IP_REQ_MUX1_CANFD3', 'DMA_IP_REQ_MUX1_LINFLEXD2_TX', 'DMA_IP_REQ_MUX1_LINFLEXD2_RX', 'DMA_IP_REQ_MUX1_QUADSPI_TX', 'DMA_IP_REQ_MUX1_QUADSPI_RX', 'DMA_IP_REQ_MUX1_DSPI3_TX', 'DMA_IP_REQ_MUX1_DSPI3_RX', 'DMA_IP_REQ_MUX1_DSPI3_CMD', 'DMA_IP_REQ_MUX1_DSPI4_TX', 'DMA_IP_REQ_MUX1_DSPI4_RX', 'DMA_IP_REQ_MUX1_DSPI4_CMD', 'DMA_IP_REQ_MUX1_DSPI5_TX', 'DMA_IP_REQ_MUX1_DSPI5_RX', 'DMA_IP_REQ_MUX1_DSPI5_CMD', 'DMA_IP_REQ_MUX1_I2C2_RX', 'DMA_IP_REQ_MUX1_I2C2_TX', 'DMA_IP_REQ_MUX1_I2C3_RX', 'DMA_IP_REQ_MUX1_I2C3_TX', 'DMA_IP_REQ_MUX1_I2C4_RX', 'DMA_IP_REQ_MUX1_I2C4_TX', 'DMA_IP_REQ_MUX1_ETHERNET0_CH0', 'DMA_IP_REQ_MUX1_ETHERNET0_CH1', 'DMA_IP_REQ_MUX1_ETHERNET0_CH2', 'DMA_IP_REQ_MUX1_SIUL0_2', 'DMA_IP_REQ_MUX1_SIUL0_3', 'DMA_IP_REQ_MUX1_SIUL0_6', 'DMA_IP_REQ_MUX1_SIUL0_7', 'DMA_IP_REQ_MUX1_SIUL0_10', 'DMA_IP_REQ_MUX1_SIUL0_11', 'DMA_IP_REQ_MUX1_SARADC1', 'DMA_IP_REQ_MUX1_ETHERNET0_CH3', 'DMA_IP_REQ_MUX1_PFE_MAC0', 'DMA_IP_REQ_MUX1_PFE_MAC1', 'DMA_IP_REQ_MUX1_PFE_MAC2', 'DMA_IP_REQ_MUX1_SIUL1_2', 'DMA_IP_REQ_MUX1_SIUL1_3', 'DMA_IP_REQ_MUX1_SIUL1_6', 'DMA_IP_REQ_MUX1_SIUL1_7', 'DMA_IP_REQ_MUX1_FTM1_CH0', 'DMA_IP_REQ_MUX1_FTM1_CH1', 'DMA_IP_REQ_MUX1_FTM1_CH2', 'DMA_IP_REQ_MUX1_FTM1_CH3', 'DMA_IP_REQ_MUX1_FTM1_CH4', 'DMA_IP_REQ_MUX1_FTM1_CH5', 'DMA_IP_REQ_MUX1_ALWAYS_ON0', 'DMA_IP_REQ_MUX1_ALWAYS_ON1', 'DMA_IP_REQ_MUX1_ALWAYS_ON2', 'DMA_IP_REQ_MUX1_ALWAYS_ON3'] |

## 4.56 Parameter dmaGlobalRequest\_Dmamux2HwRequest

Vendor specific: DMAMUX2 Source

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF       |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | DMA_IP_REQ_MUX2_DISABLED         |

| Property | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| literals | ['DMA_IP_REQ_MUX2_DISABLED', 'DMA_IP_REQ_MUX2_CANFD0', 'DMA_IP_REQ_MUX2_CANFD1', 'DMA_IP_REQ_MUX2_LINFLEXD0_TX', 'DMA_IP_REQ_MUX2_LINFLEXD0_RX', 'DMA_IP_REQ_MUX2_LINFLEXD1_TX', 'DMA_IP_REQ_MUX2_LINFLEXD1_RX', 'DMA_IP_REQ_MUX2_DSPI0_TX', 'DMA_IP_REQ_MUX2_DSPI0_RX', 'DMA_IP_REQ_MUX2_DSPI0_CMD', 'DMA_IP_REQ_MUX2_DSPI1_TX', 'DMA_IP_REQ_MUX2_DSPI1_RX', 'DMA_IP_REQ_MUX2_DSPI1_CMD', 'DMA_IP_REQ_MUX2_DSPI2_TX', 'DMA_IP_REQ_MUX2_DSPI2_RX', 'DMA_IP_REQ_MUX2_DSPI2_CMD', 'DMA_IP_REQ_MUX2_I2C0_RX', 'DMA_IP_REQ_MUX2_I2C0_TX', 'DMA_IP_REQ_MUX2_I2C1_RX', 'DMA_IP_REQ_MUX2_I2C1_TX', 'DMA_IP_REQ_MUX2_CTU_FIFO0', 'DMA_IP_REQ_MUX2_CTU_FIFO1', 'DMA_IP_REQ_MUX2_CTU_FIFO2', 'DMA_IP_REQ_MUX2_CTU_FIFO3', 'DMA_IP_REQ_MUX2_CTU_CMD_REQ', 'DMA_IP_REQ_MUX2_SUILO_0', 'DMA_IP_REQ_MUX2_SUILO_1', 'DMA_IP_REQ_MUX2_SUILO_4', 'DMA_IP_REQ_MUX2_SUILO_5', 'DMA_IP_REQ_MUX2_SUILO_8', 'DMA_IP_REQ_MUX2_SUILO_9', 'DMA_IP_REQ_MUX2_SARADC0', 'DMA_IP_REQ_MUX2_SUIL1_0', 'DMA_IP_REQ_MUX2_SUIL1_1', 'DMA_IP_REQ_MUX2_SUIL1_4', 'DMA_IP_REQ_MUX2_SUIL1_5', 'DMA_IP_REQ_MUX2_FTM0_CH0', 'DMA_IP_REQ_MUX2_FTM0_CH1', 'DMA_IP_REQ_MUX2_FTM0_CH2', 'DMA_IP_REQ_MUX2_FTM0_CH3', 'DMA_IP_REQ_MUX2_FTM0_CH4', 'DMA_IP_REQ_MUX2_FTM0_CH5', 'DMA_IP_REQ_MUX2_ALWAYS_ON0', 'DMA_IP_REQ_MUX2_ALWAYS_ON1', 'DMA_IP_REQ_MUX2_ALWAYS_ON2', 'DMA_IP_REQ_MUX2_ALWAYS_ON3'] |

## 4.57 Parameter dmaGlobalRequest\_\_Dmamux3HwRequest

Vendor specific: DMAMUX3 Source

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF       |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | DMA_IP_REQ_MUX3_DISABLED         |

| Property | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| literals | ['DMA_IP_REQ_MUX3_DISABLED', 'DMA_IP_REQ_MUX3_CANFD2', 'DMA_IP_REQ_MUX3_CANFD3', 'DMA_IP_REQ_MUX3_LINFLEXD2_TX', 'DMA_IP_REQ_MUX3_LINFLEXD2_RX', 'DMA_IP_REQ_MUX3_QUADSPI_TX', 'DMA_IP_REQ_MUX3_QUADSPI_RX', 'DMA_IP_REQ_MUX3_DSPI3_TX', 'DMA_IP_REQ_MUX3_DSPI3_RX', 'DMA_IP_REQ_MUX3_DSPI3_CMD', 'DMA_IP_REQ_MUX3_DSPI4_TX', 'DMA_IP_REQ_MUX3_DSPI4_RX', 'DMA_IP_REQ_MUX3_DSPI4_CMD', 'DMA_IP_REQ_MUX3_DSPI5_TX', 'DMA_IP_REQ_MUX3_DSPI5_RX', 'DMA_IP_REQ_MUX3_DSPI5_CMD', 'DMA_IP_REQ_MUX3_I2C2_RX', 'DMA_IP_REQ_MUX3_I2C2_TX', 'DMA_IP_REQ_MUX3_I2C3_RX', 'DMA_IP_REQ_MUX3_I2C3_TX', 'DMA_IP_REQ_MUX3_I2C4_RX', 'DMA_IP_REQ_MUX3_I2C4_TX', 'DMA_IP_REQ_MUX3_ETHERNET0_CH0', 'DMA_IP_REQ_MUX3_ETHERNET0_CH1', 'DMA_IP_REQ_MUX3_ETHERNET0_CH2', 'DMA_IP_REQ_MUX3_SIUL0_2', 'DMA_IP_REQ_MUX3_SIUL0_3', 'DMA_IP_REQ_MUX3_SIUL0_6', 'DMA_IP_REQ_MUX3_SIUL0_7', 'DMA_IP_REQ_MUX3_SIUL0_10', 'DMA_IP_REQ_MUX3_SIUL0_11', 'DMA_IP_REQ_MUX3_SARADC1', 'DMA_IP_REQ_MUX3_ETHERNET0_CH3', 'DMA_IP_REQ_MUX3_PFE_MAC0', 'DMA_IP_REQ_MUX3_PFE_MAC1', 'DMA_IP_REQ_MUX3_PFE_MAC2', 'DMA_IP_REQ_MUX3_SIUL1_2', 'DMA_IP_REQ_MUX3_SIUL1_3', 'DMA_IP_REQ_MUX3_SIUL1_6', 'DMA_IP_REQ_MUX3_SIUL1_7', 'DMA_IP_REQ_MUX3_FTM1_CH0', 'DMA_IP_REQ_MUX3_FTM1_CH1', 'DMA_IP_REQ_MUX3_FTM1_CH2', 'DMA_IP_REQ_MUX3_FTM1_CH3', 'DMA_IP_REQ_MUX3_FTM1_CH4', 'DMA_IP_REQ_MUX3_FTM1_CH5', 'DMA_IP_REQ_MUX3_ALWAYS_ON0', 'DMA_IP_REQ_MUX3_ALWAYS_ON1', 'DMA_IP_REQ_MUX3_ALWAYS_ON2', 'DMA_IP_REQ_MUX3_ALWAYS_ON3'] |

## 4.58 Parameter dmaGlobalRequest\_enDmaRequest

Vendor specific: Enable the Dma Channel Hardware Request.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | false                                                              |



## 4.59 Container dmaLogicChannelConfig\_GlobalInterruptType

Vendor specific:

TCD Global Interrupt.

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.60 Parameter dmaGlobalInterrupt\_enDmaErrorInterrupt

Vendor specific: Enable the Dma Channel Error Interrupt.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | false                                                              |

## 4.61 Container dmaLogicChannelConfig\_GlobalPriorityType

Vendor specific:

TCD Global Interrupt.

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.62 Parameter dmaGlobalPriority\_GroupPriority

Vendor specific: Set the Dma Channel Group Priority.

| Property                     | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| origin                       | NXP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| symbolicNameValue            | false                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| lowerMultiplicity            | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| upperMultiplicity            | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| postBuildVariantMultiplicity | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| multiplicityConfigClasses    | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| postBuildVariantValue        | true                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                              | VARIANT-POST-BUILD: POST-BUILD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| defaultValue                 | DMA_IP_GROUP_PRI00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| literals                     | ['DMA_IP_GROUP_PRI00', 'DMA_IP_GROUP_PRI01', 'DMA_IP_GROUP_PRI02', 'DMA_IP_GROUP_PRI03', 'DMA_IP_GROUP_PRI04', 'DMA_IP_GROUP_PRI05', 'DMA_IP_GROUP_PRI06', 'DMA_IP_GROUP_PRI07', 'DMA_IP_GROUP_PRI08', 'DMA_IP_GROUP_PRI09', 'DMA_IP_GROUP_PRI10', 'DMA_IP_GROUP_PRI11', 'DMA_IP_GROUP_PRI12', 'DMA_IP_GROUP_PRI13', 'DMA_IP_GROUP_PRI14', 'DMA_IP_GROUP_PRI15', 'DMA_IP_GROUP_PRI16', 'DMA_IP_GROUP_PRI17', 'DMA_IP_GROUP_PRI18', 'DMA_IP_GROUP_PRI19', 'DMA_IP_GROUP_PRI20', 'DMA_IP_GROUP_PRI21', 'DMA_IP_GROUP_PRI22', 'DMA_IP_GROUP_PRI23', 'DMA_IP_GROUP_PRI24', 'DMA_IP_GROUP_PRI25', 'DMA_IP_GROUP_PRI26', 'DMA_IP_GROUP_PRI27', 'DMA_IP_GROUP_PRI28', 'DMA_IP_GROUP_PRI29', 'DMA_IP_GROUP_PRI30', 'DMA_IP_GROUP_PRI31'] |

## 4.63 Parameter dmaGlobalPriority\_LevelPriority

Vendor specific: Set the Dma Channel Level Priority.

| Property                     | Value                                                                                                                                                                            |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF                                                                                                                                                       |
| origin                       | NXP                                                                                                                                                                              |
| symbolicNameValue            | false                                                                                                                                                                            |
| lowerMultiplicity            | 1                                                                                                                                                                                |
| upperMultiplicity            | 1                                                                                                                                                                                |
| postBuildVariantMultiplicity | N/A                                                                                                                                                                              |
| multiplicityConfigClasses    | N/A                                                                                                                                                                              |
| postBuildVariantValue        | true                                                                                                                                                                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE                                                                                                                                                 |
|                              | VARIANT-POST-BUILD: POST-BUILD                                                                                                                                                   |
| defaultValue                 | DMA_IP_LEVEL_PRI00                                                                                                                                                               |
| literals                     | ['DMA_IP_LEVEL_PRI00', 'DMA_IP_LEVEL_PRI01', 'DMA_IP_LEVEL_PRI02', 'DMA_IP_LEVEL_PRI03', 'DMA_IP_LEVEL_PRI04', 'DMA_IP_LEVEL_PRI05', 'DMA_IP_LEVEL_PRI06', 'DMA_IP_LEVEL_PRI07'] |

## 4.64 Parameter dmaGlobalPriority\_enPreemption

Vendor specific: Enable the Dma Channel Preemption.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

## 4.65 Parameter dmaGlobalPriority\_disPreempt

Vendor specific: Disable the Dma Channel Preempt.

| Property | Value                  |
|----------|------------------------|
| type     | ECUC-BOOLEAN-PARAM-DEF |

| Property                     | Value                            |
|------------------------------|----------------------------------|
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

## 4.66 Container dmaLogicChannel\_TransferConfigType

Vendor specific:

Logic Channel Transfer Configuration.

Included subcontainers:

- [dmaLogicChannelConfig\\_TransferControlType](#)
- [dmaLogicChannelConfig\\_TransferSourceType](#)
- [dmaLogicChannelConfig\\_TransferDestinationType](#)
- [dmaLogicChannelConfig\\_TransferMinorLoopType](#)
- [dmaLogicChannelConfig\\_TransferMajorLoopType](#)

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.67 Container dmaLogicChannelConfig\_TransferControlType

Vendor specific:

TCD Global Interrupt.

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.68 Parameter dmaLogicChannelConfig\_enDmaMajorInterrupt

Vendor specific: Enable the Dma Channel major interrupt.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | false                                                              |

## 4.69 Parameter dmaLogicChannelConfig\_enDmaHalfMajorInterrupt

Vendor specific: Enable the Dma Channel half major interrupt.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |

| Property     | Value                          |
|--------------|--------------------------------|
|              | VARIANT-POST-BUILD: POST-BUILD |
| defaultValue | false                          |

## 4.70 Parameter dmaLogicChannelConfig\_disDmaAutoHwReq

Vendor specific: Disable the Dma Channel automatic request.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | false                                                              |

## 4.71 Parameter dmaLogicChannelConfig\_bandwidthControl

Vendor specific: Set the Dma Channel bandwidth control.

| Property                     | Value                                                                                                                         |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF                                                                                                    |
| origin                       | NXP                                                                                                                           |
| symbolicNameValue            | false                                                                                                                         |
| lowerMultiplicity            | 1                                                                                                                             |
| upperMultiplicity            | 1                                                                                                                             |
| postBuildVariantMultiplicity | N/A                                                                                                                           |
| multiplicityConfigClasses    | N/A                                                                                                                           |
| postBuildVariantValue        | true                                                                                                                          |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD                                                            |
| defaultValue                 | DMA_IP_BWC_ENGINE_NO_STALL                                                                                                    |
| literals                     | ['DMA_IP_BWC_ENGINE_NO_STALL', 'DMA_IP_BWC_ENGINE_H←PE', 'DMA_IP_BWC_ENGINE_4CYCLE_STALL', 'DMA_IP_BWC_EN←GINE_8CYCLE_STALL'] |

## 4.72 Parameter dmaLogicChannelConfig\_ScatterGatherAddressType

Vendor specific: ScatterGather Address

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-STRING-PARAM-DEF                                              |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0U                                                                 |

## 4.73 Parameter dmaLogicChannelConfig\_DestinationStoreAddressType

Vendor specific: Destination Store Address

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-STRING-PARAM-DEF                                              |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0U                                                                 |

## 4.74 Container dmaLogicChannelConfig\_TransferSourceType

Vendor specific: Transfer Source

Included subcontainers:





## Tresos Configuration Plug-in

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.75 Parameter dmaLogicChannelConfig\_SourceAddressType

Vendor specific: Source Address

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-STRING-PARAM-DEF                                              |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0                                                                  |

## 4.76 Parameter dmaLogicChannelConfig\_SourceSignedOffsetType

Vendor specific: Set the Dma Channel source signed offset value.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0                                                                  |

| Property | Value  |
|----------|--------|
| max      | 32767  |
| min      | -32767 |

## 4.77 Parameter `dmaLogicChannelConfig_SourceLastAddressAdjustmentType`

Vendor specific: Set the Dma Channel source signed last address adjustment.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0                                |
| max                          | 2147483647                       |
| min                          | -2147483647                      |

## 4.78 Parameter `dmaTransferConfig_TransferSizeType`

Vendor specific: Set the Dma Channel source transfer size.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF       |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | DMA_IP_TRANSFER_SIZE_1_BYTE      |

| Property | Value                                                                                                                                                                                                                        |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| literals | ['DMA_IP_TRANSFER_SIZE_1_BYTE', 'DMA_IP_TRANSFER_SIZE_2_BYTE', 'DMA_IP_TRANSFER_SIZE_4_BYTE', 'DMA_IP_TRANSFER_SIZE_8_BYTE', 'DMA_IP_TRANSFER_SIZE_16_BYTE', 'DMA_IP_TRANSFER_SIZE_32_BYTE', 'DMA_IP_TRANSFER_SIZE_64_BYTE'] |

## 4.79 Parameter dmaLogicChannelConfig\_SourceModuloType

Vendor specific: Set the Dma Channel source modulo.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0                                                                  |
| max                          | 31                                                                 |
| min                          | 0                                                                  |

## 4.80 Container dmaLogicChannelConfig\_TransferDestinationType

Vendor specific: Transfer Destination

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.81 Parameter dmaLogicChannelConfig\_DestinationAddressType

Vendor specific: Destination Address

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-STRING-PARAM-DEF                                              |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0                                                                  |

## 4.82 Parameter dmaLogicChannelConfig\_DestinationSignedOffsetType

Vendor specific: Set the Dma Channel destination signed offset value.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0                                                                  |
| max                          | 32767                                                              |
| min                          | -32767                                                             |

## 4.83 Parameter dmaLogicChannelConfig\_DestinationLastAddressAdjustmentType

Vendor specific: Set the Dma Channel destination signed last address adjustment.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0                                |
| max                          | 2147483647                       |
| min                          | -2147483647                      |

#### 4.84 Parameter dmaTransferConfig\_\_TransferSizeType

Vendor specific: Set the Dma Channel destination transfer size.

| Property                     | Value                                                                                                                                                                                                                        |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF                                                                                                                                                                                                   |
| origin                       | NXP                                                                                                                                                                                                                          |
| symbolicNameValue            | false                                                                                                                                                                                                                        |
| lowerMultiplicity            | 1                                                                                                                                                                                                                            |
| upperMultiplicity            | 1                                                                                                                                                                                                                            |
| postBuildVariantMultiplicity | N/A                                                                                                                                                                                                                          |
| multiplicityConfigClasses    | N/A                                                                                                                                                                                                                          |
| postBuildVariantValue        | true                                                                                                                                                                                                                         |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE                                                                                                                                                                                             |
|                              | VARIANT-POST-BUILD: POST-BUILD                                                                                                                                                                                               |
| defaultValue                 | DMA_IP_TRANSFER_SIZE_1_BYTE                                                                                                                                                                                                  |
| literals                     | ['DMA_IP_TRANSFER_SIZE_1_BYTE', 'DMA_IP_TRANSFER_SIZE_2_BYTE', 'DMA_IP_TRANSFER_SIZE_4_BYTE', 'DMA_IP_TRANSFER_SIZE_8_BYTE', 'DMA_IP_TRANSFER_SIZE_16_BYTE', 'DMA_IP_TRANSFER_SIZE_32_BYTE', 'DMA_IP_TRANSFER_SIZE_64_BYTE'] |

#### 4.85 Parameter dmaLogicChannelConfig\_\_DestinationModuloType

Vendor specific: Set the Dma Channel destination modulo.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0                                |
| max                          | 31                               |
| min                          | 0                                |

## 4.86 Container dmaLogicChannelConfig\_TransferMinorLoopType

Vendor specific: Transfer Minor Loop

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.87 Parameter dmaLogicChannelConfig\_enSourceOffset

Vendor specific: Enable the Dma Channel minor loop source offset.

| Property          | Value                  |
|-------------------|------------------------|
| type              | ECUC-BOOLEAN-PARAM-DEF |
| origin            | NXP                    |
| symbolicNameValue | false                  |
| lowerMultiplicity | 1                      |
| upperMultiplicity | 1                      |

| Property                     | Value                            |
|------------------------------|----------------------------------|
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

## 4.88 Parameter dmaLogicChannelConfig\_enDestinationOffset

Vendor specific: Enable the Dma Channel minor loop destination offset.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

## 4.89 Parameter dmaLogicChannelConfig\_OffsetValueType

Vendor specific: Set the Dma Channel minor loop signed offset.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |



| Property     | Value    |
|--------------|----------|
| defaultValue | 0        |
| max          | 1048575  |
| min          | -1048575 |

## 4.90 Parameter dmaLogicChannelConfig\_enMinorLoopLinkCh

Vendor specific: Enable the Dma Channel minor loop logic channel linking.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | false                                                              |

## 4.91 Parameter dmaLogicChannelConfig\_MinorLoopSizeType

Vendor specific: Set the Dma Channel minor loop transfer size.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0                                                                  |
| max                          | 1073741823                                                         |
| min                          | 0                                                                  |

## 4.92 Reference `dynamic_dmaLogicChannelConfig_MinorLoopLinkChValueType`

Vendor specific: Set the Dma Channel minor loop logic channel link.

| Property                     | Value                                                |
|------------------------------|------------------------------------------------------|
| type                         | ECUC-REFERENCE-DEF                                   |
| origin                       | NXP                                                  |
| lowerMultiplicity            | 0                                                    |
| upperMultiplicity            | 1                                                    |
| postBuildVariantMultiplicity | false                                                |
| multiplicityConfigClasses    | VARIANT-POST-BUILD: POST-BUILD                       |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE                     |
| postBuildVariantValue        | false                                                |
| valueConfigClasses           | VARIANT-POST-BUILD: POST-BUILD                       |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE                     |
| requiresSymbolicNameValue    | true                                                 |
| destination                  | /AUTOSAR/EcucDefs/Mcl/MclConfig/dmaLogicChannel_Type |

## 4.93 Container `dmaLogicChannelConfig_TransferMajorLoopType`

Vendor specific: Transfer Major Loop

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.94 Parameter `dmaLogicChannelConfig_enMajorLoopLinkCh`

Vendor specific: Enable the Dma Channel major loop logic channel linking.

| Property | Value                  |
|----------|------------------------|
| type     | ECUC-BOOLEAN-PARAM-DEF |

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | false                                                              |

#### 4.95 Parameter dmaLogicChannelConfig\_MajorLoopCountType

Vendor specific: Sets the Dma Channel major loop count.

If minor loop channel linking is enabled, the major loop count limit is between 0 and 511.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 1                                                                  |
| max                          | 32767                                                              |
| min                          | 0                                                                  |

#### 4.96 Reference dynamic\_dmaLogicChannelConfig\_MajorLoopLink↵ ChValueType

Vendor specific: Set the Dma Channel major loop logic channel link.

| Property | Value              |
|----------|--------------------|
| type     | ECUC-REFERENCE-DEF |

| Property                     | Value                                               |
|------------------------------|-----------------------------------------------------|
| origin                       | NXP                                                 |
| lowerMultiplicity            | 0                                                   |
| upperMultiplicity            | 1                                                   |
| postBuildVariantMultiplicity | false                                               |
| multiplicityConfigClasses    | VARIANT-POST-BUILD: POST-BUILD                      |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE                    |
| postBuildVariantValue        | false                                               |
| valueConfigClasses           | VARIANT-POST-BUILD: POST-BUILD                      |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE                    |
| requiresSymbolicNameValue    | true                                                |
| destination                  | /AUTOSAR/EcuDefs/Mcl/MclConfig/dmaLogicChannel_Type |

## 4.97 Container dmaLogicChannel\_ScatterGatherConfigType

Vendor specific:

Logic Channel ScatterGather Configuration.

Included subcontainers:

- [dmaLogicChannelConfig\\_ScatterGatherArrayType](#)

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.98 Container dmaLogicChannelConfig\_ScatterGatherArrayType

Vendor specific: Logic Channel ScatterGather Configuration.

Included subcontainers:

- [dmaLogicChannelConfig\\_ScatterGatherElementConfigType](#)

| Property | Value                         |
|----------|-------------------------------|
| type     | ECUC-PARAM-CONF-CONTAINER-DEF |

| Property                     | Value                            |
|------------------------------|----------------------------------|
| lowerMultiplicity            | 0                                |
| upperMultiplicity            | 256                              |
| postBuildVariantMultiplicity | false                            |
| multiplicityConfigClasses    | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |

## 4.99 Container

### **dmaLogicChannelConfig\_ScatterGatherElementConfigType**

Vendor specific: Element

Included subcontainers:

- [dmaLogicChannelConfig\\_TransferControlType](#)
- [dmaLogicChannelConfig\\_TransferSourceType](#)
- [dmaLogicChannelConfig\\_TransferDestinationType](#)
- [dmaLogicChannelConfig\\_TransferMinorLoopType](#)
- [dmaLogicChannelConfig\\_TransferMajorLoopType](#)

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.100 Parameter

### **dmaLogicChannelConfig\_ScatterGatherElementNameType**

Vendor specific: Element Name

| Property          | Value                 |
|-------------------|-----------------------|
| type              | ECUC-STRING-PARAM-DEF |
| origin            | NXP                   |
| symbolicNameValue | false                 |
| lowerMultiplicity | 1                     |

| Property                     | Value                            |
|------------------------------|----------------------------------|
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0_SGA_ELEMENT_0                  |

#### 4.101 Parameter dmaLogicChannelConfig\_\_LastElementLink\_\_↔ ScatterGatherType

Vendor specific: For non-circular lists, the last element shall have this checkbox set.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

#### 4.102 Parameter dmaLogicChannelConfig\_\_enScatterGatherConfig

Vendor specific: Enable Scatter/Gather Configuration

| Property                     | Value                  |
|------------------------------|------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF |
| origin                       | NXP                    |
| symbolicNameValue            | false                  |
| lowerMultiplicity            | 1                      |
| upperMultiplicity            | 1                      |
| postBuildVariantMultiplicity | N/A                    |
| multiplicityConfigClasses    | N/A                    |
| postBuildVariantValue        | true                   |

| Property           | Value                            |
|--------------------|----------------------------------|
| valueConfigClasses | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                    | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue       | false                            |

#### 4.103 Reference dynamic\_dmaLogicChannelConfig\_\_BasicElement↔ Link\_\_ScatterGatherType

Vendor specific: Element Link. Elements shall be part of the same Logic Channel.

| Property                     | Value                                                                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type                         | ECUC-REFERENCE-DEF                                                                                                                                                                |
| origin                       | NXP                                                                                                                                                                               |
| lowerMultiplicity            | 0                                                                                                                                                                                 |
| upperMultiplicity            | 1                                                                                                                                                                                 |
| postBuildVariantMultiplicity | false                                                                                                                                                                             |
| multiplicityConfigClasses    | VARIANT-POST-BUILD: POST-BUILD                                                                                                                                                    |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE                                                                                                                                                  |
| postBuildVariantValue        | false                                                                                                                                                                             |
| valueConfigClasses           | VARIANT-POST-BUILD: POST-BUILD                                                                                                                                                    |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE                                                                                                                                                  |
| requiresSymbolicNameValue    | true                                                                                                                                                                              |
| destination                  | /AUTOSAR/EcucDefs/Mcl/MclConfig/dmaLogicChannel_Type/dmaLogic↔<br>Channel__ConfigType/dmaLogicChannel__ScatterGatherConfigType/dma↔<br>LogicChannelConfig__ScatterGatherArrayType |

#### 4.104 Container dmaLogicChannelConfig\_\_TransferControlType

Vendor specific:

TCD Global Interrupt.

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

#### 4.105 Parameter dmaLogicChannelConfig\_enStart

Vendor specific: Enable the Dma Channel start service request (software request).

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

#### 4.106 Parameter dmaLogicChannelConfig\_enDmaMajorInterrupt

Vendor specific: Enable the Dma Channel major interrupt.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

#### 4.107 Parameter dmaLogicChannelConfig\_enDmaHalfMajorInterrupt

Vendor specific: Enable the Dma Channel half major interrupt.



| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| default Value                | false                            |

#### 4.108 Parameter dmaLogicChannelConfig\_disDmaAutoHwReq

Vendor specific: Disable the Dma Channel automatic request.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| default Value                | false                            |

#### 4.109 Parameter dmaLogicChannelConfig\_bandwidthControl

Vendor specific: Set the Dma Channel bandwidth control.

| Property          | Value                      |
|-------------------|----------------------------|
| type              | ECUC-ENUMERATION-PARAM-DEF |
| origin            | NXP                        |
| symbolicNameValue | false                      |
| lowerMultiplicity | 1                          |
| upperMultiplicity | 1                          |

| Property                     | Value                                                                                                                       |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| postBuildVariantMultiplicity | N/A                                                                                                                         |
| multiplicityConfigClasses    | N/A                                                                                                                         |
| postBuildVariantValue        | true                                                                                                                        |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE                                                                                            |
|                              | VARIANT-POST-BUILD: POST-BUILD                                                                                              |
| defaultValue                 | DMA_IP_BWC_ENGINE_NO_STALL                                                                                                  |
| literals                     | ['DMA_IP_BWC_ENGINE_NO_STALL', 'DMA_IP_BWC_ENGINE_HPE', 'DMA_IP_BWC_ENGINE_4CYCLE_STALL', 'DMA_IP_BWC_ENGINE_8CYCLE_STALL'] |

#### 4.110 Parameter dmaLogicChannelConfig\_ScatterGatherAddressType

Vendor specific: ScatterGather Address

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-STRING-PARAM-DEF            |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0U                               |

#### 4.111 Parameter dmaLogicChannelConfig\_DestinationStoreAddressType

Vendor specific: Destination Store Address

| Property          | Value                 |
|-------------------|-----------------------|
| type              | ECUC-STRING-PARAM-DEF |
| origin            | NXP                   |
| symbolicNameValue | false                 |
| lowerMultiplicity | 1                     |
| upperMultiplicity | 1                     |

| Property                     | Value                            |
|------------------------------|----------------------------------|
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0U                               |

#### 4.112 Container dmaLogicChannelConfig\_TransferSourceType

Vendor specific: Transfer Source

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

#### 4.113 Parameter dmaLogicChannelConfig\_SourceAddressType

Vendor specific: Source Address

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-STRING-PARAM-DEF            |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0                                |

#### 4.114 Parameter dmaLogicChannelConfig\_SourceSignedOffsetType

Vendor specific: Set the Dma Channel source signed offset value.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0                                                                  |
| max                          | 32767                                                              |
| min                          | -32767                                                             |

#### 4.115 Parameter dmaLogicChannelConfig\_SourceLastAddressAdjustmentType

Vendor specific: Set the Dma Channel source signed last address adjustment.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0                                                                  |
| max                          | 2147483647                                                         |
| min                          | -2147483647                                                        |

## 4.116 Parameter dmaTransferConfig\_TransferSizeType

Vendor specific: Set the Dma Channel source transfer size.

| Property                     | Value                                                                                                                                                                                                                        |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF                                                                                                                                                                                                   |
| origin                       | NXP                                                                                                                                                                                                                          |
| symbolicNameValue            | false                                                                                                                                                                                                                        |
| lowerMultiplicity            | 1                                                                                                                                                                                                                            |
| upperMultiplicity            | 1                                                                                                                                                                                                                            |
| postBuildVariantMultiplicity | N/A                                                                                                                                                                                                                          |
| multiplicityConfigClasses    | N/A                                                                                                                                                                                                                          |
| postBuildVariantValue        | true                                                                                                                                                                                                                         |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE                                                                                                                                                                                             |
|                              | VARIANT-POST-BUILD: POST-BUILD                                                                                                                                                                                               |
| defaultValue                 | DMA_IP_TRANSFER_SIZE_1_BYTE                                                                                                                                                                                                  |
| literals                     | ['DMA_IP_TRANSFER_SIZE_1_BYTE', 'DMA_IP_TRANSFER_SIZE_2_BYTE', 'DMA_IP_TRANSFER_SIZE_4_BYTE', 'DMA_IP_TRANSFER_SIZE_8_BYTE', 'DMA_IP_TRANSFER_SIZE_16_BYTE', 'DMA_IP_TRANSFER_SIZE_32_BYTE', 'DMA_IP_TRANSFER_SIZE_64_BYTE'] |

## 4.117 Parameter dmaLogicChannelConfig\_SourceModuloType

Vendor specific: Set the Dma Channel source modulo.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0                                |
| max                          | 31                               |
| min                          | 0                                |

#### 4.118 Container dmaLogicChannelConfig\_TransferDestinationType

Vendor specific: Transfer Destination

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

#### 4.119 Parameter dmaLogicChannelConfig\_DestinationAddressType

Vendor specific: Destination Address

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-STRING-PARAM-DEF                                              |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0                                                                  |

#### 4.120 Parameter dmaLogicChannelConfig\_DestinationSignedOffsetType

Vendor specific: Set the Dma Channel destination signed offset value.

| Property | Value                  |
|----------|------------------------|
| type     | ECUC-INTEGER-PARAM-DEF |

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0                                                                  |
| max                          | 32767                                                              |
| min                          | -32767                                                             |

#### 4.121 Parameter dmaLogicChannelConfig\_DestinationLastAddressAdjustmentType

Vendor specific: Set the Dma Channel destination signed last address adjustment.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | 0                                                                  |
| max                          | 2147483647                                                         |
| min                          | -2147483647                                                        |

#### 4.122 Parameter dmaTransferConfig\_TransferSizeType

Vendor specific: Set the Dma Channel destination transfer size.

| Property | Value                      |
|----------|----------------------------|
| type     | ECUC-ENUMERATION-PARAM-DEF |

| Property                     | Value                                                                                                                                                                                                                        |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| origin                       | NXP                                                                                                                                                                                                                          |
| symbolicNameValue            | false                                                                                                                                                                                                                        |
| lowerMultiplicity            | 1                                                                                                                                                                                                                            |
| upperMultiplicity            | 1                                                                                                                                                                                                                            |
| postBuildVariantMultiplicity | N/A                                                                                                                                                                                                                          |
| multiplicityConfigClasses    | N/A                                                                                                                                                                                                                          |
| postBuildVariantValue        | true                                                                                                                                                                                                                         |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE                                                                                                                                                                                             |
|                              | VARIANT-POST-BUILD: POST-BUILD                                                                                                                                                                                               |
| defaultValue                 | DMA_IP_TRANSFER_SIZE_1_BYTE                                                                                                                                                                                                  |
| literals                     | ['DMA_IP_TRANSFER_SIZE_1_BYTE', 'DMA_IP_TRANSFER_SIZE_2_BYTE', 'DMA_IP_TRANSFER_SIZE_4_BYTE', 'DMA_IP_TRANSFER_SIZE_8_BYTE', 'DMA_IP_TRANSFER_SIZE_16_BYTE', 'DMA_IP_TRANSFER_SIZE_32_BYTE', 'DMA_IP_TRANSFER_SIZE_64_BYTE'] |

#### 4.123 Parameter dmaLogicChannelConfig\_DestinationModuloType

Vendor specific: Set the Dma Channel destination modulo.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0                                |
| max                          | 31                               |
| min                          | 0                                |

#### 4.124 Container dmaLogicChannelConfig\_TransferMinorLoopType

Vendor specific: Transfer Minor Loop

Included subcontainers:

- None



| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

#### 4.125 Parameter dmaLogicChannelConfig\_enSourceOffset

Vendor specific: Enable the Dma Channel minor loop source offset.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | false                                                              |

#### 4.126 Parameter dmaLogicChannelConfig\_enDestinationOffset

Vendor specific: Enable the Dma Channel minor loop destination offset.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | false                                                              |

#### 4.127 Parameter dmaLogicChannelConfig\_OffsetValueType

Vendor specific: Set the Dma Channel minor loop signed offset.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0                                |
| max                          | 1048575                          |
| min                          | -1048575                         |

#### 4.128 Parameter dmaLogicChannelConfig\_enMinorLoopLinkCh

Vendor specific: Enable the Dma Channel minor loop logic channel linking.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

#### 4.129 Parameter dmaLogicChannelConfig\_MinorLoopSizeType

Vendor specific: Set the Dma Channel minor loop transfer size.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0                                |
| max                          | 1073741823                       |
| min                          | 0                                |

#### 4.130 Reference `dynamic_dmaLogicChannelConfig_MinorLoopLink`↔ `ChValueType`

Vendor specific: Set the Dma Channel minor loop logic channel link.

| Property                     | Value                                                |
|------------------------------|------------------------------------------------------|
| type                         | ECUC-REFERENCE-DEF                                   |
| origin                       | NXP                                                  |
| lowerMultiplicity            | 0                                                    |
| upperMultiplicity            | 1                                                    |
| postBuildVariantMultiplicity | false                                                |
| multiplicityConfigClasses    | VARIANT-POST-BUILD: POST-BUILD                       |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE                     |
| postBuildVariantValue        | false                                                |
| valueConfigClasses           | VARIANT-POST-BUILD: POST-BUILD                       |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE                     |
| requiresSymbolicNameValue    | true                                                 |
| destination                  | /AUTOSAR/EcucDefs/Mcl/MclConfig/dmaLogicChannel_Type |

#### 4.131 Container `dmaLogicChannelConfig_TransferMajorLoopType`

Vendor specific: Transfer Major Loop

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

### 4.132 Parameter dmaLogicChannelConfig\_enMajorLoopLinkCh

Vendor specific: Enable the Dma Channel major loop logic channel linking.

| Property                     | Value                                                              |
|------------------------------|--------------------------------------------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF                                             |
| origin                       | NXP                                                                |
| symbolicNameValue            | false                                                              |
| lowerMultiplicity            | 1                                                                  |
| upperMultiplicity            | 1                                                                  |
| postBuildVariantMultiplicity | N/A                                                                |
| multiplicityConfigClasses    | N/A                                                                |
| postBuildVariantValue        | true                                                               |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE<br>VARIANT-POST-BUILD: POST-BUILD |
| defaultValue                 | false                                                              |

### 4.133 Parameter dmaLogicChannelConfig\_MajorLoopCountType

Vendor specific: Set the Dma Channel major loop count.

If minor loop channel linking is enabled, the major loop count limit is between 0 and 511.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |

| Property     | Value                          |
|--------------|--------------------------------|
|              | VARIANT-POST-BUILD: POST-BUILD |
| defaultValue | 1                              |
| max          | 32767                          |
| min          | 0                              |

#### 4.134 Reference dynamic\_dmaLogicChannelConfig\_MajorLoopLink↵ ChValueType

Vendor specific: Set the Dma Channel major loop logic channel link.

| Property                     | Value                                                |
|------------------------------|------------------------------------------------------|
| type                         | ECUC-REFERENCE-DEF                                   |
| origin                       | NXP                                                  |
| lowerMultiplicity            | 0                                                    |
| upperMultiplicity            | 1                                                    |
| postBuildVariantMultiplicity | false                                                |
| multiplicityConfigClasses    | VARIANT-POST-BUILD: POST-BUILD                       |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE                     |
| postBuildVariantValue        | false                                                |
| valueConfigClasses           | VARIANT-POST-BUILD: POST-BUILD                       |
|                              | VARIANT-PRE-COMPILE: PRE-COMPILE                     |
| requiresSymbolicNameValue    | true                                                 |
| destination                  | /AUTOSAR/EcucDefs/Mcl/MclConfig/dmaLogicChannel_Type |

#### 4.135 Container dmaLogicChannel\_CrcConfigType

Vendor specific:

Crc Configuration.

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

### 4.136 Parameter dmaLogicChannel\_HwCrcChId

Vendor specific:

Select the physical DMA CRC Channel.

NOTE: This is an Implementation Specific Parameter.

| Property                     | Value                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF                                                                                                                                                                                                                                                                                                                                                                       |
| origin                       | NXP                                                                                                                                                                                                                                                                                                                                                                                              |
| symbolicNameValue            | false                                                                                                                                                                                                                                                                                                                                                                                            |
| lowerMultiplicity            | 1                                                                                                                                                                                                                                                                                                                                                                                                |
| upperMultiplicity            | 1                                                                                                                                                                                                                                                                                                                                                                                                |
| postBuildVariantMultiplicity | N/A                                                                                                                                                                                                                                                                                                                                                                                              |
| multiplicityConfigClasses    | N/A                                                                                                                                                                                                                                                                                                                                                                                              |
| postBuildVariantValue        | true                                                                                                                                                                                                                                                                                                                                                                                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE                                                                                                                                                                                                                                                                                                                                                                 |
|                              | VARIANT-POST-BUILD: POST-BUILD                                                                                                                                                                                                                                                                                                                                                                   |
| defaultValue                 | DMA_IP_HW_CRC_0_CH_0                                                                                                                                                                                                                                                                                                                                                                             |
| literals                     | ['DMA_IP_HW_CRC_0_CH_0', 'DMA_IP_HW_CRC_0_CH_1', 'DMA_IP_HW_CRC_0_CH_2', 'DMA_IP_HW_CRC_0_CH_3', 'DMA_IP_HW_CRC_0_CH_4', 'DMA_IP_HW_CRC_0_CH_5', 'DMA_IP_HW_CRC_0_CH_6', 'DMA_IP_HW_CRC_0_CH_7', 'DMA_IP_HW_CRC_1_CH_0', 'DMA_IP_HW_CRC_1_CH_1', 'DMA_IP_HW_CRC_1_CH_2', 'DMA_IP_HW_CRC_1_CH_3', 'DMA_IP_HW_CRC_1_CH_4', 'DMA_IP_HW_CRC_1_CH_5', 'DMA_IP_HW_CRC_1_CH_6', 'DMA_IP_HW_CRC_1_CH_7'] |

### 4.137 Parameter dmaLogicChannel\_CrcMode

Vendor specific:

Select the Mode for CRC logic.

NOTE: This is an Implementation Specific Parameter.

| Property                     | Value                      |
|------------------------------|----------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF |
| origin                       | NXP                        |
| symbolicNameValue            | false                      |
| lowerMultiplicity            | 1                          |
| upperMultiplicity            | 1                          |
| postBuildVariantMultiplicity | N/A                        |

| Property                  | Value                            |
|---------------------------|----------------------------------|
| multiplicityConfigClasses | N/A                              |
| postBuildVariantValue     | true                             |
| valueConfigClasses        | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                           | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue              | NORMAL_CRC_MODE                  |
| literals                  | ['NORMAL_CRC_MODE']              |

### 4.138 Parameter dmaLogicChannel\_CrcPolynomial

Vendor specific:

Select the polynomial for CRC logic.

NOTE: This is an Implementation Specific Parameter.

| Property                     | Value                                                                                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| type                         | ECUC-ENUMERATION-PARAM-DEF                                                                                                                               |
| origin                       | NXP                                                                                                                                                      |
| symbolicNameValue            | false                                                                                                                                                    |
| lowerMultiplicity            | 1                                                                                                                                                        |
| upperMultiplicity            | 1                                                                                                                                                        |
| postBuildVariantMultiplicity | N/A                                                                                                                                                      |
| multiplicityConfigClasses    | N/A                                                                                                                                                      |
| postBuildVariantValue        | true                                                                                                                                                     |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE                                                                                                                         |
|                              | VARIANT-POST-BUILD: POST-BUILD                                                                                                                           |
| defaultValue                 | ETHERNET_CCITT32_CRC32                                                                                                                                   |
| literals                     | ['ETHERNET_CCITT32_CRC32', 'CASTAGNOLI_ISCSICRC32C_CRC32', 'E2E_PROFILE4_CRC32', 'X25_CCITT16_CRC16', 'H2F_AUTOSAR_4_↵0_CRC8', 'VDA_CAN_SAE_J1850_CRC8'] |

### 4.139 Parameter dmaLogicChannel\_CrcInitialValue

Vendor specific:

Select the initial value for CRC logic.

NOTE: This is an Implementation Specific Parameter.

| Property | Value                  |
|----------|------------------------|
| type     | ECUC-INTEGER-PARAM-DEF |

| Property                     | Value                            |
|------------------------------|----------------------------------|
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | 0                                |
| max                          | 4294967295                       |
| min                          | 0                                |

#### 4.140 Parameter dmaLogicChannel\_CrcEnInitSel

Vendor specific:

Enable/Disable initial value of CRC.

NOTE: This is an Implementation Specific Parameter.

| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | false                            |

#### 4.141 Parameter dmaLogicChannel\_CrcEnLogic

Vendor specific:

Enable/Disable CRC Logic.

NOTE: This is an Implementation Specific Parameter.



| Property                     | Value                            |
|------------------------------|----------------------------------|
| type                         | ECUC-BOOLEAN-PARAM-DEF           |
| origin                       | NXP                              |
| symbolicNameValue            | false                            |
| lowerMultiplicity            | 1                                |
| upperMultiplicity            | 1                                |
| postBuildVariantMultiplicity | N/A                              |
| multiplicityConfigClasses    | N/A                              |
| postBuildVariantValue        | true                             |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PRE-COMPILE |
|                              | VARIANT-POST-BUILD: POST-BUILD   |
| defaultValue                 | true                             |

## 4.142 Container CommonPublishedInformation

Vendor specific:

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

| Property                     | Value                         |
|------------------------------|-------------------------------|
| type                         | ECUC-PARAM-CONF-CONTAINER-DEF |
| lowerMultiplicity            | 1                             |
| upperMultiplicity            | 1                             |
| postBuildVariantMultiplicity | N/A                           |
| multiplicityConfigClasses    | N/A                           |

## 4.143 Parameter ArReleaseMajorVersion

Vendor specific:

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

| Property          | Value                  |
|-------------------|------------------------|
| type              | ECUC-INTEGER-PARAM-DEF |
| origin            | NXP                    |
| symbolicNameValue | false                  |

| Property                     | Value                                      |
|------------------------------|--------------------------------------------|
| lowerMultiplicity            | 1                                          |
| upperMultiplicity            | 1                                          |
| postBuildVariantMultiplicity | N/A                                        |
| multiplicityConfigClasses    | N/A                                        |
| postBuildVariantValue        | false                                      |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION |
|                              | VARIANT-POST-BUILD: PUBLISHED-INFORMATION  |
| defaultValue                 | 4                                          |
| max                          | 4                                          |
| min                          | 4                                          |

#### 4.144 Parameter ArReleaseMinorVersion

Vendor specific:

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

| Property                     | Value                                      |
|------------------------------|--------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                     |
| origin                       | NXP                                        |
| symbolicNameValue            | false                                      |
| lowerMultiplicity            | 1                                          |
| upperMultiplicity            | 1                                          |
| postBuildVariantMultiplicity | N/A                                        |
| multiplicityConfigClasses    | N/A                                        |
| postBuildVariantValue        | false                                      |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION |
|                              | VARIANT-POST-BUILD: PUBLISHED-INFORMATION  |
| defaultValue                 | 4                                          |
| max                          | 4                                          |
| min                          | 4                                          |

#### 4.145 Parameter ArReleaseRevisionVersion

Vendor specific:

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

| Property | Value                  |
|----------|------------------------|
| type     | ECUC-INTEGER-PARAM-DEF |

| Property                     | Value                                      |
|------------------------------|--------------------------------------------|
| origin                       | NXP                                        |
| symbolicNameValue            | false                                      |
| lowerMultiplicity            | 1                                          |
| upperMultiplicity            | 1                                          |
| postBuildVariantMultiplicity | N/A                                        |
| multiplicityConfigClasses    | N/A                                        |
| postBuildVariantValue        | false                                      |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION |
|                              | VARIANT-POST-BUILD: PUBLISHED-INFORMATION  |
| defaultValue                 | 0                                          |
| max                          | 0                                          |
| min                          | 0                                          |

#### 4.146 Parameter ModuleId

Vendor specific:

Module ID of this module from Module List.

| Property                     | Value                                      |
|------------------------------|--------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                     |
| origin                       | NXP                                        |
| symbolicNameValue            | false                                      |
| lowerMultiplicity            | 1                                          |
| upperMultiplicity            | 1                                          |
| postBuildVariantMultiplicity | N/A                                        |
| multiplicityConfigClasses    | N/A                                        |
| postBuildVariantValue        | false                                      |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION |
|                              | VARIANT-POST-BUILD: PUBLISHED-INFORMATION  |
| defaultValue                 | 255                                        |
| max                          | 255                                        |
| min                          | 255                                        |

#### 4.147 Parameter SwMajorVersion

Vendor specific:

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

| Property                     | Value                                      |
|------------------------------|--------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                     |
| origin                       | NXP                                        |
| symbolicNameValue            | false                                      |
| lowerMultiplicity            | 1                                          |
| upperMultiplicity            | 1                                          |
| postBuildVariantMultiplicity | N/A                                        |
| multiplicityConfigClasses    | N/A                                        |
| postBuildVariantValue        | false                                      |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION |
|                              | VARIANT-POST-BUILD: PUBLISHED-INFORMATION  |
| defaultValue                 | 4                                          |
| max                          | 4                                          |
| min                          | 4                                          |

#### 4.148 Parameter SwMinorVersion

Vendor specific:

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

| Property                     | Value                                      |
|------------------------------|--------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                     |
| origin                       | NXP                                        |
| symbolicNameValue            | false                                      |
| lowerMultiplicity            | 1                                          |
| upperMultiplicity            | 1                                          |
| postBuildVariantMultiplicity | N/A                                        |
| multiplicityConfigClasses    | N/A                                        |
| postBuildVariantValue        | false                                      |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION |
|                              | VARIANT-POST-BUILD: PUBLISHED-INFORMATION  |
| defaultValue                 | 0                                          |
| max                          | 0                                          |
| min                          | 0                                          |

#### 4.149 Parameter SwPatchVersion

Vendor specific:

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

| Property                     | Value                                      |
|------------------------------|--------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                     |
| origin                       | NXP                                        |
| symbolicNameValue            | false                                      |
| lowerMultiplicity            | 1                                          |
| upperMultiplicity            | 1                                          |
| postBuildVariantMultiplicity | N/A                                        |
| multiplicityConfigClasses    | N/A                                        |
| postBuildVariantValue        | false                                      |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION |
|                              | VARIANT-POST-BUILD: PUBLISHED-INFORMATION  |
| defaultValue                 | 2                                          |
| max                          | 2                                          |
| min                          | 2                                          |

## 4.150 Parameter VendorId

Vendor specific:

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

| Property                     | Value                                      |
|------------------------------|--------------------------------------------|
| type                         | ECUC-INTEGER-PARAM-DEF                     |
| origin                       | NXP                                        |
| symbolicNameValue            | false                                      |
| lowerMultiplicity            | 1                                          |
| upperMultiplicity            | 1                                          |
| postBuildVariantMultiplicity | N/A                                        |
| multiplicityConfigClasses    | N/A                                        |
| postBuildVariantValue        | false                                      |
| valueConfigClasses           | VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION |
|                              | VARIANT-POST-BUILD: PUBLISHED-INFORMATION  |
| defaultValue                 | 43                                         |
| max                          | 43                                         |
| min                          | 43                                         |

This chapter describes the Tresos configuration plug-in for the *driver* Driver. The most of the parameters are described below.



# Chapter 5

## Module Index

### 5.1 Software Specification

Here is a list of all modules:

|                           |     |
|---------------------------|-----|
| CACHE IP Driver . . . . . | 98  |
| MCL Driver . . . . .      | 104 |
| DMA IP Driver . . . . .   | 136 |
| FTM IP Driver . . . . .   | 161 |

## Chapter 6

### Module Documentation

#### 6.1 CACHE IP Driver

##### 6.1.1 Detailed Description

##### Enum Reference

- enum [Cache\\_Ip\\_Type](#)  
*This type contains the Cache Ip types.*

##### Function Reference

- Std\_ReturnType [Cache\\_Ip\\_Enable](#) (const [Cache\\_Ip\\_Type](#) CacheType)  
*This function enables the Cache Ip Driver.*
- Std\_ReturnType [Cache\\_Ip\\_Disable](#) (const [Cache\\_Ip\\_Type](#) CacheType)  
*This function disables the Cache Ip Driver.*
- Std\_ReturnType [Cache\\_Ip\\_Invalidate](#) (const [Cache\\_Ip\\_Type](#) CacheType)  
*This function Invalidates the Cache Ip Driver.*
- Std\_ReturnType [Cache\\_Ip\\_Clean](#) (const [Cache\\_Ip\\_Type](#) CacheType, const boolean EnInvalidate)  
*This function Clean the Cache Ip Driver.*
- Std\_ReturnType [Cache\\_Ip\\_InvalidateByAddr](#) (const [Cache\\_Ip\\_Type](#) CacheType, const uint32 Addr, const uint32 Length)  
*This function Invalidates By Address the Cache Ip Driver.*
- Std\_ReturnType [Cache\\_Ip\\_CleanByAddr](#) (const [Cache\\_Ip\\_Type](#) CacheType, const boolean EnInvalidate, const uint32 Addr, const uint32 Length)  
*This function Clean By Address the Cache Ip Driver.*

##### 6.1.2 Enum Reference

### 6.1.2.1 Cache\_Ip\_Type

```
enum Cache_Ip_Type
```

This type contains the Cache Ip types.

The cache types specifies what type of cache shall be used when calling the interface. The `CACHE_IP_ALL` specifies both Instruction and Data caches. The `CACHE_IP_INSTRUCTION` specifies the Instruction cache. The `CACHE_IP_DATA` specifies the Data cache.

Definition at line 107 of file `Cache_Ip.h`.

## 6.1.3 Function Reference

### 6.1.3.1 Cache\_Ip\_Enable()

```
Std_ReturnType Cache_Ip_Enable (
    const Cache_Ip_Type CacheType )
```

This function enables the Cache Ip Driver.

This service is a reentrant function that shall enable the Cache Ip driver.

Parameters

|    |                  |                           |
|----|------------------|---------------------------|
| in | <i>CacheType</i> | Specifies the cache type. |
|----|------------------|---------------------------|

Returns

`E_OK` is returned if the enable action finished correctly. `E_NOT_OK` is returned if the enable action finished incorrectly.

### 6.1.3.2 Cache\_Ip\_Disable()

```
Std_ReturnType Cache_Ip_Disable (
    const Cache_Ip_Type CacheType )
```

This function disables the Cache Ip Driver.

This service is a reentrant function that shall disables the Cache Ip driver.

Parameters

|    |                  |                           |
|----|------------------|---------------------------|
| in | <i>CacheType</i> | Specifies the cache type. |
|----|------------------|---------------------------|



### Returns

E\_OK is returned if the disable action finished correctly. E\_NOT\_OK is returned if the disable action finished incorrectly.

#### 6.1.3.3 Cache\_Ip\_Invalidate()

```
Std_ReturnType Cache_Ip_Invalidate (  
    const Cache_Ip_Type CacheType )
```

This function Invalidates the Cache Ip Driver.

This service is a reentrant function that shall Invalidates the Cache Ip driver. The Invalidate operation applies for the entire selected Cache Type.

A cache invalidate operation ensures that updates made visible by observers that access memory at the point to which the invalidate is defined are made visible to an observer that controls the cache. This might result in the loss of updates to the locations affected by the invalidate operation that have been written by observers that access the cache. If the address of an entry on which the invalidate operates does not have a Normal Cacheable attribute, or if the cache is disabled, then an invalidate operation also ensures that this address is not present in the cache.

### Parameters

|    |           |                           |
|----|-----------|---------------------------|
| in | CacheType | Specifies the cache type. |
|----|-----------|---------------------------|

### Returns

E\_OK is returned if the invalidation finished correctly. E\_NOT\_OK is returned if the invalidation finished incorrectly.

#### 6.1.3.4 Cache\_Ip\_Clean()

```
Std_ReturnType Cache_Ip_Clean (  
    const Cache_Ip_Type CacheType,  
    const boolean EnInvalidate )
```

This function Clean the Cache Ip Driver.

This service is a reentrant function that shall Clean the Cache Ip driver. The Clean operation applies for the entire selected Cache Type.

A cache clean operation ensures that updates made by an observer that controls the cache are made visible to other observers that can access memory at the point to which the operation is performed. Once the Clean has completed, the new memory values are guaranteed to be visible to the point to which the operation is performed, for example

to the point of unification. The cleaning of a cache entry from a cache can overwrite memory that has been written by another observer only if the entry contains a location that has been written to by an observer in the shareability domain of that memory location.

A cache clean and invalidate operation behaves as the execution of a clean operation followed immediately by an invalidate operation. Both operations are performed to the same location.

### Parameters

|    |                     |                                                  |
|----|---------------------|--------------------------------------------------|
| in | <i>CacheType</i>    | Specifies the cache type.                        |
| in | <i>enInvalidate</i> | Specifies to execute operation Clean&Invalidate. |

### Returns

E\_OK is returned if the cleaning finished correctly. E\_NOT\_OK is returned if the cleaning finished incorrectly.

#### 6.1.3.5 Cache\_Ip\_InvalidateByAddr()

```
Std_ReturnType Cache_Ip_InvalidateByAddr (
    const Cache_Ip_Type CacheType,
    const uint32 Addr,
    const uint32 Length )
```

This function Invalidates By Address the Cache Ip Driver.

This service is a reentrant function that shall Invalidates By Address the Cache Ip driver. The Invalidate By Address operation applies for the memory segment specified by the start Address and Length. The operation Invalidates only multiple of Cache Line Size, thus the specified memory segment shall be aligned and multiple of the Cache Line Size.

A cache invalidate operation ensures that updates made visible by observers that access memory at the point to which the invalidate is defined are made visible to an observer that controls the cache. This might result in the loss of updates to the locations affected by the invalidate operation that have been written by observers that access the cache. If the address of an entry on which the invalidate operates does not have a Normal Cacheable attribute, or if the cache is disabled, then an invalidate operation also ensures that this address is not present in the cache.

### Parameters

|    |                  |                                             |
|----|------------------|---------------------------------------------|
| in | <i>CacheType</i> | Specifies the cache type.                   |
| in | <i>Addr</i>      | Specifies the memory segment start address. |
| in | <i>Length</i>    | Specifies the memory segment length.        |

### Returns

E\_OK is returned if the invalidation finished correctly. E\_NOT\_OK is returned if the invalidation finished incorrectly.

#### 6.1.3.6 Cache\_Ip\_CleanByAddr()

```
Std_ReturnType Cache_Ip_CleanByAddr (
    const Cache_Ip_Type CacheType,
```

```

const boolean EnInvalidate,
const uint32 Addr,
const uint32 Length )

```

This function Clean By Address the Cache Ip Driver.

This service is a reentrant function that shall Clean By Address the Cache Ip driver. The Clean By Address operation applies for the memory segment specified by the start Address and Length. The operation Cleans only multiple of Cache Line Size, thus the specified memory segment shall be aligned and multiple of the Cache Line Size.

A cache clean operation ensures that updates made by an observer that controls the cache are made visible to other observers that can access memory at the point to which the operation is performed. Once the Clean has completed, the new memory values are guaranteed to be visible to the point to which the operation is performed, for example to the point of unification. The cleaning of a cache entry from a cache can overwrite memory that has been written by another observer only if the entry contains a location that has been written to by an observer in the shareability domain of that memory location.

A cache clean and invalidate operation behaves as the execution of a clean operation followed immediately by an invalidate operation. Both operations are performed to the same location.

Parameters

|    |                     |                                                  |
|----|---------------------|--------------------------------------------------|
| in | <i>CacheType</i>    | Specifies the cache type.                        |
| in | <i>enInvalidate</i> | Specifies to execute operation Clean&Invalidate. |
| in | <i>Addr</i>         | Specifies the memory segment start address.      |
| in | <i>Length</i>       | Specifies the memory segment length.             |

Returns

E\_OK is returned if the cleaning finished correctly. E\_NOT\_OK is returned if the cleaning finished incorrectly.

## 6.2 MCL Driver

### 6.2.1 Detailed Description

#### Data Structures

- struct [Mcl\\_DmaChannelGlobalListType](#)  
*This type contains the Mcl Dma Channel Global List. [More...](#)*
- struct [Mcl\\_DmaChannelTransferListType](#)  
*This type contains the Mcl Dma Channel Transfer List. [More...](#)*
- struct [Mcl\\_DmaChannelScatterGatherListType](#)  
*This type contains the Mcl Dma Channel Scatter/Gather List. [More...](#)*
- struct [Mcl\\_DmaChannelCrcListType](#)  
*This type contains the Mcl Dma Channel Crc List. [More...](#)*
- struct [Mcl\\_DmaInstanceStatusType](#)  
*This type contains the Mcl Dma Instance Status. [More...](#)*
- struct [Mcl\\_DmaChannelStatusType](#)  
*This type contains the Mcl Dma Channel Status. [More...](#)*
- struct [Mcl\\_ConfigType](#)  
*This type contains the Mcl Configuration. [More...](#)*

#### Macros

- `#define MCL_DET_INIT`  
*API service ID for Mcl\_Init function.*
- `#define MCL_DET_DMA_INSTANCE_COMMAND`  
*API service ID for Mcl\_SetDmaInstanceCommand function.*
- `#define MCL_DET_DMA_INSTANCE_STATUS`  
*API service ID for Mcl\_GetDmaInstanceStatus function.*
- `#define MCL_DET_DMA_CHANNEL_COMMAND`  
*API service ID for Mcl\_SetDmaChannelCommand function.*
- `#define MCL_DET_DMA_CHANNEL_STATUS`  
*API service ID for Mcl\_GetDmaChannelStatus function.*
- `#define MCL_DET_DMA_GLOBAL`  
*API service ID for Mcl\_SetDmaChannelGlobalList function.*
- `#define MCL_DET_DMA_TRANSFER`  
*API service ID for Mcl\_SetDmaChannelTransferList function.*
- `#define MCL_DET_DMA_SCATTER_GATHER_LIST`  
*API service ID for Mcl\_SetDmaChannelScatterGatherList function.*
- `#define MCL_DET_DMA_INFORMATION`  
*API service ID for Mcl\_GetDmaChannelParam function.*
- `#define MCL_DET_DMA_SCATTER_GATHER_CONFIG`  
*API service ID for Mcl\_SetDmaChannelScatterGatherConfig function.*
- `#define MCL_DET_DMA_CRC`

- API service ID for Mcl\_SetDmaChannelCrcList function.*

  - #define **MCL\_DET\_DEINIT**

*API service ID for Mcl\_DeInit function.*
- #define **MCL\_DET\_CACHE\_ENABLE**

*API service ID for Mcl\_CacheEnable function.*
- #define **MCL\_DET\_CACHE\_DISABLE**

*API service ID for Mcl\_CacheDisable function.*
- #define **MCL\_DET\_CACHE\_INVALIDATE**

*API service ID for Mcl\_CacheInvalidate function.*
- #define **MCL\_DET\_CACHE\_CLEAN**

*API service ID for Mcl\_CacheClean function.*
- #define **MCL\_DET\_CACHE\_INVALIDATE\_BY\_ADDRESS**

*API service ID for Mcl\_CacheInvalidateByAddr function.*
- #define **MCL\_DET\_CACHE\_CLEAN\_BY\_ADDRESS**

*API service ID for Mcl\_CacheCleanByAddr function.*
- #define **MCL\_E\_UNINIT**

*All API's having pointers as parameters shall return this error if called with with a NULL value.*
- #define **MCL\_E\_PARAM\_POINTER**

*All API's having pointers as parameters shall return this error if called with with a NULL value.*
- #define **MCL\_E\_INVALID\_INSTANCE**

*All API's called with wrong instance shall return this error.*
- #define **MCL\_E\_INVALID\_CHANNEL**

*All API's called with wrong channel shall return this error.*
- #define **MCL\_E\_INVALID\_COMMAND**

*All API's called with wrong instance shall return this error.*
- #define **MCL\_E\_INVALID\_PARAMETER**

*All API's called with wrong read parameter shall return this error.*
- #define **MCL\_E\_INVALID\_STATE**

*All API's called in wrong sequence shall return this error.*
- #define **MCL\_E\_INCONSISTENCY**

*All API's called while hardware has error status shall return this error.*
- #define **MCL\_E\_TIMEOUT**

*All API's called with a timeout value shall return this error if execution is not finished in the allocated timeframe.*
- #define **MCL\_E\_PARAM\_CONFIG**

*If DET error reporting is enabled, the MCL will check upon each API call if the requested resource is configured to be available on the current core, and in case of error will return MCL\_E\_PARAM\_CONFIG.*
- #define **MCL\_E\_PROTECTED**

*If DET error reporting is enabled, the MCL will check if registers are protected.*
- #define **MCL\_E\_INIT\_FAILED**

*If VariantPreCompile is used, the configuration pointer shall have a NULL\_PTR value. If VariantPostBuild is used, the configuration pointer shall be different from NULL\_PTR. And in case of violate will return MCL\_E\_INIT\_↔ FAILED.*

## Enum Reference

- enum [Mcl\\_DmaInstanceCmdType](#)  
*This type contains the Mcl Dma Instance Commands.*
- enum [Mcl\\_DmaChannelCmdType](#)  
*This type contains the Mcl Dma Channel Commands.*
- enum [Mcl\\_DmaChannelGlobalParamType](#)  
*This type contains the Mcl Dma Channel Global Parameters.*
- enum [Mcl\\_DmaChannelTransferParamType](#)  
*This type contains the Mcl Dma Channel Transfer Parameters.*
- enum [Mcl\\_DmaChannelStateType](#)  
*This type contains the Mcl Dma Channel State values.*
- enum [Mcl\\_DmaChannelCrcParamType](#)  
*This type contains the Mcl Dma Channel Crc Parameters.*
- enum [Mcl\\_DmaChannelInfoParamType](#)  
*This type contains the Mcl Dma Channel Information Parameters.*
- enum [Mcl\\_CacheType](#)  
*This type contains the Mcl Cache Type selection.*
- enum [Mcl\\_ReturnType](#)  
*This type contains the Mcl Return Type.*

## Function Reference

- void [Mcl\\_Init](#) (const [Mcl\\_ConfigType](#) \*const ConfigPtr)  
*This function initializes the Mcl Driver.*
- void [Mcl\\_DeInit](#) (void)  
*This function deinitializes the Mcl Driver.*
- void [Mcl\\_SetDmaInstanceCommand](#) (const uint32 Instance, const [Mcl\\_DmaInstanceCmdType](#) Command)  
*This function sets Dma Instance Command.*
- void [Mcl\\_GetDmaInstanceStatus](#) (const uint32 Instance, [Mcl\\_DmaInstanceStatusType](#) \*const Status)  
*This function gets Dma Instance Status.*
- void [Mcl\\_SetDmaChannelCommand](#) (const uint32 Channel, const [Mcl\\_DmaChannelCmdType](#) Command)  
*This function sets Dma Channel Command.*
- void [Mcl\\_GetDmaChannelStatus](#) (const uint32 Channel, [Mcl\\_DmaChannelStatusType](#) \*const Status)  
*This function gets Dma Channel Status.*
- void [Mcl\\_SetDmaChannelGlobalList](#) (const uint32 Channel, const [Mcl\\_DmaChannelGlobalListType](#) List[], const uint32 ListDimension)  
*This function sets Dma Channel Global List settings.*
- void [Mcl\\_SetDmaChannelTransferList](#) (const uint32 Channel, const [Mcl\\_DmaChannelTransferListType](#) List[], const uint32 ListDimension)  
*This function sets Dma Channel Transfer List settings.*
- void [Mcl\\_SetDmaChannelScatterGatherList](#) (const uint32 Channel, const uint32 Element, const [Mcl\\_DmaChannelScatterGatherListType](#) List[], const uint32 ListDimension)  
*This function sets Dma Channel Scatter/Gather List settings.*
- void [Mcl\\_SetDmaChannelCrcList](#) (const uint32 Channel, const [Mcl\\_DmaChannelCrcListType](#) List[], const uint32 ListDimension)

*This function sets Dma Channel Crc List settings.*

- void [Mcl\\_GetDmaChannelParam](#) (const uint32 Channel, const [Mcl\\_DmaChannelInfoParamType](#) Param, uint32 \*const Value)

*This function gets the Dma Channel Parameter value.*

- void [Mcl\\_SetDmaChannelScatterGatherConfig](#) (const uint32 Channel, const uint32 Element)

*This function configures the Dma Channel Scatter/Gather.*

- void [Mcl\\_CacheEnable](#) ([Mcl\\_CacheType](#) CacheType)

*This function enables the Cache.*

- void [Mcl\\_CacheDisable](#) ([Mcl\\_CacheType](#) CacheType)

*This function disables the Cache.*

- void [Mcl\\_CacheInvalidate](#) ([Mcl\\_CacheType](#) CacheType)

*This function Invalidates the Cache.*

- void [Mcl\\_CacheClean](#) ([Mcl\\_CacheType](#) CacheType, boolean EnInvalidate)

*This function Cleans the Cache.*

- void [Mcl\\_CacheInvalidateByAddr](#) ([Mcl\\_CacheType](#) CacheType, uint32 Addr, uint32 Length)

*This function Invalidates the Cache by address.*

- void [Mcl\\_CacheCleanByAddr](#) ([Mcl\\_CacheType](#) CacheType, boolean EnInvalidate, uint32 Addr, uint32 Length)

*This function Cleans the Cache by address.*

- void [Mcl\\_SelectCommonTimebase](#) (uint8 instance, uint16 elementSyncList)

*Implementation specific function to updates the Global Timebase bits of configured modules.*

- void [Mcl\\_GetVersionInfo](#) (Std\_VersionInfoType \*const VersionInfo)

*Returns the version information of this module.*

### 6.2.1.1 MISRA-C:2012 violations

## 6.2.2 Data Structure Documentation

### 6.2.2.1 struct [Mcl\\_DmaChannelGlobalListType](#)

This type contains the Mcl Dma Channel Global List.

The Mcl Dma Channel Global List contains a pair composed from Dma Channel Global Parameter Type and the Value of the parameter.

Definition at line 504 of file CDD\_Mcl.h.

#### Data Fields

- [Mcl\\_DmaChannelGlobalParamType](#) Param

*The Mcl Dma Channel Global Parameter Type selects a parameter form the Global Parameter enum type.*

- uint32 [Value](#)

*The Value stores the parameter's value.*

### 6.2.2.1.1 Field Documentation



### 6.2.2.1.1.1 Param [Mcl\\_DmaChannelGlobalParamType](#) Param

The Mcl Dma Channel Global Parameter Type selects a parameter form the Global Parameter enum type.

Definition at line 505 of file CDD\_Mcl.h.

### 6.2.2.1.1.2 Value [uint32](#) Value

The Value stores the parameter's value.

Definition at line 506 of file CDD\_Mcl.h.

## 6.2.2.2 struct [Mcl\\_DmaChannelTransferListType](#)

This type contains the Mcl Dma Channel Transfer List.

The Mcl Dma Channel Transfer List contains a pair composed from Dma Channel Transfer Parameter Type and the Value of the parameter.

Definition at line 516 of file CDD\_Mcl.h.

### Data Fields

- [Mcl\\_DmaChannelTransferParamType](#) Param

*The Mcl Dma Channel Transfer Parameter Type selects a parameter form the Transfer Parameter enum type.*

- [uint32](#) [Value](#)

*The Value stores the parameter's value.*

### 6.2.2.2.1 Field Documentation

#### 6.2.2.2.1.1 Param [Mcl\\_DmaChannelTransferParamType](#) Param

The Mcl Dma Channel Transfer Parameter Type selects a parameter form the Transfer Parameter enum type.

Definition at line 517 of file CDD\_Mcl.h.

#### 6.2.2.2.1.2 Value [uint32](#) Value

The Value stores the parameter's value.

Definition at line 518 of file CDD\_Mcl.h.

### 6.2.2.3 struct Mcl\_DmaChannelScatterGatherListType

This type contains the Mcl Dma Channel Scatter/Gather List.

The Mcl Dma Channel Scatter/Gather List contains a pair composed from Dma Channel Scatter/Gather Parameter Type and the Value of the parameter.

Definition at line 528 of file CDD\_Mcl.h.

#### Data Fields

- [Mcl\\_DmaChannelTransferParamType Param](#)  
*The Mcl Dma Channel Transfer Parameter Type selects a parameter form the Transfer Parameter enum type.*
- uint32 [Value](#)  
*The Value stores the parameter's value.*

#### 6.2.2.3.1 Field Documentation

##### 6.2.2.3.1.1 Param [Mcl\\_DmaChannelTransferParamType](#) Param

The Mcl Dma Channel Transfer Parameter Type selects a parameter form the Transfer Parameter enum type.

Definition at line 529 of file CDD\_Mcl.h.

##### 6.2.2.3.1.2 Value [uint32](#) Value

The Value stores the parameter's value.

Definition at line 530 of file CDD\_Mcl.h.

### 6.2.2.4 struct Mcl\_DmaChannelCrcListType

This type contains the Mcl Dma Channel Crc List.

The Mcl Dma Channel Crc List contains a pair composed from Dma Channel Crc Parameter Type and the Value of the parameter.

Definition at line 541 of file CDD\_Mcl.h.

#### Data Fields

- [Mcl\\_DmaChannelCrcParamType Param](#)  
*The Mcl Dma Channel Crc Parameter Type selects a parameter form the Crc Parameter enum type.*
- uint32 [Value](#)  
*The Value stores the parameter's value.*

### 6.2.2.4.1 Field Documentation

#### 6.2.2.4.1.1 Param `Mcl_DmaChannelCrcParamType` Param

The Mcl Dma Channel Crc Parameter Type selects a parameter form the Crc Parameter enum type.

Definition at line 542 of file CDD\_Mcl.h.

#### 6.2.2.4.1.2 Value `uint32` Value

The Value stores the parameter's value.

Definition at line 543 of file CDD\_Mcl.h.

### 6.2.2.5 struct `Mcl_DmaInstanceStatusType`

This type contains the Mcl Dma Instance Status.

The Mcl Dma Instance Status contains the Hardware Errors, Active Id and Active indication for the running Dma Channel.

Definition at line 554 of file CDD\_Mcl.h.

## Data Fields

- `uint32` [Errors](#)  
*[VALUE] The Errors value is read from the DMA Instance Error Register (ES) as it is.*
- `uint8` [ActiveId](#)  
*[VALUE] The ActiveId value is read from the DMA Instance Control Register (CR) field ACTIVE\_ID.*
- `boolean` [Active](#)  
*[BOOLEAN] The Active value is read from the DMA Instance Control Register (CR) field ACTIVE.*

### 6.2.2.5.1 Field Documentation

#### 6.2.2.5.1.1 Errors `uint32` Errors

*[VALUE]* The Errors value is read from the DMA Instance Error Register (ES) as it is.

Definition at line 555 of file CDD\_Mcl.h.

#### 6.2.2.5.1.2 **ActiveId** `uint8 ActiveId`

[VALUE] The ActiveId value is read from the DMA Instance Control Register (CR) field ACTIVE\_ID.

Definition at line 556 of file CDD\_Mcl.h.

#### 6.2.2.5.1.3 **Active** `boolean Active`

[BOOLEAN] The Active value is read from the DMA Instance Control Register (CR) field ACTIVE.

Definition at line 557 of file CDD\_Mcl.h.

### 6.2.2.6 **struct Mcl\_DmaChannelStatusType**

This type contains the Mcl Dma Channel Status.

The Mcl Dma Channel Status contains the Hardware Errors, Active status and Done indication for the running Dma Channel.

Definition at line 567 of file CDD\_Mcl.h.

#### Data Fields

- [Mcl\\_DmaChannelStateType ChannelState](#)

*[VALUE] The ChStateValue value is read from the internal DMA Driver Channel State Machine. Check UM for additional information.*

- `uint32 Errors`

*[VALUE] The Errors value is read from the DMA Channel Error Register (CHx\_ES) as it is.*

- `boolean Active`

*[BOOLEAN] The Active value is read from the DMA Channel Control and Status Register (CHx\_CSR) field ACTIVE.*

- `boolean Done`

*[BOOLEAN] The Done value is read from the DMA Channel Control and Status Register (CHx\_CSR) field DONE.*

#### 6.2.2.6.1 **Field Documentation**

##### 6.2.2.6.1.1 **ChannelState** `Mcl_DmaChannelStateType ChannelState`

[VALUE] The ChStateValue value is read from the internal DMA Driver Channel State Machine. Check UM for additional information.

Definition at line 568 of file CDD\_Mcl.h.

### 6.2.2.6.1.2 Errors `uint32 Errors`

[VALUE] The Errors value is read from the DMA Channel Error Register (CHx\_ES) as it is.

Definition at line 569 of file CDD\_Mcl.h.

### 6.2.2.6.1.3 Active `boolean Active`

[BOOLEAN] The Active value is read from the DMA Channel Control and Status Register (CHx\_CSR) field ACTIVE.

Definition at line 570 of file CDD\_Mcl.h.

### 6.2.2.6.1.4 Done `boolean Done`

[BOOLEAN] The Active value is read from the DMA Channel Control and Status Register (CHx\_CSR) field DONE.

Definition at line 571 of file CDD\_Mcl.h.

## 6.2.2.7 struct Mcl\_ConfigType

This type contains the Mcl Configuration.

The Mcl Configuration structure contains pointers to the Ip's configuration structure. Based on the available support, specific configurations shall be stored.

Definition at line 288 of file Mcl\_Types.h.

## 6.2.3 Macro Definition Documentation

### 6.2.3.1 MCL\_DET\_INIT

```
#define MCL_DET_INIT
```

API service ID for Mcl\_Init function.

Parameters used when raising an error/exception

Definition at line 126 of file CDD\_Mcl.h.

### 6.2.3.2 MCL\_DET\_DMA\_INSTANCE\_COMMAND

```
#define MCL_DET_DMA_INSTANCE_COMMAND
```

API service ID for Mcl\_SetDmaInstanceCommand function.

Parameters used when raising an error/exception

Definition at line 134 of file CDD\_Mcl.h.

### 6.2.3.3 MCL\_DET\_DMA\_INSTANCE\_STATUS

```
#define MCL_DET_DMA_INSTANCE_STATUS
```

API service ID for Mcl\_GetDmaInstanceStatus function.

Parameters used when raising an error/exception

Definition at line 140 of file CDD\_Mcl.h.

### 6.2.3.4 MCL\_DET\_DMA\_CHANNEL\_COMMAND

```
#define MCL_DET_DMA_CHANNEL_COMMAND
```

API service ID for Mcl\_SetDmaChannelCommand function.

Parameters used when raising an error/exception

Definition at line 146 of file CDD\_Mcl.h.

### 6.2.3.5 MCL\_DET\_DMA\_CHANNEL\_STATUS

```
#define MCL_DET_DMA_CHANNEL_STATUS
```

API service ID for Mcl\_GetDmaChannelStatus function.

Parameters used when raising an error/exception

Definition at line 152 of file CDD\_Mcl.h.

### 6.2.3.6 MCL\_DET\_DMA\_GLOBAL

```
#define MCL_DET_DMA_GLOBAL
```

API service ID for Mcl\_SetDmaChannelGlobalList function.

Parameters used when raising an error/exception

Definition at line 158 of file CDD\_Mcl.h.

### 6.2.3.7 MCL\_DET\_DMA\_TRANSFER

```
#define MCL_DET_DMA_TRANSFER
```

API service ID for Mcl\_SetDmaChannelTransferList function.

Parameters used when raising an error/exception

Definition at line 164 of file CDD\_Mcl.h.

### 6.2.3.8 MCL\_DET\_DMA\_SCATTER\_GATHER\_LIST

```
#define MCL_DET_DMA_SCATTER_GATHER_LIST
```

API service ID for Mcl\_SetDmaChannelScatterGatherList function.

Parameters used when raising an error/exception

Definition at line 170 of file CDD\_Mcl.h.

### 6.2.3.9 MCL\_DET\_DMA\_INFORMATION

```
#define MCL_DET_DMA_INFORMATION
```

API service ID for Mcl\_GetDmaChannelParam function.

Parameters used when raising an error/exception

Definition at line 176 of file CDD\_Mcl.h.

#### 6.2.3.10 MCL\_DET\_DMA\_SCATTER\_GATHER\_CONFIG

```
#define MCL_DET_DMA_SCATTER_GATHER_CONFIG
```

API service ID for Mcl\_SetDmaChannelScatterGatherConfig function.

Parameters used when raising an error/exception

Definition at line 182 of file CDD\_Mcl.h.

#### 6.2.3.11 MCL\_DET\_DMA\_CRC

```
#define MCL_DET_DMA_CRC
```

API service ID for Mcl\_SetDmaChannelCrcList function.

Parameters used when raising an error/exception

Definition at line 189 of file CDD\_Mcl.h.

#### 6.2.3.12 MCL\_DET\_DEINIT

```
#define MCL_DET_DEINIT
```

API service ID for Mcl\_DeInit function.

Parameters used when raising an error/exception

Definition at line 199 of file CDD\_Mcl.h.

#### 6.2.3.13 MCL\_DET\_CACHE\_ENABLE

```
#define MCL_DET_CACHE_ENABLE
```

API service ID for Mcl\_CacheEnable function.

Parameters used when raising an error/exception

Definition at line 206 of file CDD\_Mcl.h.



### 6.2.3.14 MCL\_DET\_CACHE\_DISABLE

```
#define MCL_DET_CACHE_DISABLE
```

API service ID for Mcl\_CacheDisable function.

Parameters used when raising an error/exception

Definition at line 212 of file CDD\_Mcl.h.

### 6.2.3.15 MCL\_DET\_CACHE\_INVALIDATE

```
#define MCL_DET_CACHE_INVALIDATE
```

API service ID for Mcl\_CacheInvalidate function.

Parameters used when raising an error/exception

Definition at line 218 of file CDD\_Mcl.h.

### 6.2.3.16 MCL\_DET\_CACHE\_CLEAN

```
#define MCL_DET_CACHE_CLEAN
```

API service ID for Mcl\_CacheClean function.

Parameters used when raising an error/exception

Definition at line 224 of file CDD\_Mcl.h.

### 6.2.3.17 MCL\_DET\_CACHE\_INVALIDATE\_BY\_ADDRESS

```
#define MCL_DET_CACHE_INVALIDATE_BY_ADDRESS
```

API service ID for Mcl\_CacheInvalidateByAddr function.

Parameters used when raising an error/exception

Definition at line 230 of file CDD\_Mcl.h.

### 6.2.3.18 MCL\_DET\_CACHE\_CLEAN\_BY\_ADDRESS

```
#define MCL_DET_CACHE_CLEAN_BY_ADDRESS
```

API service ID for Mcl\_CacheCleanByAddr function.

Parameters used when raising an error/exception

Definition at line 236 of file CDD\_Mcl.h.

### 6.2.3.19 MCL\_E\_UNINIT

```
#define MCL_E_UNINIT
```

All API's having pointers as parameters shall return this error if called with with a NULL value.

Definition at line 251 of file CDD\_Mcl.h.

### 6.2.3.20 MCL\_E\_PARAM\_POINTER

```
#define MCL_E_PARAM_POINTER
```

All API's having pointers as parameters shall return this error if called with with a NULL value.

Definition at line 258 of file CDD\_Mcl.h.

### 6.2.3.21 MCL\_E\_INVALID\_INSTANCE

```
#define MCL_E_INVALID_INSTANCE
```

All API's called with wrong instance shall return this error.

Definition at line 264 of file CDD\_Mcl.h.

### 6.2.3.22 MCL\_E\_INVALID\_CHANNEL

```
#define MCL_E_INVALID_CHANNEL
```

All API's called with wrong channel shall return this error.

Definition at line 270 of file CDD\_Mcl.h.

### 6.2.3.23 MCL\_E\_INVALID\_COMMAND

```
#define MCL_E_INVALID_COMMAND
```

All API's called with wrong instance shall return this error.

Definition at line 276 of file CDD\_Mcl.h.

### 6.2.3.24 MCL\_E\_INVALID\_PARAMETER

```
#define MCL_E_INVALID_PARAMETER
```

All API's called with wrong read parameter shall return this error.

Definition at line 282 of file CDD\_Mcl.h.

### 6.2.3.25 MCL\_E\_INVALID\_STATE

```
#define MCL_E_INVALID_STATE
```

All API's called in wrong sequence shall return this error.

Definition at line 288 of file CDD\_Mcl.h.

### 6.2.3.26 MCL\_E\_INCONSISTENCY

```
#define MCL_E_INCONSISTENCY
```

All API's called while hardware has error status shall return this error.

Definition at line 294 of file CDD\_Mcl.h.

### 6.2.3.27 MCL\_E\_TIMEOUT

```
#define MCL_E_TIMEOUT
```

All API's called with a timeout value shall return this error if execution is not finished in the allocated timeframe.

Definition at line 301 of file CDD\_Mcl.h.

6.2.3.28 MCL\_E\_PARAM\_CONFIG

```
#define MCL_E_PARAM_CONFIG
```

If DET error reporting is enabled, the MCL will check upon each API call if the requested resource is configured to be available on the current core, and in case of error will return MCL\_E\_PARAM\_CONFIG.

Definition at line 309 of file CDD\_Mcl.h.

6.2.3.29 MCL\_E\_PROTECTED

```
#define MCL_E_PROTECTED
```

If DET error reporting is enabled, the MCL will check if registers are protected.

Definition at line 314 of file CDD\_Mcl.h.

6.2.3.30 MCL\_E\_INIT\_FAILED

```
#define MCL_E_INIT_FAILED
```

If VariantPreCompile is used, the configuration pointer shall have a NULL\_PTR value. If VariantPostBuild is used, the configuration pointer shall be different from NULL\_PTR. And in case of violate will return MCL\_E\_INIT\_FAILED.

Definition at line 321 of file CDD\_Mcl.h.

6.2.4 Enum Reference

6.2.4.1 Mcl\_DmaInstanceCmdType

```
enum Mcl_DmaInstanceCmdType
```

This type contains the Mcl Dma Instance Commands.  
The Commands trigger specific actions in the Dma Instance.

Enumerator

|                         |                                                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|
| MCL_DMA_INST_STOP       | The Stop Command stops the executing channel and forces the Minor Loop to finish.                                    |
| MCL_DMA_INST_STOP_ERROR | The StopError Command stops the executing channel, forces the Minor Loop to finish and generates an error interrupt. |
| MCL_DMA_INST_PAUSE      | The Pause Command allows the ongoing transfer to finish and pauses any new transfer.                                 |
| MCL_DMA_INST_RESUME     | The Resume Command allows the transfer to continue.                                                                  |

Definition at line 333 of file CDD\_Mcl.h.

#### 6.2.4.2 Mcl\_DmaChannelCmdType

```
enum Mcl_DmaChannelCmdType
```

This type contains the Mcl Dma Channel Commands.

The Commands trigger specific actions in the Dma Channel.

Enumerator

|                          |                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------|
| MCL_DMA_CH_START_REQUEST | The Start Request Command enables the Dma Channel to be triggered by hardware requests. |
| MCL_DMA_CH_STOP_REQUEST  | The Stop Request Command disables the Dma Channel to be triggered by hardware requests. |
| MCL_DMA_CH_START_SERVICE | The Start Service Command sends a start request to the Dma Channel.                     |
| MCL_DMA_CH_ACK_DONE      | The Ack Done Command resets the Dma Channel Done status.                                |
| MCL_DMA_CH_ACK_ERROR     | The Ack Error Command resets the Dma Channel Error status.                              |

Definition at line 346 of file CDD\_Mcl.h.

#### 6.2.4.3 Mcl\_DmaChannelGlobalParamType

```
enum Mcl_DmaChannelGlobalParamType
```

This type contains the Mcl Dma Channel Global Parameters.

The Parameters set specific functionalities.

Enumerator

|                                         |                                                                                                                                                                  |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MCL_DMA_CH_SET_EN_MASTER_ID_REPLICATION | [BOOLEAN] The EnMasterIdReplication Parameter sets the Dma Channel to use the same protection level and system bus ID of the master programming the Dma Channel. |
| MCL_DMA_CH_SET_EN_BUFFERED_WRITES       | [BOOLEAN] The EnBufferedWrites Parameter sets the Dma Channel writes to be bufferable.                                                                           |
| MCL_DMA_CH_SET_EN_MUX_SOURCE_REQ        | [BOOLEAN] The EnMuxSource Parameter enables the Dma Channel Mux Source.                                                                                          |
| MCL_DMA_CH_SET_MUX_SOURCE_REQ           | [VALUE] The MuxSource Parameter sets the Dma Channel Mux Source value.                                                                                           |

Enumerator

|                                       |                                                                           |
|---------------------------------------|---------------------------------------------------------------------------|
| MCL_DMA_CH_SET_EN_MUX_TRIGGER         | [BOOLEAN] The EnMuxTrigger Parameter enables the Dma Channel Mux Trigger. |
| MCL_DMA_CH_SET_EN_HARDWARE_REQ        | [BOOLEAN] The EnRequest Parameter enables the Dma Channel Request.        |
| MCL_DMA_CH_SET_EN_ERROR_INTERRUPT     | [BOOLEAN] The EnError Parameter enables the Dma Channel Error Interrupt.  |
| MCL_DMA_CH_SET_GROUP_PRIORITY         | [VALUE] The Group Parameter sets the Dma Channel Group Priority.          |
| MCL_DMA_CH_SET_LEVEL_PRIORITY         | [VALUE] The Level Parameter sets the Dma Channel Level Priority.          |
| MCL_DMA_CH_SET_EN_PREEMPTION_PRIORITY | [BOOLEAN] The EnPreemption Parameter enables the Dma Channel Preemption.  |
| MCL_DMA_CH_SET_DIS_PREEMPT_PRIORITY   | [BOOLEAN] The DisPreempt Parameter disables the Dma Channel Preempt.      |

Definition at line 360 of file CDD\_Mcl.h.

#### 6.2.4.4 Mcl\_DmaChannelTransferParamType

```
enum Mcl_DmaChannelTransferParamType
```

This type contains the Mcl Dma Channel Transfer Parameters.

The Parameters set specific functionalities.

Enumerator

|                                                           |                                                                                                                                   |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| MCL_DMA_CH_SET_SOURCE_ADDRESS                             | [VALUE] The Source Address Parameter sets the Dma Channel source address value.                                                   |
| MCL_DMA_CH_SET_SOURCE_SIGNED_OFFSET                       | [VALUE] The Source Signed Offset Parameter sets the Dma Channel source signed offset value.                                       |
| MCL_DMA_CH_SET_SOURCE_SIGNED_LAST_ADDRESS_ADJUSTMENT      | [VALUE] The Source Signed Last Address Adjustment Parameter sets the Dma Channel source signed last address adjustment.           |
| MCL_DMA_CH_SET_SOURCE_TRANSFER_SIZE                       | [VALUE] The Source Transfer Size Parameter sets the Dma Channel source transfer size.                                             |
| MCL_DMA_CH_SET_SOURCE_MODULO                              | [VALUE] The Source Modulo Parameter sets the Dma Channel source modulo.                                                           |
| MCL_DMA_CH_SET_DESTINATION_ADDRESS                        | [VALUE] The Destination Address Parameter sets the Dma Channel destination address value.                                         |
| MCL_DMA_CH_SET_DESTINATION_SIGNED_OFFSET                  | [VALUE] The Destination Signed Offset Parameter sets the Dma Channel destination signed offset value.                             |
| MCL_DMA_CH_SET_DESTINATION_SIGNED_LAST_ADDRESS_ADJUSTMENT | [VALUE] The Destination Signed Last Address Adjustment Parameter sets the Dma Channel destination signed last address adjustment. |

## Enumerator

|                                                |                                                                                                                     |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| MCL_DMA_CH_SET_DESTINATION_TRANSFER_SIZE       | [VALUE] The Destination Transfer Size Parameter sets the Dma Channel destination transfer size.                     |
| MCL_DMA_CH_SET_DESTINATION_MODULO              | [VALUE] The Destination Modulo Parameter sets the Dma Channel destination modulo.                                   |
| MCL_DMA_CH_SET_MINORLOOP_EN_SRC_OFFSET         | [BOOLEAN] The Minor Loop Enable Source Offset Parameter enables the Dma Channel minor loop source offset.           |
| MCL_DMA_CH_SET_MINORLOOP_EN_DST_OFFSET         | [BOOLEAN] The Minor Loop Enable Destination Offset Parameter enables the Dma Channel minor loop destination offset. |
| MCL_DMA_CH_SET_MINORLOOP_SIGNED_OFFSET         | [VALUE] The Minor Loop Signed Offset Parameter sets the Dma Channel minor loop signed offset.                       |
| MCL_DMA_CH_SET_MINORLOOP_EN_LINK               | [BOOLEAN] The Minor Loop Enable Link Parameter enables the Dma Channel minor loop logic channel linking.            |
| MCL_DMA_CH_SET_MINORLOOP_LOGIC_LINK_CH         | [VALUE] The Minor Loop Logic Channel Link Parameter sets the Dma Channel minor loop logic channel link.             |
| MCL_DMA_CH_SET_MINORLOOP_SIZE                  | [VALUE] The Minor Loop Size Parameter sets the Dma Channel minor loop transfer size.                                |
| MCL_DMA_CH_SET_MAJORLOOP_EN_LINK               | [BOOLEAN] The Major Loop Enable Link Parameter enables the Dma Channel major loop logic channel linking.            |
| MCL_DMA_CH_SET_MAJORLOOP_LOGIC_LINK_CH         | [VALUE] The Major Loop Logic Channel Link Parameter sets the Dma Channel major loop logic channel link.             |
| MCL_DMA_CH_SET_MAJORLOOP_COUNT                 | [VALUE] The Major Loop Count Parameter sets the Dma Channel major loop count.                                       |
| MCL_DMA_CH_SET_CONTROL_STORE_DEST_ADDR         | [VALUE] The Store Destination Address Parameter saves the final destination address in system memory.               |
| MCL_DMA_CH_SET_CONTROL_SOFTWARE_REQUEST        | [BOOLEAN] The Enable Start Parameter enables the Dma Channel start service request.                                 |
| MCL_DMA_CH_SET_CONTROL_EN_MAJOR_INTERRUPT      | [BOOLEAN] The Enable Major Interrupt Parameter enables the Dma Channel major interrupt.                             |
| MCL_DMA_CH_SET_CONTROL_EN_HALF_MAJOR_INTERRUPT | [BOOLEAN] The Enable Half Interrupt Parameter enables the Dma Channel half major interrupt.                         |
| MCL_DMA_CH_SET_CONTROL_DISABLE_AUTO_REQUEST    | [BOOLEAN] The Disable Automatic Request Parameter disables the Dma Channel automatic request.                       |
| MCL_DMA_CH_SET_CONTROL_BANDWIDTH               | [VALUE] The Bandwidth Control Parameter sets the Dma Channel bandwidth control.                                     |

Definition at line 390 of file CDD\_Mcl.h.

#### 6.2.4.5 Mcl\_DmaChannelStateType

enum `Mcl_DmaChannelStateType`

This type contains the Mcl Dma Channel State values.

The states represent the Channel status during runtime.

Definition at line 429 of file `CDD_Mcl.h`.

#### 6.2.4.6 Mcl\_DmaChannelCrcParamType

enum `Mcl_DmaChannelCrcParamType`

This type contains the Mcl Dma Channel Crc Parameters.

The Parameters set specific functionalities.

Definition at line 444 of file `CDD_Mcl.h`.

#### 6.2.4.7 Mcl\_DmaChannelInfoParamType

enum `Mcl_DmaChannelInfoParamType`

This type contains the Mcl Dma Channel Information Parameters.

The Parameters get specific information.

Enumerator

|                                                        |                                                                                                  |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <code>MCL_DMA_CH_GET_SOURCE_ADDRESS</code>             | [VALUE] The Source Address Parameter gets the Dma Channel source address.                        |
| <code>MCL_DMA_CH_GET_DESTINATION_ADDR←<br/>ESS</code>  | [VALUE] The Destination Address Parameter gets the Dma Channel destination address.              |
| <code>MCL_DMA_CH_GET_BEGIN_ITER_COUNT</code>           | [VALUE] The Begin Iteration Count Parameter gets the Dma Channel begin iteration count.          |
| <code>MCL_DMA_CH_GET_CURRENT_ITER_COU←<br/>NT</code>   | [VALUE] The Current Iteration Count Parameter gets the Dma Channel current iteration count.      |
| <code>MCL_DMA_CH_GET_STORE_DST_ADDR</code>             | [VALUE] The Store Destination Address Parameter gets the Dma Channel stored destination address. |
| <code>MCL_DMA_CH_GET_MASTER_ID</code>                  | [VALUE] The Master Id Parameter gets the Dma Channel master id.                                  |
| <code>MCL_DMA_CH_GET_MAJOR_INTERRUPT</code>            | [BOOLEAN] The Major Interrupt Parameter gets the Dma Channel major interrupt.                    |
| <code>MCL_DMA_CH_GET_HALF_MAJOR_INTER←<br/>RUPT</code> | [BOOLEAN] The Half Major Interrupt Parameter gets the Dma Channel half major interrupt.          |



Definition at line 459 of file CDD\_Mcl.h.

### 6.2.4.8 Mcl\_CacheType

enum `Mcl_CacheType`

This type contains the Mcl Cache Type selection.

The Cache Types select specific cache memory types.

Enumerator

|                                    |                                                                 |
|------------------------------------|-----------------------------------------------------------------|
| <code>MCL_CACHE_ALL</code>         | The Cache All Parameter selects all cache types.                |
| <code>MCL_CACHE_INSTRUCTION</code> | The Cache Instruction Parameter selects instruction cache type. |
| <code>MCL_CACHE_DATA</code>        | The Cache Data Parameter selects data cache type.               |

Definition at line 485 of file CDD\_Mcl.h.

### 6.2.4.9 Mcl\_ReturnType

enum `Mcl_ReturnType`

This type contains the Mcl Return Type.

The Return Type give information for the execution of interfaces.

Definition at line 258 of file Mcl\_Types.h.

## 6.2.5 Function Reference

### 6.2.5.1 Mcl\_Init()

```
void Mcl_Init (
    const Mcl_ConfigType *const ConfigPtr )
```

This function initializes the Mcl Driver.

This service is a non reentrant function that shall initialize the Mcl driver. The initialization is applied for the enabled IPs, configured statically.

Parameters

|    |                  |                                         |
|----|------------------|-----------------------------------------|
| in | <i>ConfigPtr</i> | Pointer to the configuration structure. |
|----|------------------|-----------------------------------------|

Returns

void

#### 6.2.5.2 Mcl\_DeInit()

```
void Mcl_DeInit (
    void )
```

This function deinitializes the Mcl Driver.

This service is a non reentrant function that shall deinitialize the Mcl driver. The deinitialization is applied for the enabled IPs, configured statically.

Returns

void

#### 6.2.5.3 Mcl\_SetDmaInstanceCommand()

```
void Mcl_SetDmaInstanceCommand (
    const uint32 Instance,
    const Mcl_DmaInstanceCmdType Command )
```

This function sets Dma Instance Command.

This service is a reentrant function that shall command the Dma Instance. The command shall trigger specific functionalities of the Dma Instance.

Parameters

|    |                 |                                        |
|----|-----------------|----------------------------------------|
| in | <i>Instance</i> | Selection value of the Logic Instance. |
| in | <i>Command</i>  | The command for the Logic Instance.    |

Returns

void

### 6.2.5.4 Mcl\_GetDmaInstanceStatus()

```
void Mcl_GetDmaInstanceStatus (
    const uint32 Instance,
    Mcl_DmaInstanceStatusType *const Status )
```

This function gets Dma Instance Status.

This service is a reentrant function that shall get the Dma Instance status. The command shall read specific functionalities of the Dma Instance.

Parameters

|     |                 |                                        |
|-----|-----------------|----------------------------------------|
| in  | <i>Instance</i> | Selection value of the Logic Instance. |
| out | <i>Status</i>   | Pointer to the Dma Instance status.    |

Returns

void

### 6.2.5.5 Mcl\_SetDmaChannelCommand()

```
void Mcl_SetDmaChannelCommand (
    const uint32 Channel,
    const Mcl_DmaChannelCmdType Command )
```

This function sets Dma Channel Command.

This service is a reentrant function that shall command the Dma Channel. The command shall trigger specific functionalities of the Dma Channel.

Parameters

|    |                |                                                      |
|----|----------------|------------------------------------------------------|
| in | <i>Channel</i> | Specifies the Logic Channel Tag defined by the user. |
| in | <i>Command</i> | The command for the Logic Channel.                   |

Returns

void

### 6.2.5.6 Mcl\_GetDmaChannelStatus()

```
void Mcl_GetDmaChannelStatus (
    const uint32 Channel,
    Mcl_DmaChannelStatusType *const Status )
```

This function gets Dma Channel Status.

This service is a reentrant function that shall get the Dma Channel status. The command shall read specific functionalities of the Dma Channel.

Parameters

|     |                |                                                      |
|-----|----------------|------------------------------------------------------|
| in  | <i>Channel</i> | Specifies the Logic Channel Tag defined by the user. |
| out | <i>Status</i>  | Pointer to the Dma Channel status.                   |

Returns

void

### 6.2.5.7 Mcl\_SetDmaChannelGlobalList()

```
void Mcl_SetDmaChannelGlobalList (
    const uint32 Channel,
    const Mcl_DmaChannelGlobalListType List[],
    const uint32 ListDimension )
```

This function sets Dma Channel Global List settings.

This service is a reentrant function that shall set the Dma Channel global parameters list. The list is composed of an array of Dma Channel global parameters settings. The settings list(array) is defined by the user needs: it contains the desired parameters to be configured, in any desired order.

How to use this interface:

1. Use the "Mcl\_DmaChannelGlobalListType" to create a list(array) with the desired paramaters to configure (see parameters: "Mcl\_DmaChannelGlobalParamType") The list can declared globally or locally: Global example: `Mcl_DmaChannelGlobalListType global_Mcl_DmaChannelGlobalList0[NUMBER_OF_PARAMETERS] = {...}`; Local example: `Mcl_DmaChannelGlobalListType Mcl_DmaChannelGlobalList[NUMBER_OF_PARAMETERS]`; `Mcl_DmaChannelGlobalList[PARAMETER0].Param = MCL_DMA_CH_SET_EN_BUFFERED_WRITES`; `Mcl_DmaChannelGlobalList[PARAMETER0].Value = TRUE`; `Mcl_DmaChannelGlobalList[PARAMETER1].Param = ...`; `Mcl_DmaChannelGlobalList[PARAMETER1].Value = ...`;
2. Call the "Mcl\_SetDmaChannelGlobalList()" interface: `Mcl_SetDmaChannelGlobalList(LOGIC_CHANNEL_Lx, Mcl_DmaChannelGlobalList, NUMBER_OF_PARAMETERS)`;

### Parameters

|    |                      |                                                      |
|----|----------------------|------------------------------------------------------|
| in | <i>Channel</i>       | Specifies the Logic Channel Tag defined by the user. |
| in | <i>List</i>          | Pointer to the Global List Array.                    |
| in | <i>ListDimension</i> | Number of entries in the List.                       |

### Returns

void

#### 6.2.5.8 Mcl\_SetDmaChannelTransferList()

```
void Mcl_SetDmaChannelTransferList (
    const uint32 Channel,
    const Mcl_DmaChannelTransferListType List[],
    const uint32 ListDimension )
```

This function sets Dma Channel Transfer List settings.

This service is a reentrant function that shall set the Dma Channel transfer parameters list. The list is composed of an array of Dma Channel transfer parameters settings. The settings array is defined by the user needs: it contains entries for each desired parameter to be configured, in any suitable order.

How to use this interface:

1. Use the "Mcl\_DmaChannelTransferListType" to create a list(array) with the desired paramaters to configure (see parameters: "Mcl\_DmaChannelTransferParamType") The list can declared globally or locally: Global example: `Mcl_DmaChannelTransferListType global_Mcl_DmaChannelTransferList0[NUMBER_OF_PARAMETERS] = {...}`; Local example: `Mcl_DmaChannelTransferListType Mcl_DmaChannelTransferList[NUMBER_OF_PARAMETERS]; Mcl_DmaChannelTransferList[PARAMETER0].Param = MCL_DMA_CH_SET_SOURCE_ADDRESS; Mcl_DmaChannelTransferList[PARAMETER0].Value = Mcl_DmaChannelTransferList[PARAMETER1].Param = MCL_DMA_CH_SET_DESTINATION_ADDRESS; Mcl_DmaChannelTransferList[PARAMETER1].Value =`
2. Call the "Mcl\_SetDmaChannelTransferList()" interface: `Mcl_SetDmaChannelTransferList(LOGIC_CHANNELx, Mcl_DmaChannelTransferList, NUMBER_OF_PARAMETERS);`

### Parameters

|    |                      |                                                      |
|----|----------------------|------------------------------------------------------|
| in | <i>Channel</i>       | Specifies the Logic Channel Tag defined by the user. |
| in | <i>List</i>          | Pointer to the Transfer List Array.                  |
| in | <i>ListDimension</i> | Number of entries in the List.                       |

Returns

void

### 6.2.5.9 Mcl\_SetDmaChannelScatterGatherList()

```
void Mcl_SetDmaChannelScatterGatherList (
    const uint32 Channel,
    const uint32 Element,
    const Mcl_DmaChannelScatterGatherListType List[],
    const uint32 ListDimension )
```

This function sets Dma Channel Scatter/Gather List settings.

This service is a reentrant function that shall set the Dma Channel scatter/gather parameters list. The Scatter/Gather List configures Logic Elements belonging to the same Dma Logic Channel. The settings array is defined by the user needs: it contains entries for each desired parameter to be configured, in any suitable order.

How to use this interface:

1. Use the "Mcl\_DmaChannelScatterGatherListType" to create a list(array) with the desired parameters to configure (see parameters: "Mcl\_DmaChannelTransferParamType") The list can be declared globally or locally: Global example: `Mcl_DmaChannelScatterGatherListType global_Mcl_DmaChannelScatterGatherList0[NUMBER_OF_PARAMETERS] = {...}`; Local example: `Mcl_DmaChannelScatterGatherListType Mcl_DmaChannelScatterGatherList[NUMBER_OF_PARAMETERS]`; `Mcl_DmaChannelScatterGatherList[PARAMETER0].Param = MCL_DMA_CH_SET_SOURCE_ADDRESS`; `Mcl_DmaChannelScatterGatherList[PARAMETER0].Value = Mcl_DmaChannelScatterGatherList[PARAMETER1].Param = MCL_DMA_CH_SET_DESTINATION_ADDRESS`; `Mcl_DmaChannelScatterGatherList[PARAMETER1].Value =`
2. Call the "Mcl\_SetDmaChannelScatterGatherList()" interface: `Mcl_SetDmaChannelScatterGatherList(LOGIC_CHANNELx, LOGIC_ELEMENTy, Mcl_DmaChannelScatterGatherList, NUMBER_OF_PARAMETERS)`;

Parameters

|    |                      |                                                      |
|----|----------------------|------------------------------------------------------|
| in | <i>Channel</i>       | Specifies the Logic Channel Tag defined by the user. |
| in | <i>Element</i>       | Specifies the Logic Element Id.                      |
| in | <i>List</i>          | Pointer to the Scatter/Gather List Array.            |
| in | <i>ListDimension</i> | Number of entries in the List.                       |

Returns

void

### 6.2.5.10 Mcl\_SetDmaChannelCrcList()

```
void Mcl_SetDmaChannelCrcList (
    const uint32 Channel,
    const Mcl_DmaChannelCrcListType List[],
    const uint32 ListDimension )
```

This function sets Dma Channel Crc List settings.

This service is a reentrant function that shall set the Dma Channel Crc parameters list. The list is composed of an array of Dma Channel Crc parameters settings. The settings array is defined by the user needs: it contains entries for each desired parameter to be configured, in any suitable order.

How to use this interface:

1. Use the "Mcl\_DmaChannelCrcListType" to create a list(array) with the desired parameters to configure (see parameters: "Mcl\_DmaChannelCrcParamType") The list can be declared globally or locally: Global example: `Mcl_DmaChannelCrcListType global_Mcl_DmaChannelCrcList0[NUMBER_OF_PARAMETERS] = {...}`; Local example: `Mcl_DmaChannelCrcListType Mcl_DmaChannelCrcList[NUMBER_OF_PARAMETERS]`; `Mcl_DmaChannelCrcList[PARAMETER0].Param = MCL_DMA_CH_SET_CRC_POLYNOMIAL`; `Mcl_DmaChannelCrcList[PARAMETER0].Value = ISCSICRC`; `Mcl_DmaChannelCrcList[PARAMETER1].Param = MCL_DMA_CH_SET_CRC_EN_LOGIC`; `Mcl_DmaChannelCrcList[PARAMETER1].Value = TRUE`;
2. Call the "Mcl\_SetDmaChannelCrcList()" interface: `Mcl_SetDmaChannelCrcList(LOGIC_CHANNELx, Mcl_DmaChannelCrcList, NUMBER_OF_PARAMETERS)`;

Parameters

|    |                      |                                                      |
|----|----------------------|------------------------------------------------------|
| in | <i>Channel</i>       | Specifies the Logic Channel Tag defined by the user. |
| in | <i>List</i>          | Pointer to the Crc List Array.                       |
| in | <i>ListDimension</i> | Number of entries in the List.                       |

Returns

void

### 6.2.5.11 Mcl\_GetDmaChannelParam()

```
void Mcl_GetDmaChannelParam (
    const uint32 Channel,
    const Mcl_DmaChannelInfoParamType Param,
    uint32 *const Value )
```

This function gets the Dma Channel Parameter value.

This service is a reentrant function that shall get the Dma Channel parameters value.

## Parameters

|     |                |                                                      |
|-----|----------------|------------------------------------------------------|
| in  | <i>Channel</i> | Specifies the Logic Channel Tag defined by the user. |
| in  | <i>Param</i>   | Selection parameter.                                 |
| out | <i>Value</i>   | Pointer to the parameter value.                      |

## Returns

void

**6.2.5.12 Mcl\_SetDmaChannelScatterGatherConfig()**

```
void Mcl_SetDmaChannelScatterGatherConfig (
    const uint32 Channel,
    const uint32 Element )
```

This function configures the Dma Channel Scatter/Gather.

This service is a reentrant function that shall configure the Dma Channel scatter/gather functionality. The Scatter/Gather settings, for the specified Dma Logic Channel, are loaded into the Software TCDs. Each software TCD corresponds to a Logic Element. The specified Logic Element shall be loaded into the Dma Logic Channel's Hardware TCD. The Logic Elements (describing the Software TCDs) form a simple chained list, the "Element" function parameter representing the lists's head.

## Parameters

|    |                |                                                              |
|----|----------------|--------------------------------------------------------------|
| in | <i>Channel</i> | Specifies the Logic Channel Tag defined by the user.         |
| in | <i>Element</i> | Specifies the Logic Element Id representing the list's head. |

## Returns

void

**6.2.5.13 Mcl\_CacheEnable()**

```
void Mcl_CacheEnable (
    Mcl_CacheType CacheType )
```

This function enables the Cache.

This service is a reentrant function that shall enable the Cache functionality.



### Parameters

|    |                  |                                    |
|----|------------------|------------------------------------|
| in | <i>CacheType</i> | Selection value of the Cache Type. |
|----|------------------|------------------------------------|

### Returns

void

#### 6.2.5.14 Mcl\_CacheDisable()

```
void Mcl_CacheDisable (
    Mcl_CacheType CacheType )
```

This function disables the Cache.

This service is a reentrant function that shall disable the Cache functionality.

### Parameters

|    |                  |                                    |
|----|------------------|------------------------------------|
| in | <i>CacheType</i> | Selection value of the Cache Type. |
|----|------------------|------------------------------------|

### Returns

void

#### 6.2.5.15 Mcl\_CacheInvalidate()

```
void Mcl_CacheInvalidate (
    Mcl_CacheType CacheType )
```

This function Invalidates the Cache.

This service is a reentrant function that shall Invalidate the Cache functionality. The Invalidation applies to the entire Cache.

### Parameters

|    |                  |                                    |
|----|------------------|------------------------------------|
| in | <i>CacheType</i> | Selection value of the Cache Type. |
|----|------------------|------------------------------------|

Returns

void

#### 6.2.5.16 Mcl\_CacheClean()

```
void Mcl_CacheClean (
    Mcl_CacheType CacheType,
    boolean EnInvalidate )
```

This function Cleans the Cache.

This service is a reentrant function that shall Clean the Cache functionality. The Clean applies to the entire Cache. By enabling the Invalidation, the function shall execute specific Cache Clean&Invalidate function.

Parameters

|    |                     |                                                 |
|----|---------------------|-------------------------------------------------|
| in | <i>CacheType</i>    | Selection value of the Cache Type.              |
| in | <i>EnInvalidate</i> | Enable the Invalidation specific functionality. |

Returns

void

#### 6.2.5.17 Mcl\_CacheInvalidateByAddr()

```
void Mcl_CacheInvalidateByAddr (
    Mcl_CacheType CacheType,
    uint32 Addr,
    uint32 Length )
```

This function Invalidates the Cache by address.

This service is a reentrant function that shall Invalidate the Cache by address. The Invalidation applies to the area in Cache specified by the address and length. The buffer shall be aligned to the Cache Line size.

Parameters

|    |                  |                                    |
|----|------------------|------------------------------------|
| in | <i>CacheType</i> | Selection value of the Cache Type. |
| in | <i>Addr</i>      | Address value of the buffer.       |
| in | <i>Length</i>    | Length value of the buffer.        |

Returns

void

### 6.2.5.18 Mcl\_CacheCleanByAddr()

```
void Mcl_CacheCleanByAddr (
    Mcl_CacheType CacheType,
    boolean EnInvalidate,
    uint32 Addr,
    uint32 Length )
```

This function Cleans the Cache by address.

This service is a reentrant function that shall Clean the Cache by address. The Clean applies to the area in Cache specified by the address and length. By enabling the Invalidation, the function shall execute specific Cache Clean&Invalidate function.

Parameters

|    |                     |                                                 |
|----|---------------------|-------------------------------------------------|
| in | <i>CacheType</i>    | Selection value of the Cache Type.              |
| in | <i>EnInvalidate</i> | Enable the Invalidation specific functionality. |
| in | <i>Addr</i>         | Address value of the buffer.                    |
| in | <i>Length</i>       | Length value of the buffer.                     |

Returns

void

### 6.2.5.19 Mcl\_SelectCommonTimebase()

```
void Mcl_SelectCommonTimebase (
    uint8 instance,
    uint16 elementSyncList )
```

Implementation specific function to updates the Global Timebase bits of configured modules.

This function is used to set the global timebase bits for modules that support the global timebase feature. The function selects the module that gives the common timebase and the modules that are use this timebase (as bits in elementSyncList). Then it synchronizes the modules. example: elementSyncList is 0x0003 - modules 0 and 1 use the timebase given by instance elementSyncList is 0x0005 - modules 0 and 2 use the timebase given by instance

Parameters

|    |                        |                       |
|----|------------------------|-----------------------|
| in | <i>instance</i>        | FTM module id         |
| in | <i>elementSyncList</i> | FTM module mask value |

Returns

void

#### 6.2.5.20 Mcl\_GetVersionInfo()

```
void Mcl_GetVersionInfo (
    Std_VersionInfoType *const VersionInfo )
```

Returns the version information of this module.

Returns the version information of MCL module.

Parameters

|     |                    |                                                           |
|-----|--------------------|-----------------------------------------------------------|
| out | <i>VersionInfo</i> | A pointer to a structure used to get version information. |
|-----|--------------------|-----------------------------------------------------------|

Returns

void

## 6.3 DMA IP Driver

### 6.3.1 Detailed Description

#### Data Structures

- struct [Dma\\_Ip\\_LogicChannelGlobalListType](#)  
*This type contains the Dma Ip Logic Channel Global List. [More...](#)*
- struct [Dma\\_Ip\\_LogicChannelTransferListType](#)  
*This type contains the Dma Ip Channel Transfer List. [More...](#)*
- struct [Dma\\_Ip\\_LogicChannelCrcListType](#)  
*This type contains the Dma Ip Channel ScatterGather List. [More...](#)*
- struct [Dma\\_Ip\\_LogicInstanceStatusType](#)  
*This type contains the Dma Ip Instance Status. [More...](#)*
- struct [Dma\\_Ip\\_LogicChannelStatusType](#)  
*This type contains the Dma Ip Channel Status. [More...](#)*
- struct [Dma\\_Ip\\_LogicChannelIdType](#)  
*This type contains the Dma Ip Logic Channel Identification. [More...](#)*
- struct [Dma\\_Ip\\_GlobalConfigType](#)  
*This type contains the Dma Ip Global Configuration. [More...](#)*
- struct [Dma\\_Ip\\_TransferConfigType](#)  
*This type contains the Dma Ip Transfer Configuration. [More...](#)*
- struct [Dma\\_Ip\\_ScatterGatherConfigType](#)  
*This type contains the Dma Ip Scatter/Gather Configuration. [More...](#)*
- struct [Dma\\_Ip\\_LogicChannelConfigType](#)  
*This type contains the Dma Ip Logic Channel Configuration. [More...](#)*
- struct [Dma\\_Ip\\_LogicInstanceIdType](#)  
*This type contains the Dma Ip Logic Instance Identification. [More...](#)*
- struct [Dma\\_Ip\\_LogicInstanceConfigType](#)  
*This type contains the Dma Ip Logic Instance Configuration. [More...](#)*
- struct [Dma\\_Ip\\_HwChannelStateType](#)  
*This type contains the Dma Ip Hardware Channel State. [More...](#)*
- struct [Dma\\_Ip\\_MultiCoreConfigType](#)  
*This type contains the Dma Ip Multicore configuration Type. [More...](#)*
- struct [Dma\\_Ip\\_VirtualSectionConfigType](#)  
*This type contains the Dma Ip Virtual memory section configuration Type. [More...](#)*
- struct [Dma\\_Ip\\_VirtualMemoryConfigType](#)  
*This type contains the Dma Ip Virtual memory configuration Type. [More...](#)*
- struct [Dma\\_Ip\\_InitType](#)  
*This type contains the Dma Ip Initialization. [More...](#)*

#### Types Reference

- typedef void(\* [Dma\\_Ip\\_Callback](#)) (void)  
*This type contains the Dma Ip Callback interface.*

## Enum Reference

- enum [Dma\\_Ip\\_LogicInstanceCmdType](#)  
*This type contains the Dma Ip Logic Instance Commands.*
- enum [Dma\\_Ip\\_LogicChannelCmdType](#)  
*This type contains the Dma Ip Logic Channel Commands.*
- enum [Dma\\_Ip\\_LogicChannelGlobalParamType](#)  
*This type contains the Dma Ip Logic Channel Global Parameters.*
- enum [Dma\\_Ip\\_LogicChannelTransferParamType](#)  
*This type contains the Dma Ip Logic Channel Transfer Parameters.*
- enum [Dma\\_Ip\\_LogicChannelCrcParamType](#)  
*This type contains the Dma Ip Logic Channel Crc Parameters.*
- enum [Dma\\_Ip\\_LogicChannelInfoParamType](#)  
*This type contains the Dma Ip Logic Channel Information Parameters.*
- enum [Dma\\_Ip\\_ReturnType](#)  
*This type contains the Dma Ip Return Type.*
- enum [Dma\\_Ip\\_HwChannelStateValueType](#)  
*This type contains the Dma Ip Channel State Value Type.*

## Function Reference

- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_Init](#) (const [Dma\\_Ip\\_InitType](#) \*const DmaInit)  
*This function initializes the Dma Ip Driver.*
- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_Deinit](#) (void)  
*This function deinitializes the Dma Ip Driver.*
- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_SetLogicInstanceCommand](#) (const uint32 LogicInst, const [Dma\\_Ip\\_LogicInstanceCmdType](#) Command)  
*This function sets Dma Ip Instance Command.*
- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_GetLogicInstanceStatus](#) (const uint32 LogicInst, [Dma\\_Ip\\_LogicInstanceStatusType](#) \*const Status)  
*This function gets Dma Ip Instance Status.*
- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_LogicChannelInit](#) (const uint32 LogicCh)  
*This function initializes the Dma Ip Logic Channel.*
- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_LogicChannelDeinit](#) (const uint32 LogicCh)  
*This function deinitializes the Dma Ip Logic Channel.*
- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_SetLogicChannelCommand](#) (const uint32 LogicCh, const [Dma\\_Ip\\_LogicChannelCmdType](#) Command)  
*This function sets Dma Ip Logic Channel Command.*
- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_GetLogicChannelStatus](#) (const uint32 LogicCh, [Dma\\_Ip\\_LogicChannelStatusType](#) \*const ChStatus)  
*This function gets Dma Ip Logic Channel Status.*
- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_SetLogicChannelGlobalList](#) (const uint32 LogicCh, const [Dma\\_Ip\\_LogicChannelGlobalListType](#) List[], const uint32 ListDimension)  
*This function sets Dma Ip Logic Channel Global List settings.*
- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_SetLogicChannelTransferList](#) (const uint32 LogicCh, const [Dma\\_Ip\\_LogicChannelTransferParamType](#) List[], const uint32 ListDimension)

*This function sets Dma Ip Logic Channel Transfer List settings.*

- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_SetLogicChannelScatterGatherList](#) (const uint32 LogicCh, const uint32 Element, const [Dma\\_Ip\\_LogicChannelScatterGatherListType](#) List[], const uint32 ListDimension)

*This function sets Dma Ip Logic Channel Scatter/Gather List settings.*

- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_SetLogicChannelCrcList](#) (const uint32 LogicCh, const [Dma\\_Ip\\_LogicChannelCrcListType](#) List[], const uint32 ListDimension)

*This function sets Dma Ip Logic Channel Crc List settings.*

- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_GetLogicChannelParam](#) (const uint32 LogicCh, const [Dma\\_Ip\\_LogicChannelInfoParamType](#) Param, uint32 \*const Value)

*This function gets the Dma Ip Logic Channel Parameter value.*

- [Dma\\_Ip\\_ReturnType Dma\\_Ip\\_SetLogicChannelScatterGatherConfig](#) (const uint32 LogicCh, const uint32 Element)

*This function configures the Dma Ip Logic Channel Scatter/Gather.*

### 6.3.2 Data Structure Documentation

#### 6.3.2.1 struct Dma\_Ip\_LogicChannelGlobalListType

This type contains the Dma Ip Logic Channel Global List.

The Dma Ip Channel Global List contains a pair composed from Dma Channel Global Parameter Type and the Value of the parameter. The Dma Ip Channel Global Parameter Type selects a parameter form the Global Parameter enum type. The Value stores the parameter's value.

Definition at line 240 of file Dma\_Ip.h.

#### 6.3.2.2 struct Dma\_Ip\_LogicChannelTransferListType

This type contains the Dma Ip Channel Transfer List.

The Dma Ip Channel Transfer List contains a pair composed from Dma Channel Transfer Parameter Type and the Value of the parameter. The Dma Ip Channel Transfer Parameter Type selects a parameter form the Transfer Parameter enum type. The Value stores the parameter's value.

This type contains the Dma Ip Channel ScatterGather List.

The Dma Ip Channel Transfer List contains a pair composed from Dma Channel ScatterGather Parameter Type and the Value of the parameter. The Dma Ip Channel ScatterGather Parameter Type selects a parameter form the ScatterGather Parameter enum type. The Value stores the parameter's value.

Definition at line 263 of file Dma\_Ip.h.

#### 6.3.2.3 struct Dma\_Ip\_LogicChannelCrcListType

This type contains the Dma Ip Channel ScatterGather List.

The Dma Ip Channel Transfer List contains a pair composed from Dma Channel ScatterGather Parameter Type and the Value of the parameter. The Dma Ip Channel ScatterGather Parameter Type selects a parameter form the ScatterGather Parameter enum type. The Value stores the parameter's value.

Definition at line 278 of file Dma\_Ip.h.

### 6.3.2.4 struct Dma\_Ip\_LogicInstanceStatusType

This type contains the Dma Ip Instance Status.

The Dma Ip Instance Status contains the Hardware Errors, Active Id and Active indication for the running Dma Channel. The Errors shall contain the Hardware Errors. The ActiveId shall contain the running Dma Channel Id. The Active shall contain the running Dma Channel Active status.

Definition at line 294 of file Dma\_Ip.h.

#### Data Fields

- uint32 [Errors](#)  
[VALUE] The Errors value is read from the DMA Instance Error Register (ES) as it is.
- uint8 [ActiveId](#)  
[VALUE] The ActiveId value is read from the DMA Instance Control Register (CR) field ACTIVE\_ID.
- boolean [Active](#)  
[BOOLEAN] The Active value is read from the DMA Instance Control Register (CR) field ACTIVE.

#### 6.3.2.4.1 Field Documentation

##### 6.3.2.4.1.1 Errors uint32 Errors

[VALUE] The Errors value is read from the DMA Instance Error Register (ES) as it is.

Definition at line 295 of file Dma\_Ip.h.

##### 6.3.2.4.1.2 ActiveId uint8 ActiveId

[VALUE] The ActiveId value is read from the DMA Instance Control Register (CR) field ACTIVE\_ID.

Definition at line 296 of file Dma\_Ip.h.

##### 6.3.2.4.1.3 Active boolean Active

[BOOLEAN] The Active value is read from the DMA Instance Control Register (CR) field ACTIVE.

Definition at line 297 of file Dma\_Ip.h.



### 6.3.2.5 struct Dma\_Ip\_LogicChannelStatusType

This type contains the Dma Ip Channel Status.

The Dma Ip Channel Status contains the Hardware Errors, Active status and Done indication for the running Dma Channel. The Channel State Value shall contain the internal driver state of the Dma Channel. The Errors shall contain the Hardware Dma Channel Errors. The Active shall contain the running Dma Channel Id. The Done shall contain the running Dma Channel Active status.

Definition at line 311 of file Dma\_Ip.h.

#### Data Fields

- [Dma\\_Ip\\_HwChannelStateValueType ChStateValue](#)  
*[VALUE] The ChStateValue value is read from the internal DMA Driver Channel State Machine. Check UM for additional information.*
- [uint32 Errors](#)  
*[VALUE] The Errors value is read from the DMA Channel Error Register (CHx\_ES) as it is.*
- [boolean Active](#)  
*[BOOLEAN] The Active value is read from the DMA Channel Control and Status Register (CHx\_CSR) field ACTIVE.*
- [boolean Done](#)  
*[BOOLEAN] The Active value is read from the DMA Channel Control and Status Register (CHx\_CSR) field DONE.*

#### 6.3.2.5.1 Field Documentation

##### 6.3.2.5.1.1 ChStateValue [Dma\\_Ip\\_HwChannelStateValueType ChStateValue](#)

*[VALUE]* The ChStateValue value is read from the internal DMA Driver Channel State Machine. Check UM for additional information.

Definition at line 312 of file Dma\_Ip.h.

##### 6.3.2.5.1.2 Errors [uint32 Errors](#)

*[VALUE]* The Errors value is read from the DMA Channel Error Register (CHx\_ES) as it is.

Definition at line 314 of file Dma\_Ip.h.

#### 6.3.2.5.1.3 Active `boolean Active`

[BOOLEAN] The Active value is read from the DMA Channel Control and Status Register (CHx\_CSR) field ACTIVE.

Definition at line 315 of file Dma\_Ip.h.

#### 6.3.2.5.1.4 Done `boolean Done`

[BOOLEAN] The Active value is read from the DMA Channel Control and Status Register (CHx\_CSR) field DONE.

Definition at line 316 of file Dma\_Ip.h.

### 6.3.2.6 struct Dma\_Ip\_LogicChannelIdType

This type contains the Dma Ip Logic Channel Identification.

The Logic Channel is identified by the following structure: The Logic Channel Id contains the ID value. The Hardware Version Id contains the DMA Hardware Ip Block Version. The Hardware Instance Id contains the DMA Hardware Instance identification. The Hardware Channel Id contains the DMA Hardware Channel identification. The Hardware Crc Instance Id contains the DMA Hardware Crc Instance identification if CRC\_IP is available. The Hardware Crc Channel Id contains the DMA Hardware Crc Channel identification if CRC\_IP is available. The Interrupt Callback stores pointer to the user defined interrupt callback. The Error Interrupt Callback stores pointer to the user defined error interrupt callback.

Definition at line 178 of file Dma\_Ip\_Types.h.

#### Data Fields

- uint32 [LogicChId](#)  
*DMA logic channel number.*
- uint8 [HwVersId](#)  
*DMA hardware version.*
- uint8 [HwInstId](#)  
*DMA hardware instance number.*
- uint8 [HwChId](#)  
*DMA hardware channel number.*
- [Dma\\_Ip\\_Callback](#) [IntCallback](#)  
*The channel callback is installed in the interrupt and is called automatically from the interrupt every time it is triggered.*
- [Dma\\_Ip\\_Callback](#) [ErrIntCallback](#)  
*The channel error callback is installed in the error interrupt and is called automatically from the interrupt every time it is triggered.*

#### 6.3.2.6.1 Field Documentation

### 6.3.2.6.1.1 LogicChId `uint32 LogicChId`

DMA logic channel number.

Definition at line 179 of file `Dma_Ip_Types.h`.

### 6.3.2.6.1.2 HwVersId `uint8 HwVersId`

DMA hardware version.

Definition at line 180 of file `Dma_Ip_Types.h`.

### 6.3.2.6.1.3 HwInstId `uint8 HwInstId`

DMA hardware instance number.

Definition at line 181 of file `Dma_Ip_Types.h`.

### 6.3.2.6.1.4 HwChId `uint8 HwChId`

DMA hardware channel number.

Definition at line 182 of file `Dma_Ip_Types.h`.

### 6.3.2.6.1.5 IntCallback `Dma_Ip_Callback IntCallback`

The channel callback is installed in the interrupt and is called automatically from the interrupt every time it is triggered.

Definition at line 183 of file `Dma_Ip_Types.h`.

### 6.3.2.6.1.6 ErrIntCallback `Dma_Ip_Callback ErrIntCallback`

The channel error callback is installed in the error interrupt and is called automatically from the interrupt every time it is triggered.

Definition at line 185 of file `Dma_Ip_Types.h`.

### 6.3.2.7 struct Dma\_Ip\_GlobalConfigType

This type contains the Dma Ip Global Configuration.

The Global Configuration of an Logic Channel, contains all the information describing a channel, but are not present in the DMA TCD. It contains configuration split in four main areas: Control defines functionality covering the channel's bus control. Request defines functionality covering the channel's request interface, including DMAMUX. Interrupt defines functionality covering special channel interrupts. Priority defines functionality covering the channel's priority mechanism.

Definition at line 201 of file Dma\_Ip\_Types.h.

### 6.3.2.8 struct Dma\_Ip\_TransferConfigType

This type contains the Dma Ip Transfer Configuration.

The Transfer Configuration of an Logic Channel, contains all the information describing a channel transfer functionalities and are present in the DMA TCD. It contains configuration split in five main areas: Control defines functionality covering the channel control functions. It includes additionally the Scatter/Gather Address and Destination Store Address.

Definition at line 236 of file Dma\_Ip\_Types.h.

### 6.3.2.9 struct Dma\_Ip\_ScatterGatherConfigType

This type contains the Dma Ip Scatter/Gather Configuration.

The Scatter/Gather Configuration of a Logic Channel, contains all the information describing the channel's needed resources for a Scatter/Gather element. It contains a pointer to the transfer configuration, a pointer to a Software TCD, a pointer to the next Scatter/Gather configuration and a status flag for loading the transfer configuration into the Software TCD.

Definition at line 294 of file Dma\_Ip\_Types.h.

Data Fields

| Type                                                    | Name           | Description                                                                                                                                                                                               |
|---------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Dma_Ip_TransferConfigType</a> *             | TransferConfig | Configuration that shall be loaded into the Software TCD.                                                                                                                                                 |
| <a href="#">Dma_Ip_SwTcdRegType</a> *                   | Stcd           | Software TCD shall be loaded with own TCD configuration.                                                                                                                                                  |
| boolean                                                 | Loaded         |                                                                                                                                                                                                           |
| struct <a href="#">Dma_Ip_ScatterGatherConfigType</a> * | NextConfig     | Next Logic Channel configuration. If the address is not NULL, then the ESG flag is enabled (Scatter/Gather address) and the destination adjustment is disabled. The next configuration address is stored. |

**6.3.2.10 struct Dma\_Ip\_LogicChannelConfigType**

This type contains the Dma Ip Logic Channel Configuration.

The Logic Channel Configuration consists of the Logic Channel Identifier, pointer to the Global Configuration, pointer to the Transfer Configuration and pointer to the Scatter/Gather configuration. The Logic Channel Configuration contains all data needed to initialize a Logic Channel.

Definition at line 323 of file Dma\_Ip\_Types.h.

Data Fields

| Type                                              | Name                        | Description                                                                                                                |
|---------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Dma_Ip_LogicChannelIdType</a>         | LogicChId                   | The Logic Channel ID contains configuration information and identification.                                                |
| const <a href="#">Dma_Ip_GlobalConfigType</a> *   | pxGlobalConfig              | The Global Configuration pointer shall contain the global data.                                                            |
| const <a href="#">Dma_Ip_TransferConfigType</a> * | pxTransferConfig            | The Transfer Configuration pointer shall contain the transfer data.                                                        |
| <a href="#">Dma_Ip_ScatterGatherConfigType</a> ** | ppxScatterGatherConfigArray | The Scatter/Gather Configuration pointer shall contain a pointer to an array containing all Scatter/Gather Logic Elements. |
| const Dma_Ip_CrcConfigType *                      | pxCrcConfig                 |                                                                                                                            |

**6.3.2.11 struct Dma\_Ip\_LogicInstanceIdType**

This type contains the Dma Ip Logic Instance Identification.

The Logic Instance is identified by the following structure: The Logic Instance Id contains the ID value. The Hardware Version Id contains the DMA Hardware Ip Block Version. The Hardware Instance Id contains the DMA Hardware Instance identification.

Definition at line 342 of file Dma\_Ip\_Types.h.

Data Fields

| Type   | Name        | Description                   |
|--------|-------------|-------------------------------|
| uint32 | LogicInstId | DMA logic instance number.    |
| uint8  | HwVersId    | DMA hardware version number.  |
| uint8  | HwInstId    | DMA hardware instance number. |

**6.3.2.12 struct Dma\_Ip\_LogicInstanceConfigType**

This type contains the Dma Ip Logic Instance Configuration.

The Logic Instance Configuration contains all the information describing an instance functionality.

Definition at line 354 of file Dma\_Ip\_Types.h.

Data Fields

| Type                                       | Name                    | Description                                                                                                                                                                                              |
|--------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Dma_Ip_LogicInstanceIdType</a> | LogicInstId             | DMA logic instance number.                                                                                                                                                                               |
| boolean                                    | EnDebug                 | When in debug mode, the DMA stalls the start of a new channel. Executing channels are allowed to complete. DMA resumes channel execution when the system exits debug mode or clears the EDBG field to 0. |
| boolean                                    | EnRoundRobin            | Enable Round Robin Channel Arbitration.                                                                                                                                                                  |
| boolean                                    | EnHaltAfterError        | When this field is set to 1, any error causes the HALT field to be set to 1. Then all service requests are ignored until the HALT field is cleared to 0.                                                 |
| boolean                                    | EnChLinking             | Global Channel Linking Control.                                                                                                                                                                          |
| boolean                                    | EnGIMasterIdReplication | If master ID replication is disabled, the privileged protection level (Supervisor mode) for DMA transfers is used.                                                                                       |
| boolean                                    | EnSwapBit               | Enable DMA CRC instance swap bit option.                                                                                                                                                                 |
| boolean                                    | EnSwapByte              | Enable DMA CRC instance swap byte option.                                                                                                                                                                |
| boolean                                    | EnGlobal                | Enable DMA CRC instance global enable option.                                                                                                                                                            |

### 6.3.2.13 struct Dma\_Ip\_HwChannelStateType

This type contains the Dma Ip Hardware Channel State.

The Hardware Channel State contains the channel's state based on runtime actions. The structure links the hardware channel state with the Logic Channel.

Definition at line 377 of file Dma\_Ip\_Types.h.

### 6.3.2.14 struct Dma\_Ip\_MultiCoreConfigType

This type contains the Dma Ip Multicore configuration Type.

Definition at line 388 of file Dma\_Ip\_Types.h.

### 6.3.2.15 struct Dma\_Ip\_VirtualSectionConfigType

This type contains the Dma Ip Virtual memory section configuration Type.

Definition at line 401 of file Dma\_Ip\_Types.h.

### 6.3.2.16 struct Dma\_Ip\_VirtualMemoryConfigType

This type contains the Dma Ip Virtual memory configuration Type.

Definition at line 413 of file Dma\_Ip\_Types.h.

### 6.3.2.17 struct Dma\_Ip\_InitType

This type contains the Dma Ip Initialization.

The Dma Ip Initialization contains all the information required to initialize the Dma Peripheral. Each pointer shall be loaded with a specific configuration used by the Dma.

Definition at line 427 of file Dma\_Ip\_Types.h.

## 6.3.3 Types Reference

### 6.3.3.1 Dma\_Ip\_Callback

```
typedef void(* Dma_Ip_Callback) (void)
```

This type contains the Dma Ip Callback interface.

The Callback is defined by the user and installed by the driver in the corresponding IRQ.

Returns

void

Definition at line 162 of file Dma\_Ip\_Types.h.

## 6.3.4 Enum Reference

### 6.3.4.1 Dma\_Ip\_LogicInstanceCmdType

```
enum Dma_Ip_LogicInstanceCmdType
```

This type contains the Dma Ip Logic Instance Commands.

The Commands trigger specific actions in the Dma Logic Instance.

Enumerator

|                                        |                                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| DMA_IP_INST_CANCEL_TRANSFER            | The Cancel Transfer cancels the executing channel and forces the Minor Loop to finish.                                                  |
| DMA_IP_INST_CANCEL_TRANSFER_WITH_ERROR | The Cancel Transfer With Error Command cancels the executing channel, forces the Minor Loop to finish and generates an error interrupt. |
| DMA_IP_INST_HALT                       | The Halt Command allows the ongoing transfer to finish and halts any new transfer.                                                      |
| DMA_IP_INST_RESUME                     | The Resume Command allows the transfer to continue.                                                                                     |

Definition at line 104 of file Dma\_Ip.h.

#### 6.3.4.2 Dma\_Ip\_LogicChannelCmdType

```
enum Dma_Ip_LogicChannelCmdType
```

This type contains the Dma Ip Logic Channel Commands.

The Commands trigger specific actions in the Dma Ip Logic Channel.

Enumerator

|                                  |                                                                                                   |
|----------------------------------|---------------------------------------------------------------------------------------------------|
| DMA_IP_CH_SET_HARDWARE_REQUEST   | The Set Hardware Request Command enables the Dma Channel to be triggered by hardware requests.    |
| DMA_IP_CH_CLEAR_HARDWARE_REQUEST | The Clear Hardware Request Command disables the Dma Channel to be triggered by hardware requests. |
| DMA_IP_CH_SET_SOFTWARE_REQUEST   | The Set Software Request Command sends a soft start request to the Dma Channel.                   |
| DMA_IP_CH_CLEAR_DONE             | The Clear Done Command resets the Dma Channel Done status.                                        |
| DMA_IP_CH_CLEAR_ERROR            | The Clear Error Command resets the Dma Channel Error status.                                      |

Definition at line 116 of file Dma\_Ip.h.

#### 6.3.4.3 Dma\_Ip\_LogicChannelGlobalParamType

```
enum Dma_Ip_LogicChannelGlobalParamType
```

This type contains the Dma Ip Logic Channel Global Parameters.

The Parameters set specific functionalities for the Dma Ip Logic Channel.



Enumerator

|                                        |                                                                                                                                                                  |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DMA_IP_CH_SET_EN_MASTER_ID_REPLICATION | [BOOLEAN] The EnMasterIdReplication Parameter sets the Dma Channel to use the same protection level and system bus ID of the master programming the Dma Channel. |
| DMA_IP_CH_SET_EN_BUFFERED_WRITES       | [BOOLEAN] The EnBufferedWrites Parameter sets the Dma Channel writes to be bufferable.                                                                           |
| DMA_IP_CH_SET_EN_MUX_SOURCE_REQ        | [BOOLEAN] The EnMuxSource Parameter enables the Dma Channel Mux Source.                                                                                          |
| DMA_IP_CH_SET_MUX_SOURCE_REQ           | [VALUE] The MuxSource Parameter sets the Dma Channel Mux Source value.                                                                                           |
| DMA_IP_CH_SET_EN_MUX_TRIGGER           | [BOOLEAN] The EnMuxTrigger Parameter enables the Dma Channel Mux Trigger.                                                                                        |
| DMA_IP_CH_SET_EN_HARDWARE_REQ          | [BOOLEAN] The EnRequest Parameter enables the Dma Channel Request.                                                                                               |
| DMA_IP_CH_SET_EN_ERROR_INTERRUPT       | [BOOLEAN] The EnError Parameter enables the Dma Channel Error Interrupt.                                                                                         |
| DMA_IP_CH_SET_GROUP_PRIORITY           | [VALUE] The Group Parameter sets the Dma Channel Group Priority.                                                                                                 |
| DMA_IP_CH_SET_LEVEL_PRIORITY           | [VALUE] The Level Parameter sets the Dma Channel Level Priority.                                                                                                 |
| DMA_IP_CH_SET_EN_PREEMPTION_PRIORITY   | [BOOLEAN] The EnPreemption Parameter enables the Dma Channel Preemption.                                                                                         |
| DMA_IP_CH_SET_DIS_PREEMPT_PRIORITY     | [BOOLEAN] The DisPreempt Parameter disables the Dma Channel Preempt.                                                                                             |

Definition at line 129 of file Dma\_Ip.h.

#### 6.3.4.4 Dma\_Ip\_LogicChannelTransferParamType

```
enum Dma_Ip_LogicChannelTransferParamType
```

This type contains the Dma Ip Logic Channel Transfer Parameters.

The Parameters set specific functionalities.

Enumerator

|                                           |                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| DMA_IP_CH_SET_SOURCE_ADDRESS              | [VALUE] The Source Address Parameter sets the Dma Channel source address value.                                         |
| DMA_IP_CH_SET_SOURCE_SIGNED_OFFSET        | [VALUE] The Source Signed Offset Parameter sets the Dma Channel source signed offset value.                             |
| DMA_IP_CH_SET_SOURCE_SIGNED_LAST_ADDR_ADJ | [VALUE] The Source Signed Last Address Adjustment Parameter sets the Dma Channel source signed last address adjustment. |

## Enumerator

|                                                     |                                                                                                                                                |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| DMA_IP_CH_SET_SOURCE_TRANSFER_SIZE                  | [VALUE] The Source Transfer Size Parameter sets the Dma Channel source transfer size.                                                          |
| DMA_IP_CH_SET_SOURCE_MODULO                         | [VALUE] The Source Modulo Parameter sets the Dma Channel source modulo.                                                                        |
| DMA_IP_CH_SET_DESTINATION_ADDRESS                   | [VALUE] The Destination Address Parameter sets the Dma Channel destination address value.                                                      |
| DMA_IP_CH_SET_DESTINATION_SIGNED_↔<br>OFFSET        | [VALUE] The Destination Signed Offset Parameter sets the Dma Channel destination signed offset value.                                          |
| DMA_IP_CH_SET_DESTINATION_SIGNED_↔<br>LAST_ADDR_ADJ | [VALUE] The Destination Signed Last Address Adjustment Parameter sets the Dma Channel destination signed last address adjustment.              |
| DMA_IP_CH_SET_DESTINATION_TRANSFER_↔<br>R_SIZE      | [VALUE] The Destination Transfer Size Parameter sets the Dma Channel destination transfer size.                                                |
| DMA_IP_CH_SET_DESTINATION_MODULO                    | [VALUE] The Destination Modulo Parameter sets the Dma Channel destination modulo.                                                              |
| DMA_IP_CH_SET_MINORLOOP_EN_SRC_O↔<br>FFSET          | [BOOLEAN] The Minor Loop Enable Source Offset Parameter enables the Dma Channel minor loop source offset.                                      |
| DMA_IP_CH_SET_MINORLOOP_EN_DST_O↔<br>FFSET          | [BOOLEAN] The Minor Loop Enable Destination Offset Parameter enables the Dma Channel minor loop destination offset.                            |
| DMA_IP_CH_SET_MINORLOOP_SIGNED_O↔<br>FFSET          | [VALUE] The Minor Loop Signed Offset Parameter sets the Dma Channel minor loop signed offset.                                                  |
| DMA_IP_CH_SET_MINORLOOP_EN_LINK                     | [BOOLEAN] The Minor Loop Enable Link Parameter enables the Dma Channel minor loop logic channel linking.                                       |
| DMA_IP_CH_SET_MINORLOOP_LOGIC_LINK_↔<br>K_CH        | [VALUE] The Minor Loop Logic Channel Link Parameter sets the Dma Channel minor loop logic channel link.                                        |
| DMA_IP_CH_SET_MINORLOOP_SIZE                        | [VALUE] The Minor Loop Size Parameter sets the Dma Channel minor loop transfer size.                                                           |
| DMA_IP_CH_SET_MAJORLOOP_EN_LINK                     | [BOOLEAN] The Major Loop Enable Link Parameter enables the Dma Channel major loop logic channel linking.                                       |
| DMA_IP_CH_SET_MAJORLOOP_LOGIC_LINK_↔<br>NK_CH       | [VALUE] The Major Loop Logic Channel Link Parameter sets the Dma Channel major loop logic channel link.                                        |
| DMA_IP_CH_SET_MAJORLOOP_COUNT                       | [VALUE] The Major Loop Count Parameter sets the Dma Channel major loop count.                                                                  |
| DMA_IP_CH_SET_CONTROL_STORE_DST_↔<br>ADDR           | [VALUE] The Store Destination Address Parameter saves the final destination address in system memory.                                          |
| DMA_IP_CH_SET_CONTROL_SOFTWARE_↔<br>REQUEST         | [BOOLEAN] The Enable Start Parameter enables the Dma Channel start service request. The main usage is for ScatterGather Element configuration. |
| DMA_IP_CH_SET_CONTROL_EN_MAJOR_I↔<br>NTERRUPT       | [BOOLEAN] The Enable Major Interrupt Parameter enables the Dma Channel major interrupt.                                                        |
| DMA_IP_CH_SET_CONTROL_EN_HALF_M↔<br>AJOR_INTERRUPT  | [BOOLEAN] The Enable Half Interrupt Parameter enables the Dma Channel half major interrupt.                                                    |

Enumerator

|                                        |                                                                                               |
|----------------------------------------|-----------------------------------------------------------------------------------------------|
| DMA_IP_CH_SET_CONTROL_DIS_AUTO_REQUEST | [BOOLEAN] The Disable Automatic Request Parameter disables the Dma Channel automatic request. |
| DMA_IP_CH_SET_CONTROL_BANDWIDTH        | [VALUE] The Bandwidth Control Parameter sets the Dma Channel bandwidth control.               |

Definition at line 158 of file Dma\_Ip.h.

6.3.4.5 Dma\_Ip\_LogicChannelCrcParamType

enum Dma\_Ip\_LogicChannelCrcParamType

This type contains the Dma Ip Logic Channel Crc Parameters.

The Parameters set specific CRC information.

Definition at line 197 of file Dma\_Ip.h.

6.3.4.6 Dma\_Ip\_LogicChannelInfoParamType

enum Dma\_Ip\_LogicChannelInfoParamType

This type contains the Dma Ip Logic Channel Information Parameters.

The Parameters get specific information.

Enumerator

|                                    |                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------|
| DMA_IP_CH_GET_SOURCE_ADDRESS       | [VALUE] The Source Address Parameter gets the Dma Channel source address.                        |
| DMA_IP_CH_GET_DESTINATION_ADDRESS  | [VALUE] The Destination Address Parameter gets the Dma Channel destination address.              |
| DMA_IP_CH_GET_BEGIN_ITER_COUNT     | [VALUE] The Begin Iteration Count Parameter gets the Dma Channel begin iteration count.          |
| DMA_IP_CH_GET_CURRENT_ITER_COUNT   | [VALUE] The Current Iteration Count Parameter gets the Dma Channel current iteration count.      |
| DMA_IP_CH_GET_STORE_DST_ADDR       | [VALUE] The Store Destination Address Parameter gets the Dma Channel stored destination address. |
| DMA_IP_CH_GET_MASTER_ID            | [VALUE] The Master Id Parameter gets the Dma Channel master id.                                  |
| DMA_IP_CH_GET_MAJOR_INTERRUPT      | [BOOLEAN] The Major Interrupt Parameter gets the Dma Channel major interrupt.                    |
| DMA_IP_CH_GET_HALF_MAJOR_INTERRUPT | [BOOLEAN] The Half Major Interrupt Parameter gets the Dma Channel half major interrupt.          |

150S32MCLDRDmaNXP Semiconductors

Definition at line 211 of file Dma\_Ip.h.

#### 6.3.4.7 Dma\_Ip\_ReturnType

```
enum Dma_Ip_ReturnType
```

This type contains the Dma Ip Return Type.

The Return Type give information for the execution of interfaces.

Definition at line 126 of file Dma\_Ip\_Types.h.

#### 6.3.4.8 Dma\_Ip\_HwChannelStateValueType

```
enum Dma_Ip_HwChannelStateValueType
```

This type contains the Dma Ip Channel State Value Type.

The Channel State type provides information about the channel's general state. The Reset State is present when the Dma Channel is uninitialized. The Ready State is present when the Dma Channel is initialized and without error. The Error State is present when the Dma Channel is initialized and with error.

Definition at line 143 of file Dma\_Ip\_Types.h.

### 6.3.5 Function Reference

#### 6.3.5.1 Dma\_Ip\_Init()

```
Dma_Ip_ReturnType Dma_Ip_Init (
    const Dma_Ip_InitType *const DmaInit )
```

This function initializes the Dma Ip Driver.

This service is a non reentrant function that shall initialize the Dma Ip driver.

Parameters

|    |                |                                         |
|----|----------------|-----------------------------------------|
| in | <i>DmaInit</i> | Pointer to the configuration structure. |
|----|----------------|-----------------------------------------|

Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the initialization finished ok. DMA\_IP\_STATUS\_WRONG\_STATE is returned if the Dma Ip Channel state is not Dma\_Ip\_Ch\_ResetState.

6.3.5.2 Dma\_Ip\_Deinit()

```
Dma_Ip_ReturnType Dma_Ip_Deinit (
    void )
```

This function deinitializes the Dma Ip Driver.

This service is a non reentrant function that shall deinitialize the Dma Ip driver.

Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the deinitialization finished ok

6.3.5.3 Dma\_Ip\_SetLogicInstanceCommand()

```
Dma_Ip_ReturnType Dma_Ip_SetLogicInstanceCommand (
    const uint32 LogicInst,
    const Dma_Ip_LogicInstanceCmdType Command )
```

This function sets Dma Ip Instance Command.

This service is a reentrant function that shall command the Dma Instance. The command shall trigger specific functionalities of the Dma Instance.

Parameters

|    |                  |                                        |
|----|------------------|----------------------------------------|
| in | <i>LogicInst</i> | Selection value of the Logic Instance. |
| in | <i>Command</i>   | The command for the Logic Instance.    |

Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the command finished ok.

#### 6.3.5.4 Dma\_Ip\_GetLogicInstanceStatus()

```
Dma_Ip_ReturnType Dma_Ip_GetLogicInstanceStatus (
    const uint32 LogicInst,
    Dma_Ip_LogicInstanceStatusType *const Status )
```

This function gets Dma Ip Instance Status.

This service is a reentrant function that shall get the Dma Instance status. The command shall read specific functionalities of the Dma Instance.

Parameters

|     |                  |                                        |
|-----|------------------|----------------------------------------|
| in  | <i>LogicInst</i> | Selection value of the Logic Instance. |
| out | <i>Status</i>    | Pointer to the Dma Instance status.    |

Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the get status finished ok.

#### 6.3.5.5 Dma\_Ip\_LogicChannelInit()

```
Dma_Ip_ReturnType Dma_Ip_LogicChannelInit (
    const uint32 LogicCh )
```

This function initializes the Dma Ip Logic Channel.

This service is a non reentrant function that shall initialize the Dma Ip Logic Channel.

Parameters

|    |                |                                       |
|----|----------------|---------------------------------------|
| in | <i>LogicCh</i> | Selection value of the Logic Channel. |
|----|----------------|---------------------------------------|

Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the initialization finished ok. DMA\_IP\_STATUS\_WRONG\_STATE is returned if the Dma Ip Channel state is not Dma\_Ip\_Ch\_ResetState.

#### 6.3.5.6 Dma\_Ip\_LogicChannelDeinit()

```
Dma_Ip_ReturnType Dma_Ip_LogicChannelDeinit (
    const uint32 LogicCh )
```

## Module Documentation

This function deinitializes the Dma Ip Logic Channel.

This service is a non reentrant function that shall deinitialize the Dma Ip Logic Channel.

Parameters

|    |                |                                       |
|----|----------------|---------------------------------------|
| in | <i>LogicCh</i> | Selection value of the Logic Channel. |
|----|----------------|---------------------------------------|

Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the deinitialization finished ok.

#### 6.3.5.7 Dma\_Ip\_SetLogicChannelCommand()

```
Dma_Ip_ReturnType Dma_Ip_SetLogicChannelCommand (
    const uint32 LogicCh,
    const Dma_Ip_LogicChannelCmdType Command )
```

This function sets Dma Ip Logic Channel Command.

This service is a reentrant function that shall command the Dma Channel. The command shall trigger specific functionalities of the Dma Channel.

Parameters

|    |                |                                       |
|----|----------------|---------------------------------------|
| in | <i>LogicCh</i> | Selection value of the Logic Channel. |
| in | <i>Command</i> | The command for the Logic Channel.    |

Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the initialization finished ok. DMA\_IP\_STATUS\_WRONG\_STATE is returned if the Dma Ip Channel state is not Dma\_Ip\_Ch\_ReadyState.

#### 6.3.5.8 Dma\_Ip\_GetLogicChannelStatus()

```
Dma_Ip_ReturnType Dma_Ip_GetLogicChannelStatus (
    const uint32 LogicCh,
    Dma_Ip_LogicChannelStatusType *const ChStatus )
```

This function gets Dma Ip Logic Channel Status.

This service is a reentrant function that shall get the Dma Channel status. The command shall read specific functionalities of the Dma Channel.



### Parameters

|     |                |                                       |
|-----|----------------|---------------------------------------|
| in  | <i>LogicCh</i> | Selection value of the Logic Channel. |
| out | <i>Status</i>  | Pointer to the Dma Channel status.    |

### Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the get status finished ok.

### 6.3.5.9 Dma\_Ip\_SetLogicChannelGlobalList()

```
Dma_Ip_ReturnType Dma_Ip_SetLogicChannelGlobalList (
    const uint32 LogicCh,
    const Dma_Ip_LogicChannelGlobalListType List[],
    const uint32 ListDimension )
```

This function sets Dma Ip Logic Channel Global List settings.

This service is a reentrant function that shall set the Dma Ip Logic Channel global parameters list. The list is composed of an array of Dma Ip Logic Channel global parameters settings. The settings list(array) is defined by the user needs: it contains the desired parameters to be configured, in any desired order.

How to use this interface:

1. Use the "Dma\_Ip\_LogicChannelGlobalListType" to create a list(array) with the desired paramaters to configure (see parameters: "Dma\_Ip\_LogicChannelGlobalParamType") The list can declared globally or locally: A. Global example: `Dma_Ip_LogicChannelGlobalListType global_Dma_Ip_ChannelGlobalList0[NUMBER_OF_PARAMETERS] = {...}`; B. Local example: `Dma_Ip_LogicChannelGlobalListType Dma_Ip_ChannelGlobalList[NUMBER_OF_PARAMETERS]; Dma_Ip_ChannelGlobalList[PARAMETER0].Param = DMA_IP_CH_SET_EN_PREEMPTION_PRIORITY; Dma_Ip_ChannelGlobalList[PARAMETER0].Value = TRUE; Dma_Ip_ChannelGlobalList[PARAMETER1].Param = ...; Dma_Ip_ChannelGlobalList[PARAMETER1].Value = ...;`
2. Call the "Dma\_Ip\_SetLogicChannelGlobalList()" interface: `Dma_Ip_SetLogicChannelGlobalList(LOGIC_CHANNELx, Dma_Ip_ChannelGlobalList, NUMBER_OF_PARAMETERS);`

### Parameters

|    |                      |                                   |
|----|----------------------|-----------------------------------|
| in | <i>Channel</i>       | Specifies the Logic Channel Id.   |
| in | <i>List</i>          | Pointer to the Global List Array. |
| in | <i>ListDimension</i> | Number of entries in the List.    |

## Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the initialization finished ok. DMA\_IP\_STATUS\_WRONG\_STATE is returned if the Dma Ip Channel state is not Dma\_Ip\_Ch\_ReadyState.

### 6.3.5.10 Dma\_Ip\_SetLogicChannelTransferList()

```
Dma_Ip_ReturnType Dma_Ip_SetLogicChannelTransferList (
    const uint32 LogicCh,
    const Dma_Ip_LogicChannelTransferListType List[],
    const uint32 ListDimension )
```

This function sets Dma Ip Logic Channel Transfer List settings.

-> This service is a reentrant function that shall set the Dma Ip Logic Channel transfer parameters list. -> The "Transfer List" loads the configuration directly into the Hardware TCD and disables the ScatterGather for the Hardware TCD. -> The list is composed of an array of Dma Ip Logic Channel transfer parameters settings. -> The settings array is defined by the user needs: it contains entries for each desired parameter to be configured, in any suitable order.

-> How to use this interface: <-

1. Use the "Dma\_Ip\_LogicChannelTransferListType" to create a list(array) with the desired parameters to configure (see parameters: "Dma\_Ip\_LogicChannelTransferParamType") The list can be declared globally or locally: Global example: #define DMA\_IP\_TRANSFER\_LIST0\_DIMENSION ((uint32)2U) `Dma_Ip_LogicChannelTransferListType` global Dma\_Ip\_ChannelTransferList0[DMA\_IP\_TRANSFER\_LIST0\_DIMENSION] = {...}; Local example: #define DMA\_IP\_TRANSFER\_LIST0\_DIMENSION ((uint32)2U) `Dma_Ip_LogicChannelTransferListType` Dma\_Ip\_ChannelTransferList0[DMA\_IP\_TRANSFER\_LIST0\_DIMENSION]; Dma\_Ip\_ChannelTransferList0[PARAMETER0].Param = DMA\_IP\_CH\_SET\_VAL\_SOURCE\_ADDRESS; Dma\_Ip\_ChannelTransferList0[PARAMETER0].Value = Dma\_Ip\_ChannelTransferList0[PARAMETER1].Param = DMA\_IP\_CH\_SET\_DESTINATION\_ADDRESS; Dma\_Ip\_ChannelTransferList0[PARAMETER1].Value =
2. Call the "Dma\_Ip\_SetLogicChannelTransferList()" interface: `Dma_Ip_SetLogicChannelTransferList(LOGIC_CHANNELx, Dma_Ip_ChannelTransferList0, DMA_IP_TRANSFER_LIST0_DIMENSION);`

-> Coding Example: <- -> The user shall create the desired configuration list for his specific application. "UserDefinedFileName.h" #define DMA\_IP\_TRANSFER\_LIST0\_DIMENSION ((uint32)8U) #define DMA\_IP\_SET\_TRANSFER\_TYPE0(CHANNEL, DIMENSION, SADDR, SOFF, SSIZE, DADDR, DOFF, DSIZE, MINOR\_SIZE, MAJOR\_COUNT) \ `Dma_Ip_LogicChannelTransferListType` Dma\_Ip\_ChannelTransferList0[DIMENSION]; \ Dma\_Ip\_ChannelTransferList0[0U].Param = DMA\_IP\_CH\_SET\_VAL\_SOURCE\_ADDRESS; \ Dma\_Ip\_ChannelTransferList0[0U].Value = SADDR; \ Dma\_Ip\_ChannelTransferList0[1U].Param = DMA\_IP\_CH\_SET\_SOURCE\_SIGNED\_OFFSET; \ Dma\_Ip\_ChannelTransferList0[1U].Value = SOFF; \ Dma\_Ip\_ChannelTransferList0[2U].Param = DMA\_IP\_CH\_SET\_SOURCE\_TRANSFER\_SIZE; \ Dma\_Ip\_ChannelTransferList0[2U].Value = SSIZE; \ Dma\_Ip\_ChannelTransferList0[3U].Param = DMA\_IP\_CH\_SET\_DESTINATION\_ADDRESS; \ Dma\_Ip\_ChannelTransferList0[3U].Value = DADDR; \ Dma\_Ip\_ChannelTransferList0[4U].Param = DMA\_IP\_CH\_SET\_DESTINATION\_SIGNED\_OFFSET; \ Dma\_Ip\_ChannelTransferList0[4U].Value = DOFF; \ Dma\_Ip\_ChannelTransferList0[5U].Param = DMA\_IP\_CH\_SET\_DESTINATION\_TRANSFER\_SIZE;

```
\ Dma_Ip_ChannelTransferList0[5U].Value = DSIZE; \ Dma_Ip_ChannelTransferList0[6U].Param = DMA_IP_CH_SET_MINORLOOP_SIZE; \ Dma_Ip_ChannelTransferList0[6U].Value = MINOR_SIZE; \ Dma_Ip_ChannelTransferList0[7U].Param = DMA_IP_CH_SET_MAJORLOOP_COUNT; \ Dma_Ip_ChannelTransferList0[7U].Value = MAJOR_COUNT; \ Dma_Ip_SetLogicChannelTransferList(CHANNEL, Dma_Ip_ChannelTransferList0, DIMENSION);
```

"ApplicationFileName.c" void ConfigureDmaChannel(ConfigType \* pxConfig) { uint32 MinorLoopSize = 2U; uint32 MajorLoopCount; if(pxConfig->MajorLoopCondition == TRUE) { MajorLoopCount = 8U; } else { MajorLoopCount = 24U; } DMA\_IP\_SET\_TRANSFER\_TYPE0(pxConfig->LogicChannel, DMA\_IP\_TRANSFER\_LIST0\_DIMENSION, pxConfig->SourceBuffer, 2U, DMA\_IP\_TRANSFER\_SIZE\_2\_BYTE, &RegisterAddress, 0U, DMA\_IP\_TRANSFER\_SIZE\_2\_BYTE, MinorLoopSize, MajorLoopCount); }

Parameters

|    |                      |                                     |
|----|----------------------|-------------------------------------|
| in | <i>Channel</i>       | Specifies the Logic Channel Id.     |
| in | <i>List</i>          | Pointer to the Transfer List Array. |
| in | <i>ListDimension</i> | Number of entries in the List.      |

Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the initialization finished ok. DMA\_IP\_STATUS\_WRONG\_STATE is returned if the Dma Ip Channel state is not Dma\_Ip\_Ch\_ReadyState.

### 6.3.5.11 Dma\_Ip\_SetLogicChannelScatterGatherList()

```
Dma_Ip_ReturnType Dma_Ip_SetLogicChannelScatterGatherList (
    const uint32 LogicCh,
    const uint32 Element,
    const Dma_Ip_LogicChannelScatterGatherListType List[],
    const uint32 ListDimension )
```

This function sets Dma Ip Logic Channel Scatter/Gather List settings.

-> This service is a reentrant function that shall set the Dma Ip Logic Channel scatter/gather parameters list. -> The "Scatter/Gather List" configures Logic Elements belonging to the same Dma Logic Channel. -> The "Scatter/Gather List" loads the configuration into the Software TCD. The Software TCD has the Scatter/Gather Enable set (ESG bit) and the Next Software TCD Address already loaded during the configuration generation process. -> The "Scatter/Gather List" shall not be able to modify the Scatter/Gather Element linkage (reorder of elements in the chain). The linkage of the elements is set only during the configuration process. -> The settings array is defined by the user needs: it contains entries for each desired parameter to be configured, in any suitable order. -> This service does not load the Logic Element into the Hardware TCD. This functionality is covered by Dma\_Ip\_SetLogicChannelScatterGatherConfig.

How to use this interface:

1. Use the "Dma\_Ip\_LogicChannelScatterGatherListType" to create a list(array) with the desired parameters to configure (see parameters: "Dma\_Ip\_LogicChannelTransferParamType") The list can be declared globally or locally: Global example: `Dma_Ip_LogicChannelScatterGatherListType global_Dma_Ip_ChannelScatterGatherList0[NUMBER_OF_PARAMETERS] = {...};` Local example: `Dma_Ip_LogicChannelScatterGatherListType Dma_Ip_ChannelScatterGatherList[NUMBER_OF_PARAMETERS]; Dma_Ip_ChannelScatterGatherList[PARAMETER0].Param = DMA_IP_CH_SET_VAL_SOURCE_ADDRESS; Dma_Ip_ChannelScatterGatherList[PARAMETER0].Value = Dma_Ip_ChannelScatterGatherList[PARAMETER1].Param = DMA_IP_CH_SET_DESTINATION_ADDRESS; Dma_Ip_ChannelScatterGatherList[PARAMETER1].Value =`
2. Call the "Dma\_Ip\_SetLogicChannelScatterGatherList()" interface: `Dma_Ip_SetLogicChannelScatterGatherList(LOGIC_CHANNELx, LOGIC_ELEMENTy, Dma_Ip_ChannelScatterGatherList, NUMBER_OF_PARAMETERS);`

#### Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the initialization finished ok. DMA\_IP\_STATUS\_WRONG\_STATE is returned if the Dma Ip Channel state is not Dma\_Ip\_Ch\_ReadyState.

#### 6.3.5.12 Dma\_Ip\_SetLogicChannelCrcList()

```
Dma_Ip_ReturnType Dma_Ip_SetLogicChannelCrcList (
    const uint32 LogicCh,
    const Dma_Ip_LogicChannelCrcListType List[],
    const uint32 ListDimension )
```

This function sets Dma Ip Logic Channel Crc List settings.

This service is a reentrant function that shall set the Dma Ip Logic Channel crc parameters list. The settings array is defined by the user needs: it contains entries for each desired parameter to be configured, in any suitable order.

How to use this interface:

1. Use the "Dma\_Ip\_LogicChannelCrcListType" to create a list(array) with the desired parameters to configure (see parameters: "Dma\_Ip\_LogicChannelCrcParamType") The list can be declared globally or locally: Global example: `Dma_Ip_LogicChannelCrcListType global_Dma_Ip_ChannelCrcList0[NUMBER_OF_PARAMETERS] = {...};` Local example: `Dma_Ip_LogicChannelCrcListType Dma_Ip_ChannelCrcList[NUMBER_OF_PARAMETERS]; Dma_Ip_ChannelCrcList[PARAMETER0].Param = DMA_IP_CH_SET_CRC_POLYNOMIAL; Dma_Ip_ChannelCrcList[PARAMETER0].Value = ISCSICRC-32C; Dma_Ip_ChannelCrcList[PARAMETER1].Param = DMA_IP_CH_SET_CRC_EN_LOGIC; Dma_Ip_ChannelCrcList[PARAMETER1].Value = TRUE;`
2. Call the "Dma\_Ip\_SetLogicChannelCrcList()" interface: `Dma_Ip_SetLogicChannelCrcList(LOGIC_CHANNELx, Dma_Ip_ChannelCrcList, NUMBER_OF_PARAMETERS);`

#### Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the initialization finished ok. DMA\_IP\_STATUS\_WRONG\_STATE is returned if the Dma Ip Channel state is not Dma\_Ip\_Ch\_ReadyState.

### 6.3.5.13 Dma\_Ip\_GetLogicChannelParam()

```
Dma_Ip_ReturnType Dma_Ip_GetLogicChannelParam (
    const uint32 LogicCh,
    const Dma_Ip_LogicChannelInfoParamType Param,
    uint32 *const Value )
```

This function gets the Dma Ip Logic Channel Parameter value.

This service is a reentrant function that shall get the Dma Channel parameters value.

Parameters

|     |                |                                       |
|-----|----------------|---------------------------------------|
| in  | <i>LogicCh</i> | Selection value of the Logic Channel. |
| in  | <i>Param</i>   | Selection parameter.                  |
| out | <i>Value</i>   | Pointer to the parameter value.       |

Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the get information finished ok.

### 6.3.5.14 Dma\_Ip\_SetLogicChannelScatterGatherConfig()

```
Dma_Ip_ReturnType Dma_Ip_SetLogicChannelScatterGatherConfig (
    const uint32 LogicCh,
    const uint32 Element )
```

This function configures the Dma Ip Logic Channel Scatter/Gather.

This service is a reentrant function that shall configure the Dma Channel scatter/gather functionality. The specified Logic Element (corresponding to a Software TCD) shall be loaded into the Dma Logic Channel's Hardware TCD. The Logic Elements (describing the Software TCDs) form a simple chained list and the "Element" parameter represents the lists's head.

Parameters

|    |                |                                                                    |
|----|----------------|--------------------------------------------------------------------|
| in | <i>LogicCh</i> | Selection value of the Logic Channel.                              |
| in | <i>Element</i> | Selection value of the Logic Element representing the list's head. |

Returns

Dma\_Ip\_ReturnType DMA\_IP\_STATUS\_SUCCESS is returned if the initialization finished ok. DMA\_IP\_STATUS\_WRONG\_STATE is returned if the Dma Ip Channel state is not Dma\_Ip\_Ch\_ReadyState. DMA\_IP\_STATUS\_ERROR is returned if the Dma Ip Channel contains an error.

## 6.4 FTM IP Driver

### 6.4.1 Detailed Description

#### Function Reference

- void [Ftm\\_Mcl\\_Ip\\_SelectCommonTimebase](#) (uint8 Instance, uint32 SyncList)  
*Implementation specific function to updates the Global Timebase bits of configured modules.*

### 6.4.2 Function Reference

#### 6.4.2.1 Ftm\_Mcl\_Ip\_SelectCommonTimebase()

```
void Ftm_Mcl_Ip_SelectCommonTimebase (
    uint8 Instance,
    uint32 SyncList )
```

Implementation specific function to updates the Global Timebase bits of configured modules.

This function is used to set the global timebase bits for modules that support the global timebase feature. The function selects the module that gives the common timebase and the modules that are use this timebase (as bits in syncList). Then it synchronizes the modules. example: syncList is 0x0003 - modules 0 and 1 use the timebase given by instance syncList is 0x0005 - modules 0 and 2 use the timebase given by instance

Parameters

|    |                 |                       |
|----|-----------------|-----------------------|
| in | <i>Instance</i> | FTM module id         |
| in | <i>SyncList</i> | FTM module mask value |

Returns

void

**How to Reach Us:**

**Home Page:**

[nxp.com](http://nxp.com)

**Web Support:**

[nxp.com/support](http://nxp.com/support)

Information in this document is provided solely to enable system and software implementers to use NXP products. There are no express or implied copyright licenses granted hereunder to design or fabricate any integrated circuits based on the information in this document. NXP reserves the right to make changes without further notice to any products herein.

NXP makes no warranty, representation, or guarantee regarding the suitability of its products for any particular purpose, nor does NXP assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation consequential or incidental damages. "Typical" parameters that may be provided in NXP data sheets and/or specifications can and do vary in different applications, and actual performance may vary over time. All operating parameters, including "typicals," must be validated for each customer application by customer's technical experts. NXP does not convey any license under its patent rights nor the rights of others. NXP sells products pursuant to standard terms and conditions of sale, which can be found at the following address: [nxp.com/SalesTermsandConditions](http://nxp.com/SalesTermsandConditions).

NXP, the NXP logo, NXP SECURE CONNECTIONS FOR A SMARTER WORLD, COOLFLUX, EMBRACE, GREENCHIP, HITAG, I2C BUS, ICODE, JCOP, LIFE VIBES, MIFARE, MIFARE CLASSIC, MIFARE DESFire, MIFARE PLUS, MIFARE FLEX, MANTIS, MIFARE ULTRALIGHT, MIFARE4MOBILE, MIGLO, NTAG, ROADLINK, SMARTLX, SMARTMX, STARPLUG, TOPFET, TRENCHMOS, UCODE, Freescale, the Freescale logo, Altivec, C-5, CodeTEST, CodeWarrior, ColdFire, ColdFire+, C-Ware, the Energy Efficient Solutions logo, Kinetis, Layerscape, MagniV, mobileGT, PEG, PowerQUICC, Processor Expert, QorIQ, QorIQ Qonverge, Ready Play, SafeAssure, the SafeAssure logo, StarCore, Symphony, VortiQa, Vybrid, Airfast, BeeKit, BeeStack, CoreNet, Flexis, MXC, Platform in a Package, QUICC Engine, SMARTMOS, Tower, TurboLink, and UMEMS are trademarks of NXP B.V. All other product or service names are the property of their respective owners. ARM, AMBA, ARM Powered, Artisan, Cortex, Jazelle, Keil, SecurCore, Thumb, TrustZone, and Vision are registered trademarks of ARM Limited (or its subsidiaries) in the EU and/or elsewhere. ARM7, ARM9, ARM11, big.LITTLE, CoreLink, CoreSight, DesignStart, Mali, mbed, NEON, POP, Sensinode, Socrates, ULINK and Versatile are trademarks of ARM Limited (or its subsidiaries) in the EU and/or elsewhere. All rights reserved. Oracle and Java are registered trademarks of Oracle and/or its affiliates. The Power Architecture and Power.org word marks and the Power and Power.org logos and related marks are trademarks and service marks licensed by Power.org.

© 2023 NXP B.V.

